Yandex Research

University of Amsterdam

Edinburgh University of Edinburgh Natural Language Processing NLP

# Seq2seq and Attention

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# Plan

- Machine translation task
- Encoder-decoder framework
- Decoding
- Bahdanau attention model
- Attention is all you need: Transformer
- What do encoder self-attention heads learn?
- Attention: other use cases
- Hack of the day!

# Machine translation task (seq2seq)

$\boldsymbol{x} = (x_1, x_2, \ldots, x_{Tx})$ - source sentence

$\boldsymbol{y} = (y_1, y_2, \ldots, y_{Ty})$ - target sentence

Any type of machine translation system can be defined as a function

$$\hat{\boldsymbol{y}} = \text{mt}(\boldsymbol{x})$$

Translation is equivalent to finding a target sentence that maximizes the conditional probability of $\boldsymbol{y}$ given a source sentence $\boldsymbol{x}$, i.e.

$$\arg\max_y p(\boldsymbol{y}|\boldsymbol{x})$$

# Machine translation task (seq2seq)

Machine translation systems create a probabilistic model for the probability of $\boldsymbol{y}$ given $\boldsymbol{x}$,

$$p(\boldsymbol{y}|\boldsymbol{x}, \theta),$$

and find the target sentence that maximizes this probability:

$$\hat{\boldsymbol{y}} = \arg\max_y p(\boldsymbol{y}|\boldsymbol{x}, \theta).$$

($\theta$ – the parameters of the model specifying the probability distribution)

# Machine translation task (seq2seq)

- Modeling

What the model $p(y|x, \theta)$ will look like?

- Learning

We need a method to learn appropriate values for parameters $\theta$ from training data.

- Search

We need to solve the problem of finding the most probable sentence (solving “argmax”)

# Conditional Language Model

Seq2seq directly models probability $P(y|x)$:

$$P(y|x) = P(y_1|x)\, P(y_2|y_1, x)\, P(y_3|y_1, y_2, x) \ldots, P(y_T|y_1, \ldots, y_{T-1}, x)$$

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Probability of next target words so far and source sentence *x*

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Why conditional language model?

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Probability of next target words so far and source sentence *x*

Why conditional language model?

- LM – predicts next word
- Conditional – predictions conditioned on source sentence

# Encoder-Decoder Framework

# RECAP: RNN Language Models

Output distribution

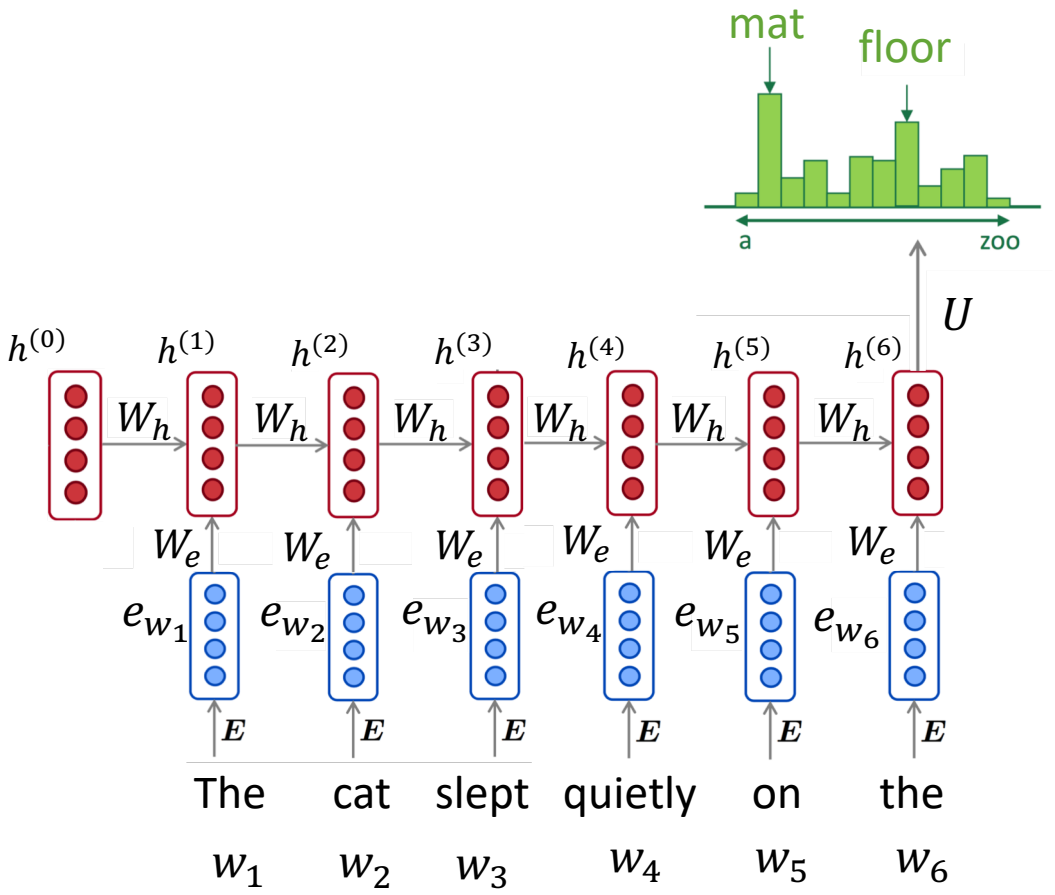
$$\hat{y}_t = \text{softmax}\left(Uh^{(t)} + b_2\right) \in \mathbb{R}^{|V|}$$

Hidden states

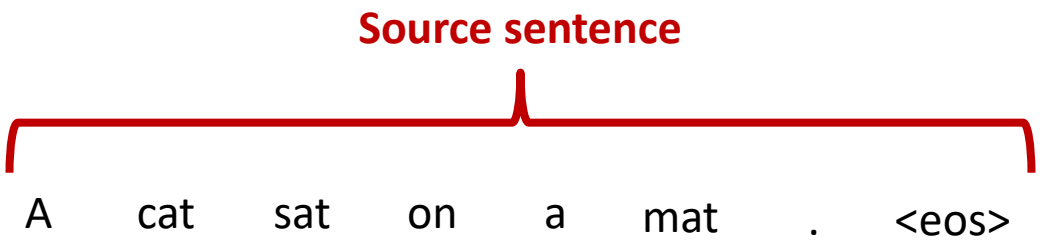
$$h^{(t)} = \sigma\left(W_h h^{(t-1)} + W_e e_{w_{t-1}} + b_1\right)$$

Word embeddings

Words/tokens



# RNN Encoder-Decoder (Vanilla)



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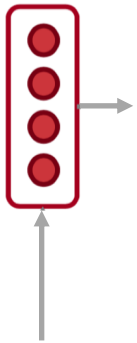

A cat sat on a mat . <eos>

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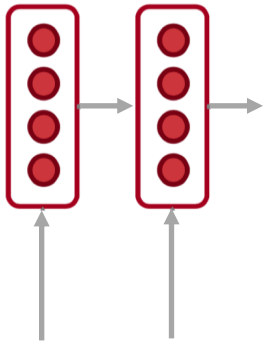

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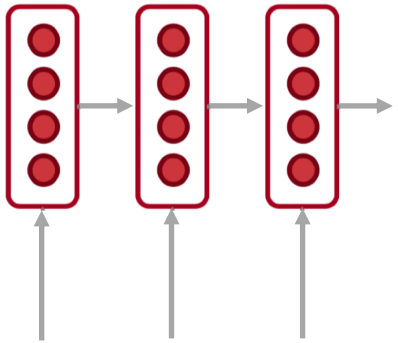

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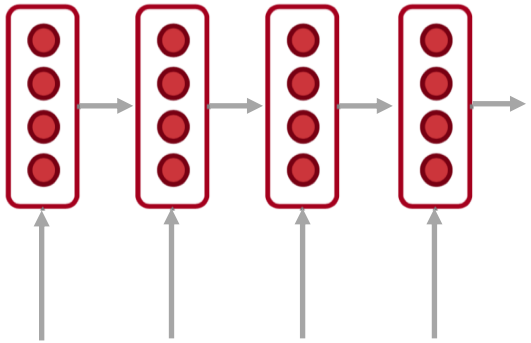

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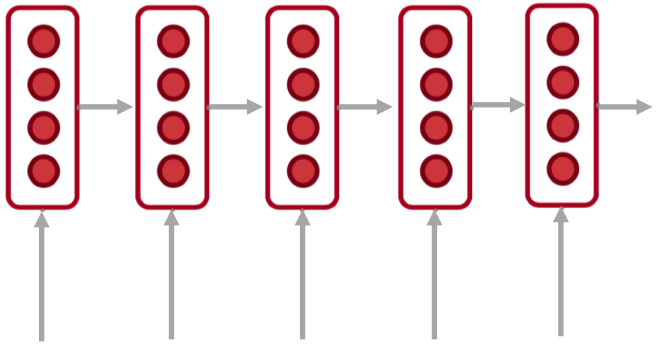

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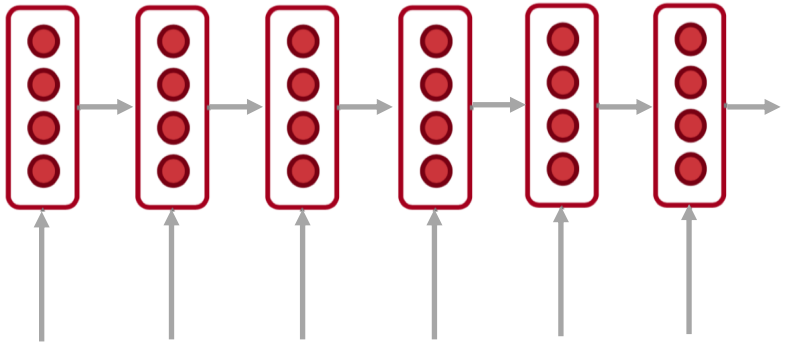

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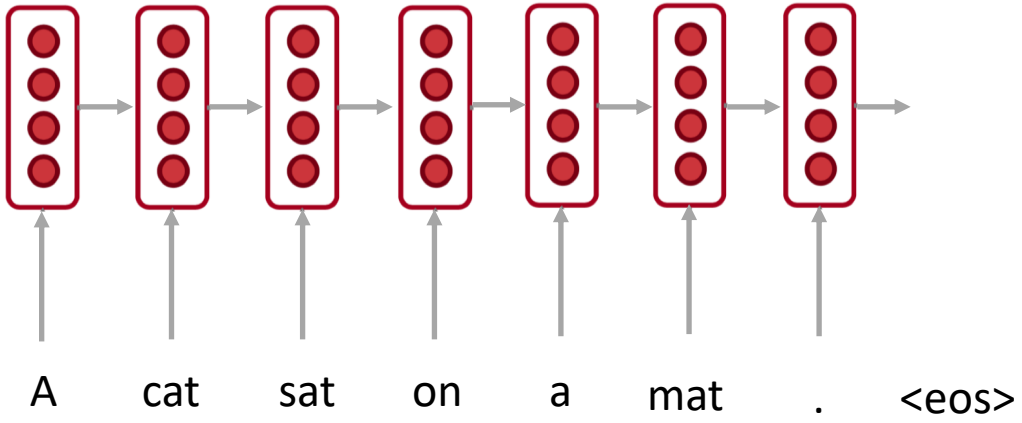

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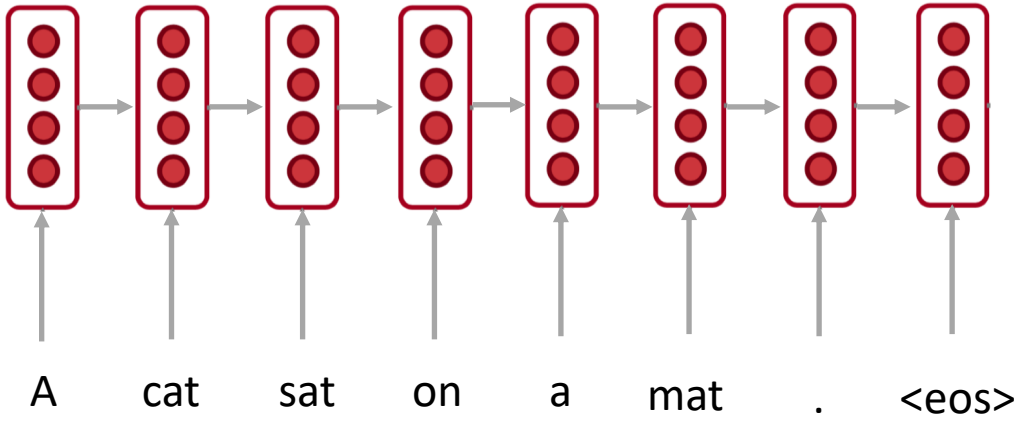

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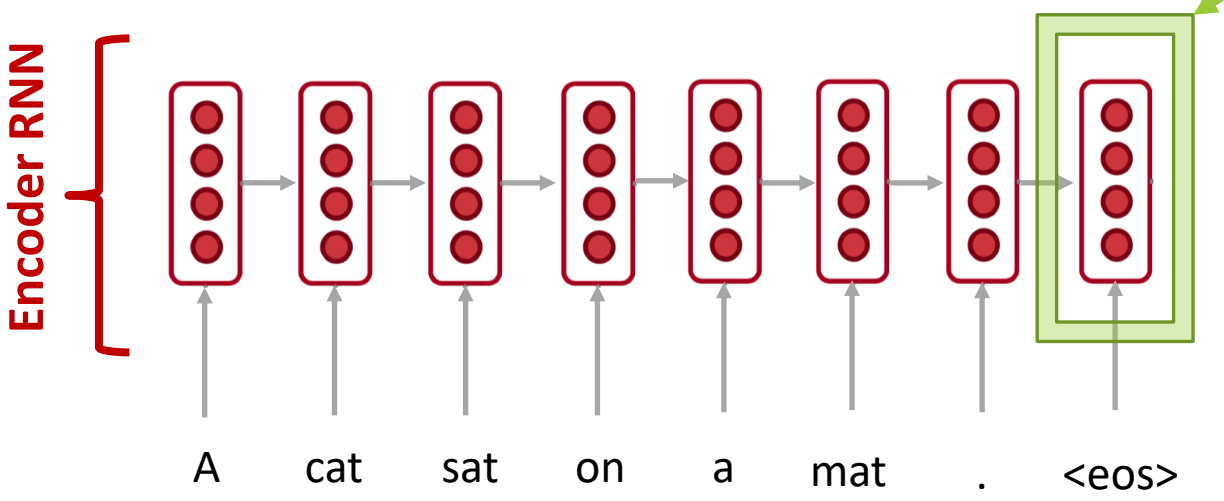
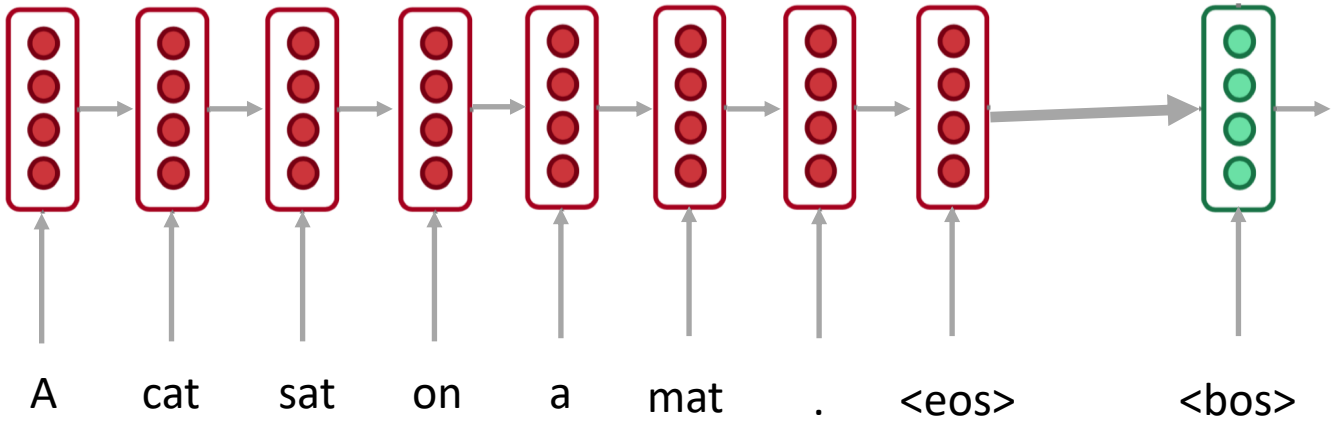



Encoding of a source sentence.
Provides an initial state for decoder RNN

# RNN Encoder-Decoder (Vanilla)



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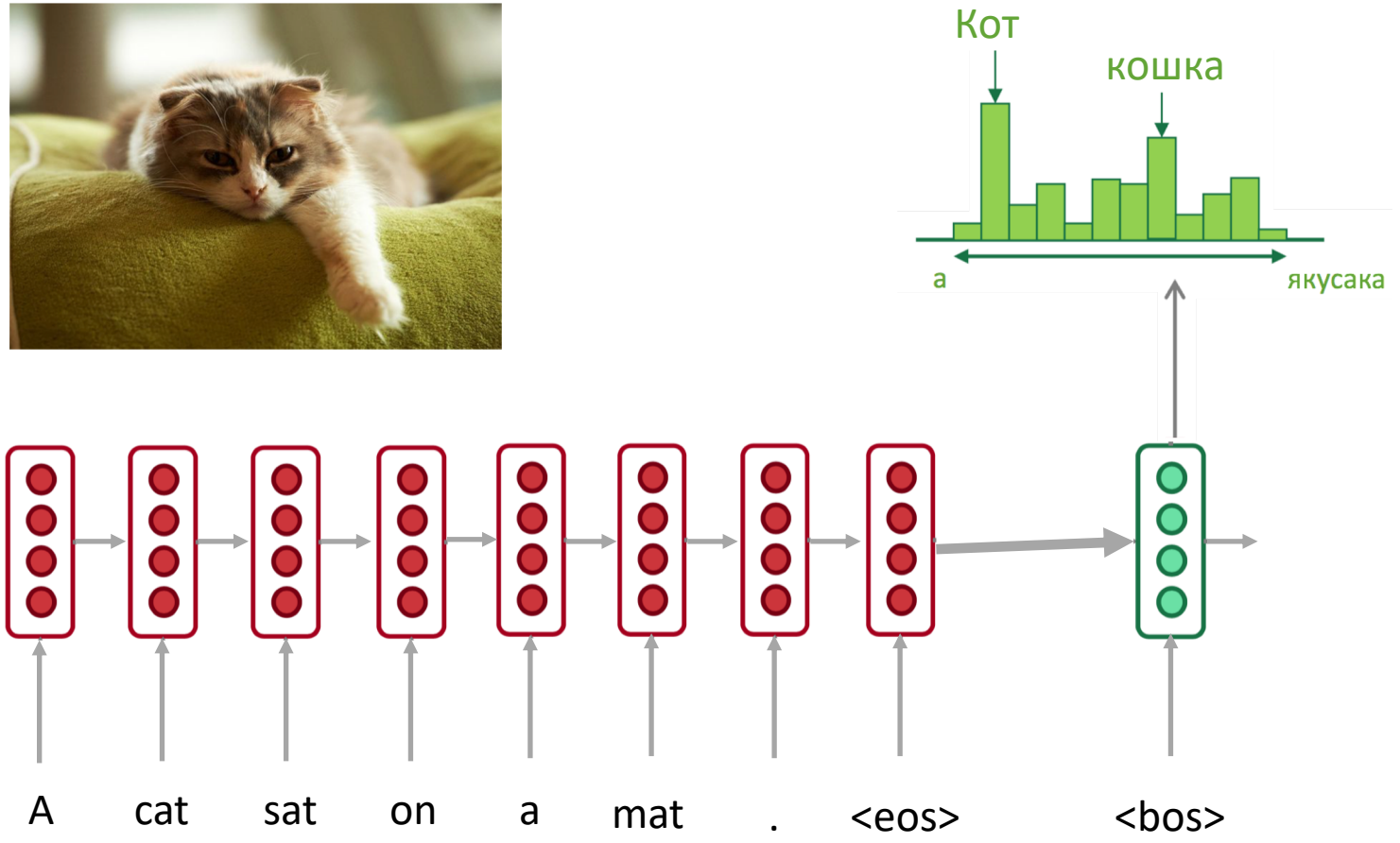

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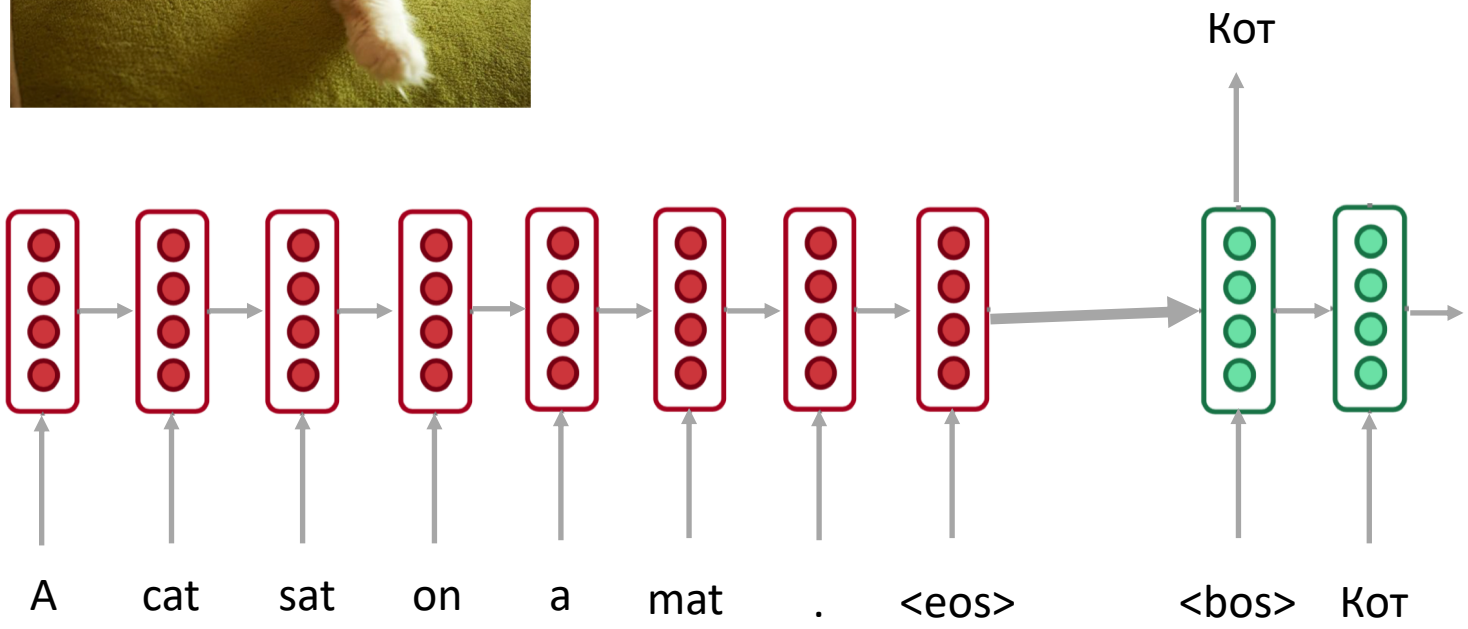

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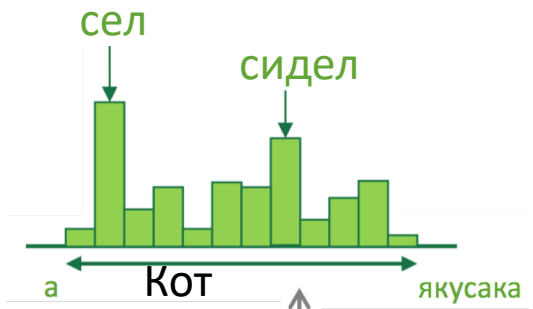


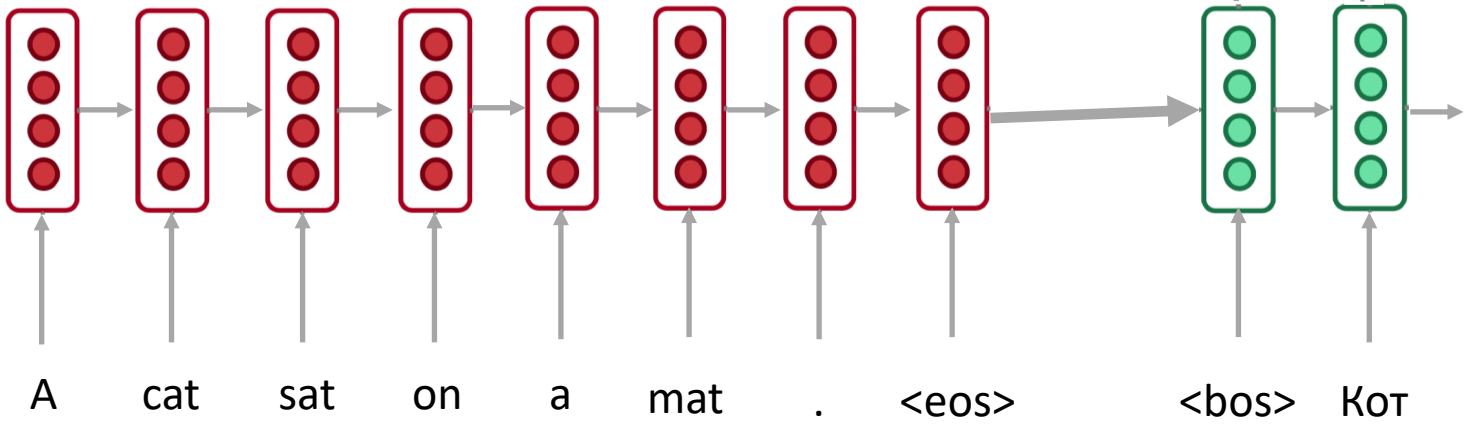

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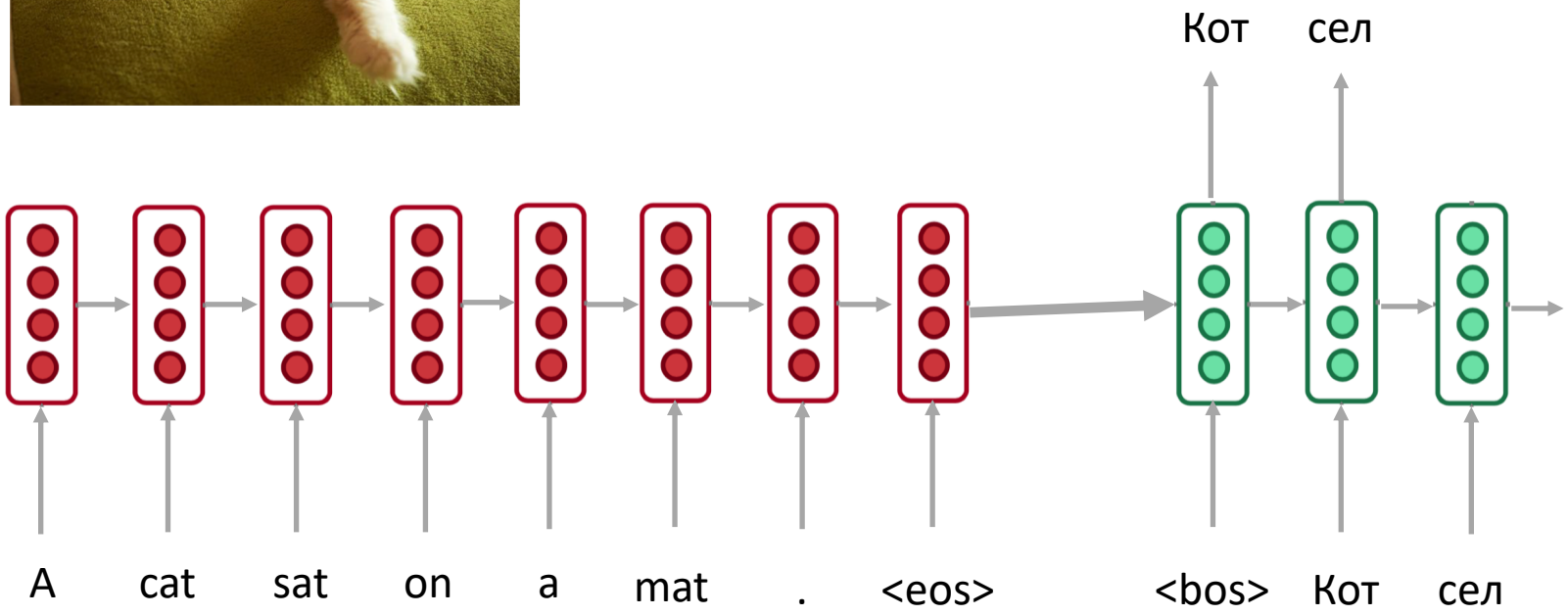

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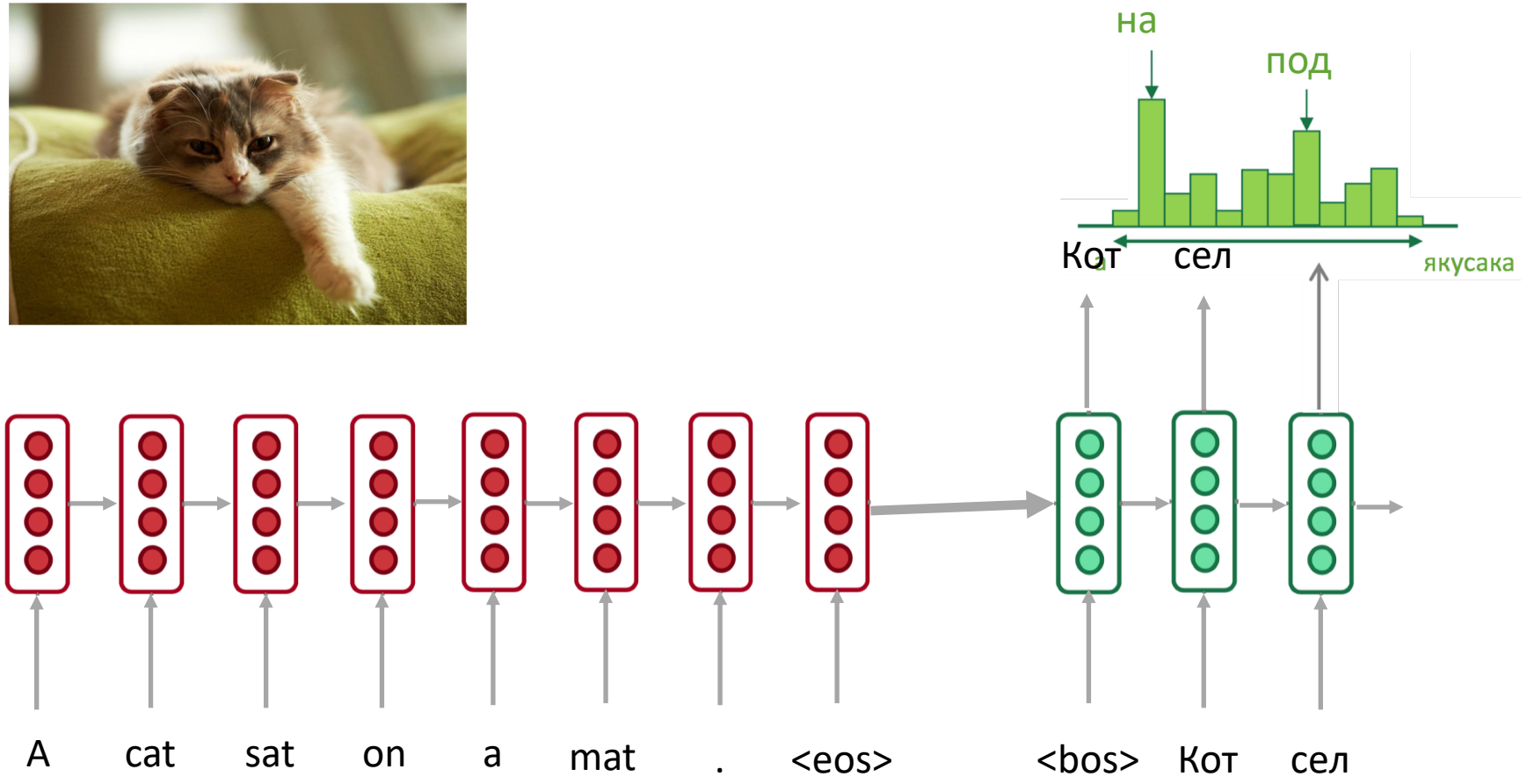

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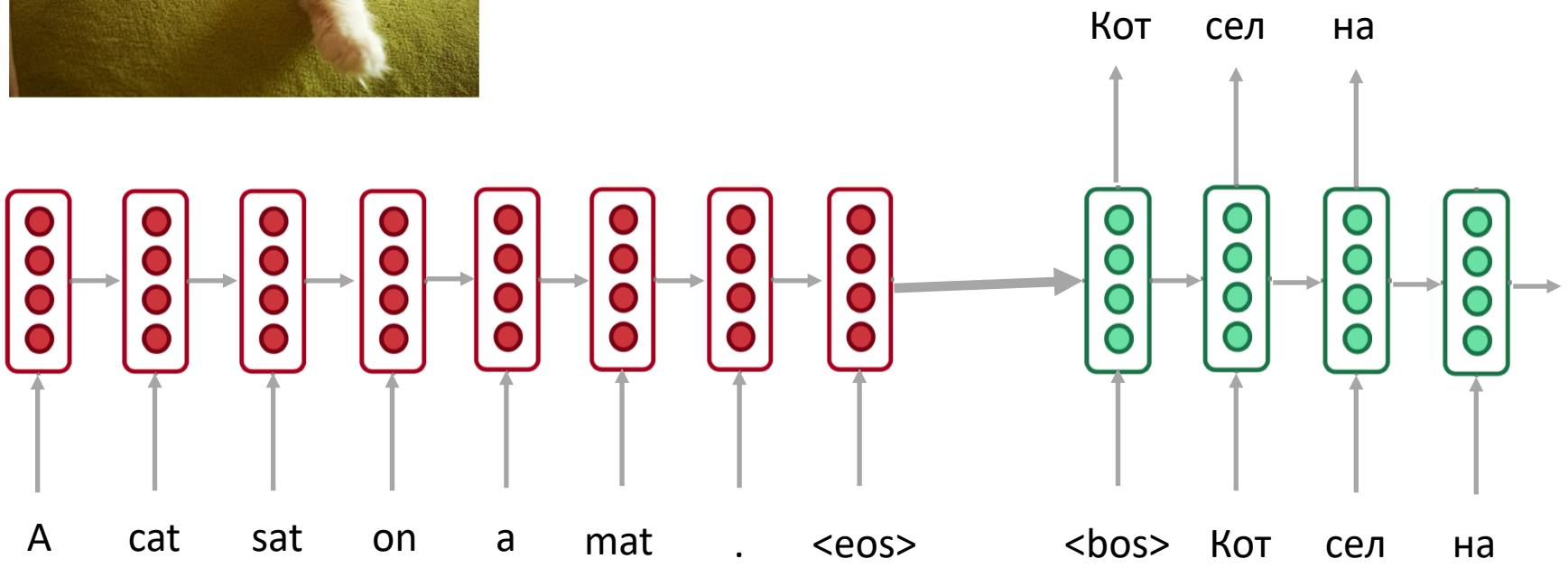

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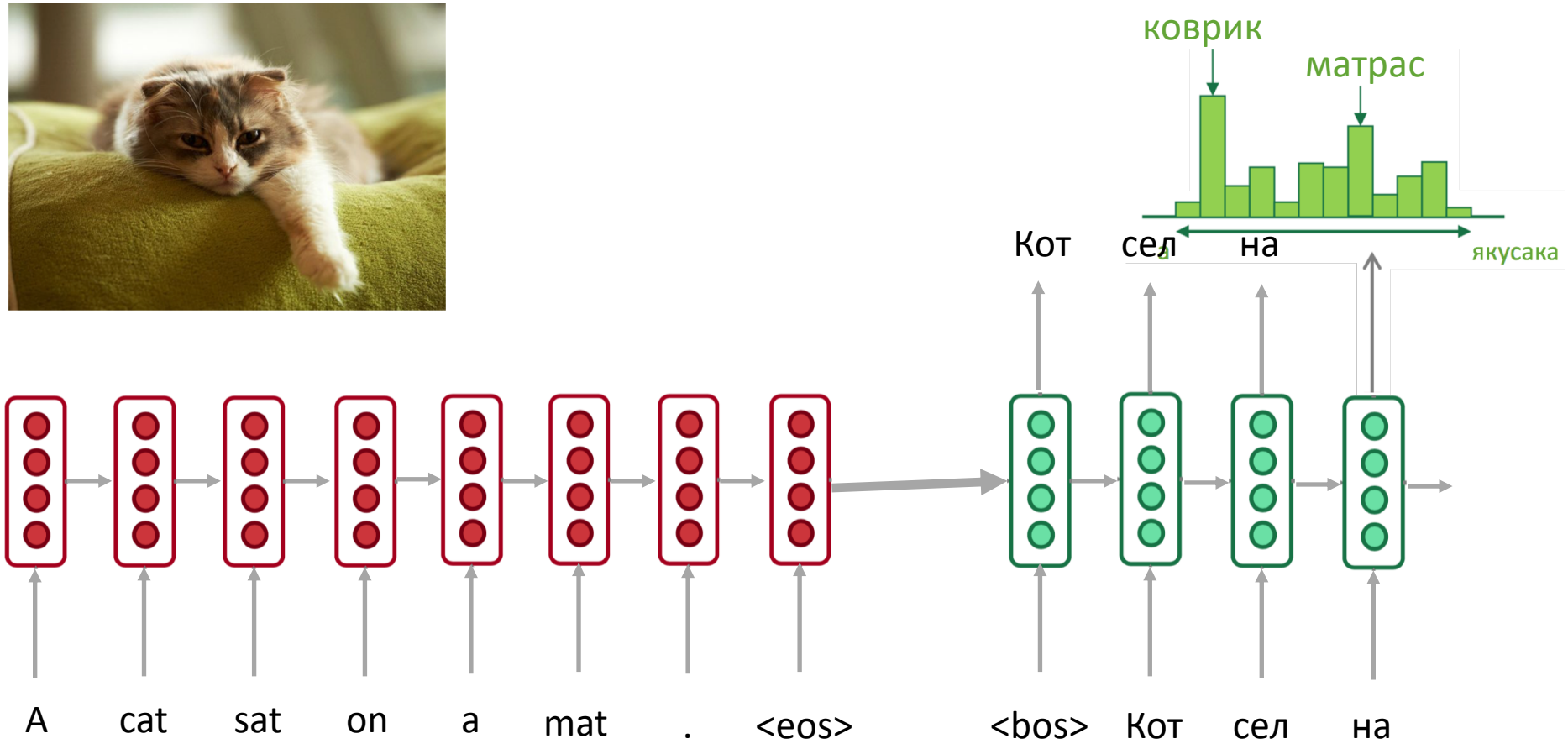

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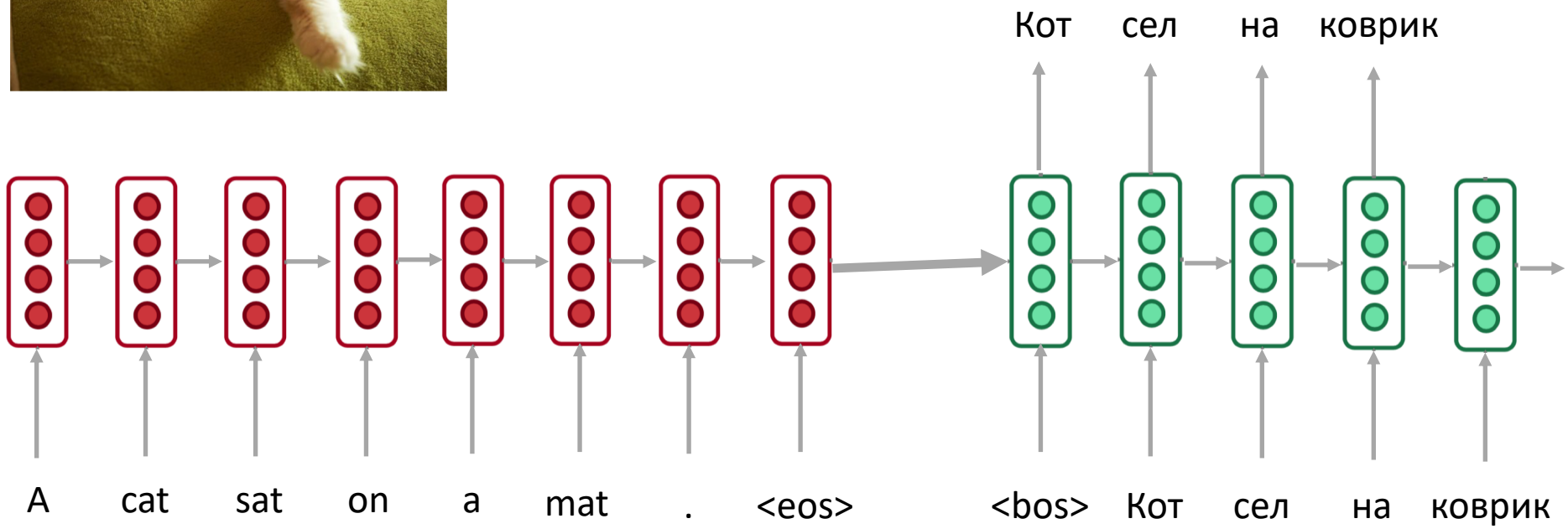

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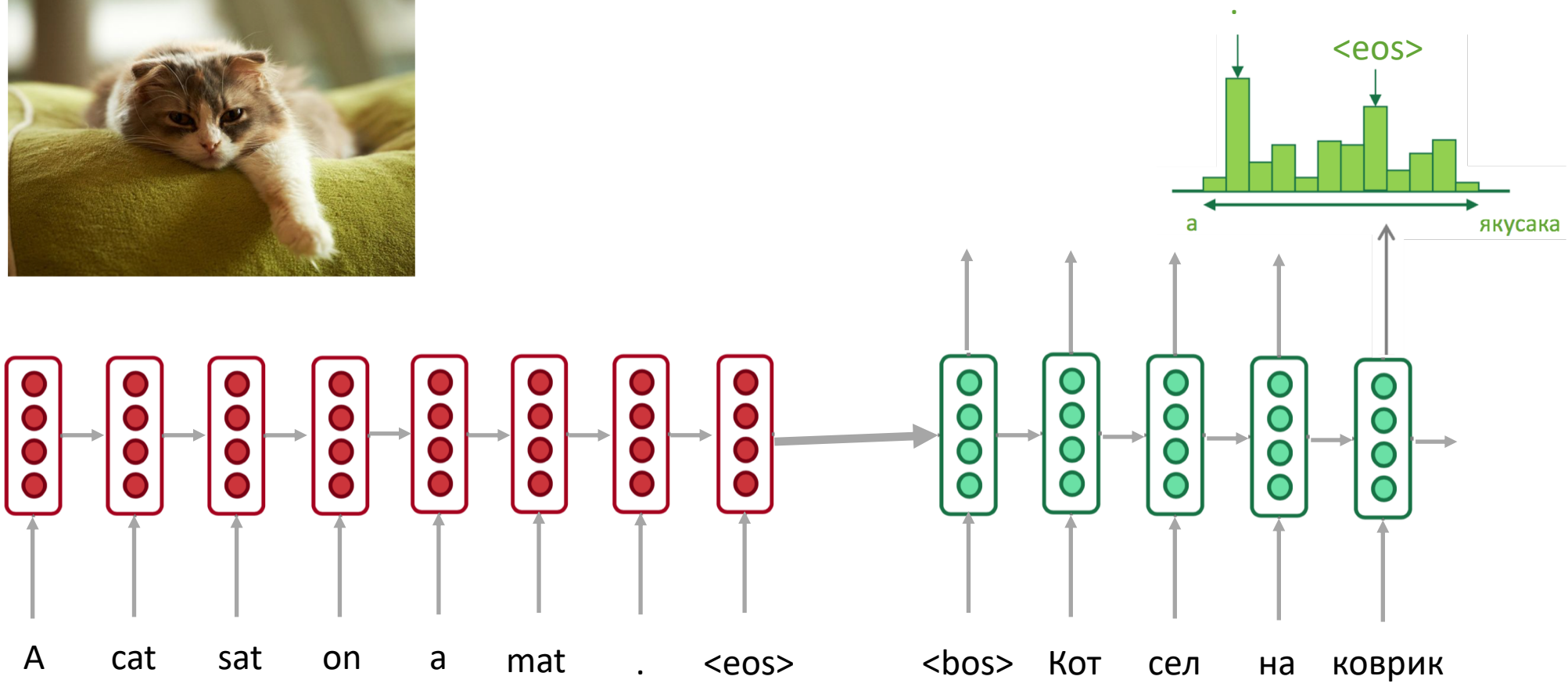

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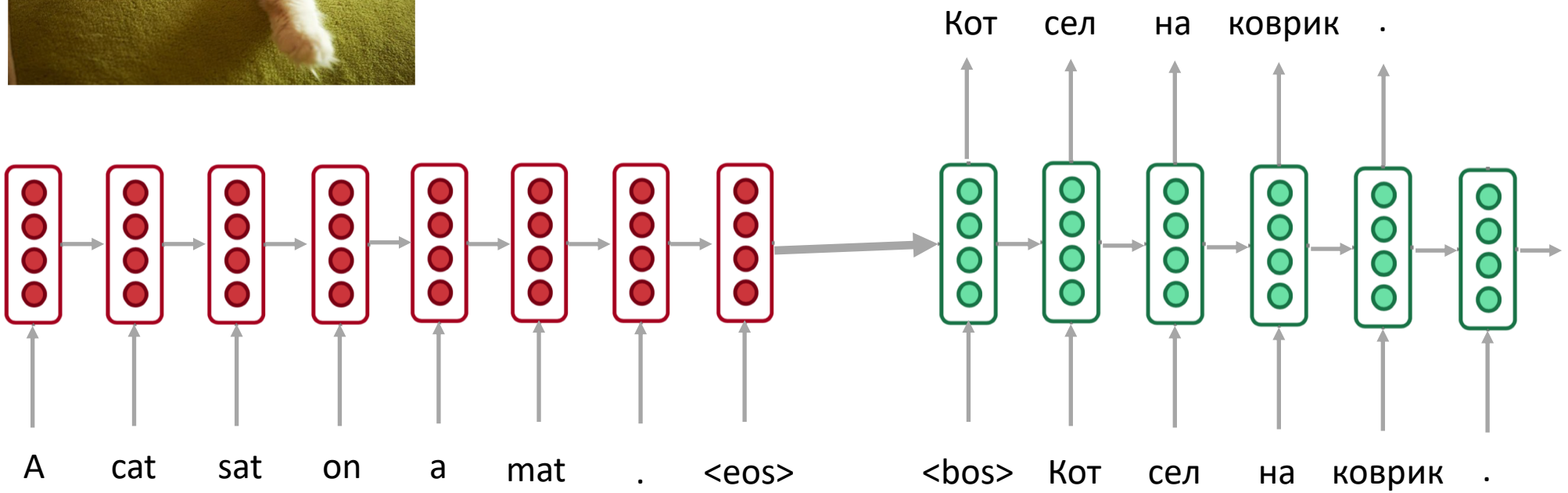

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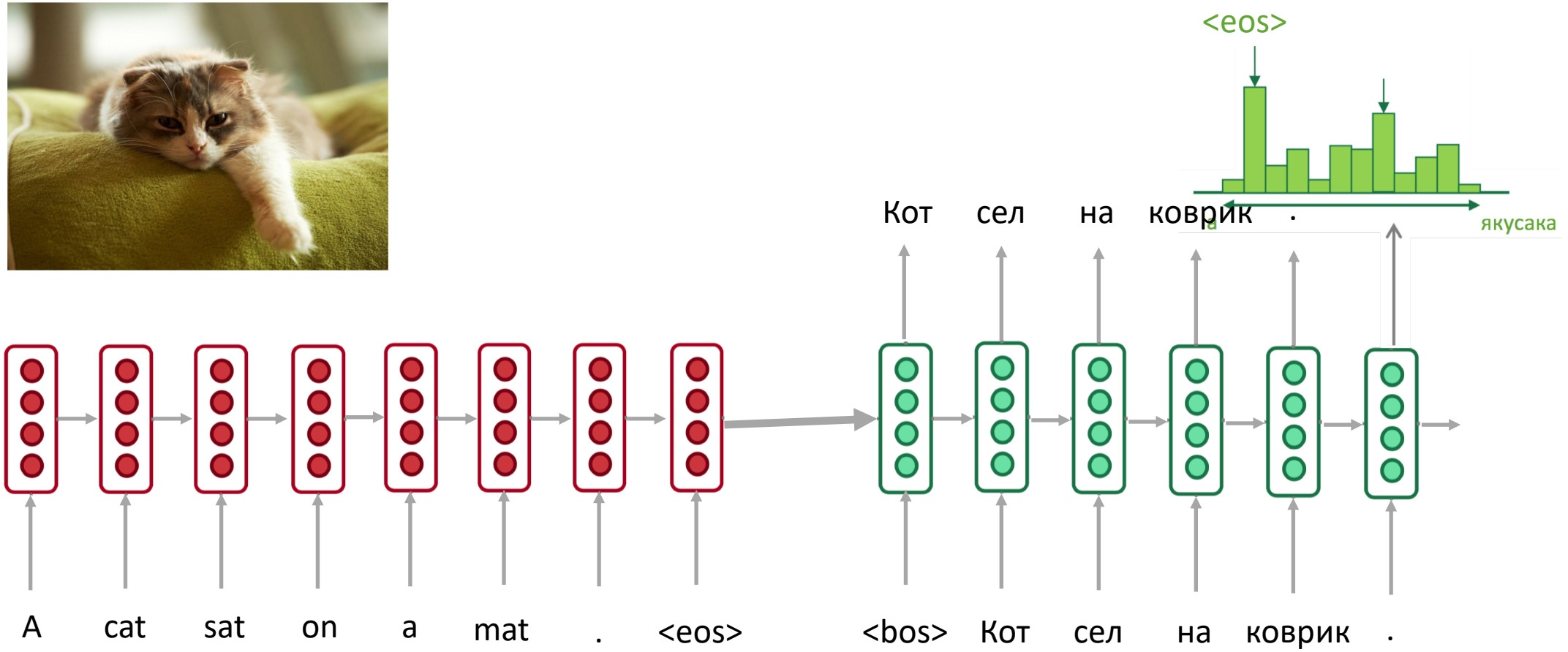

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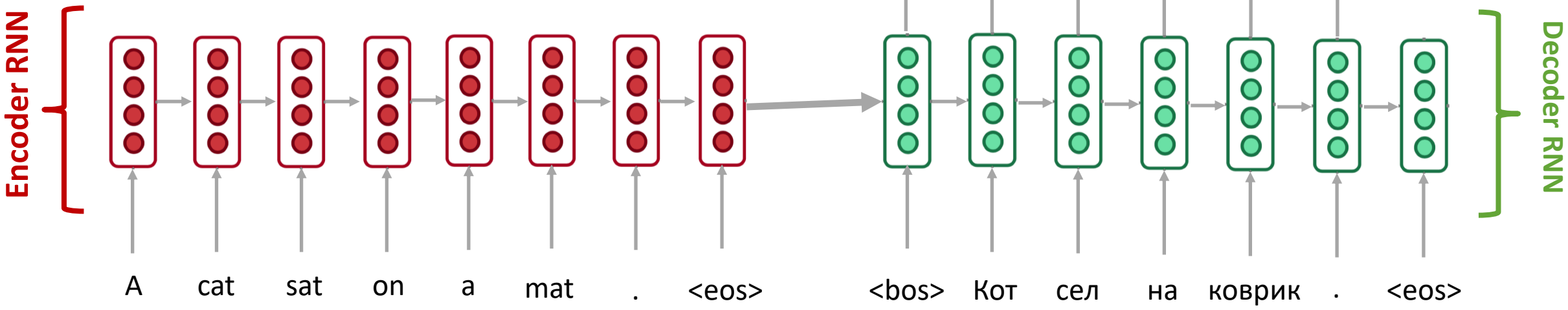

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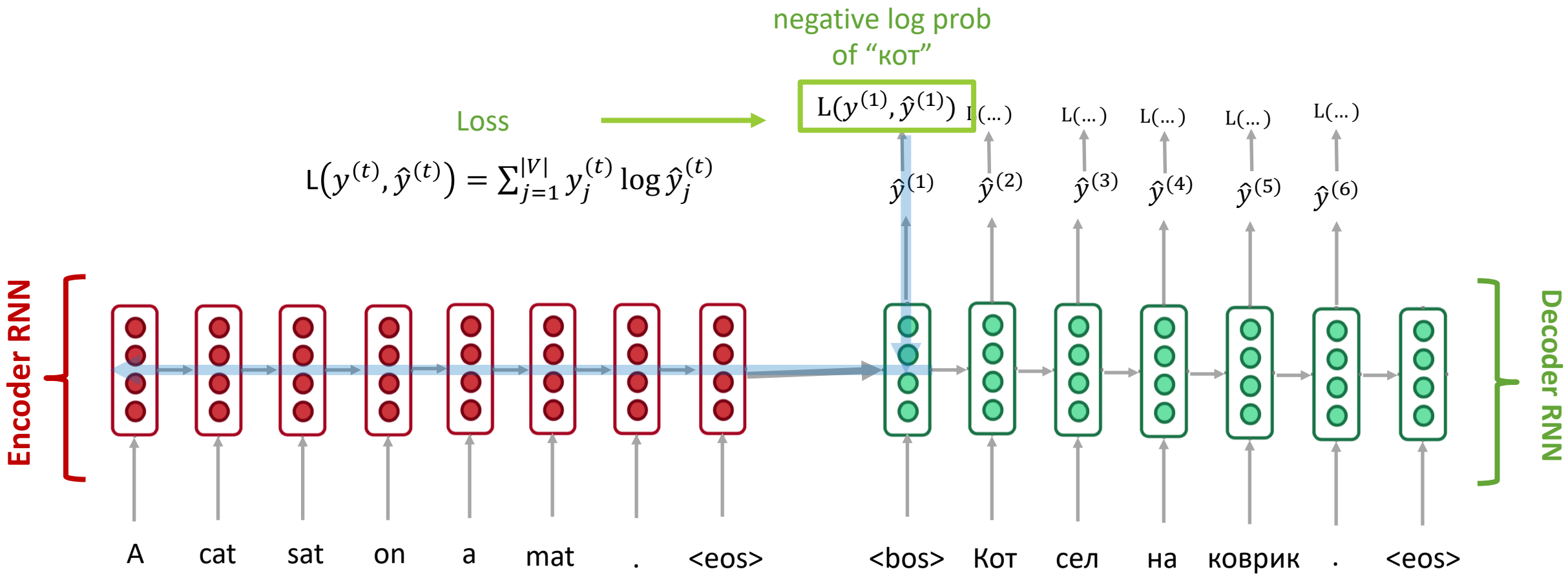

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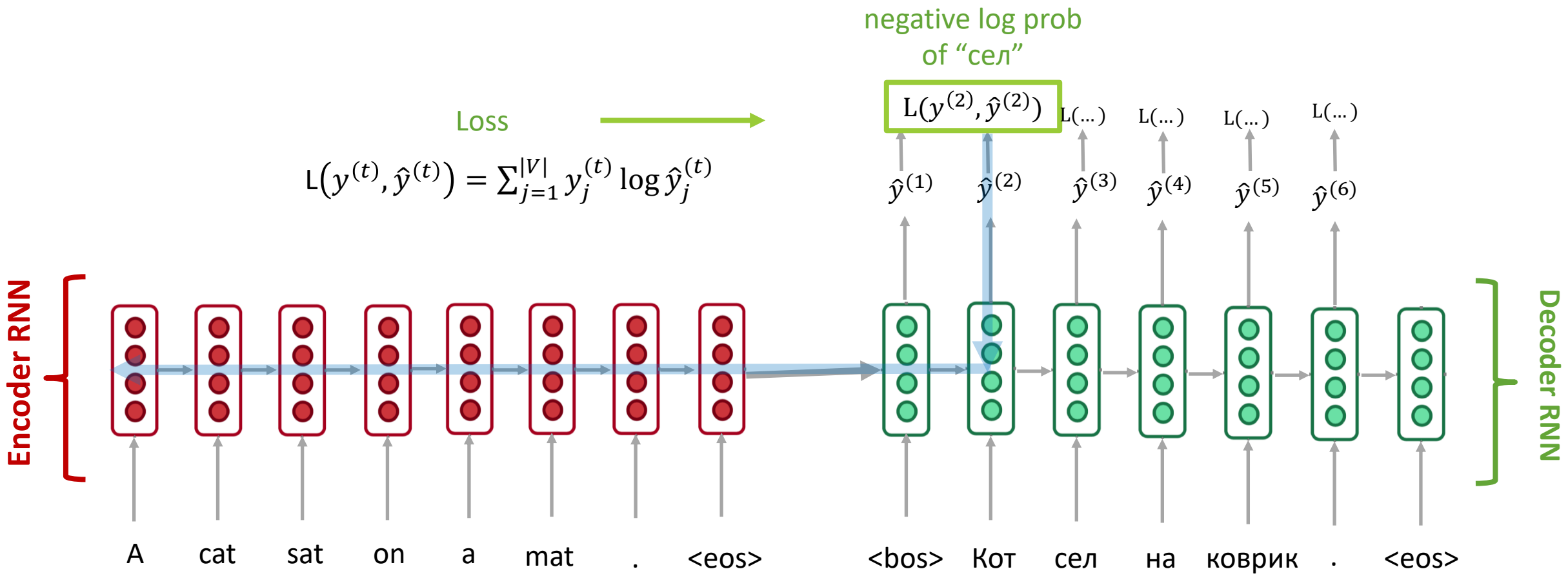

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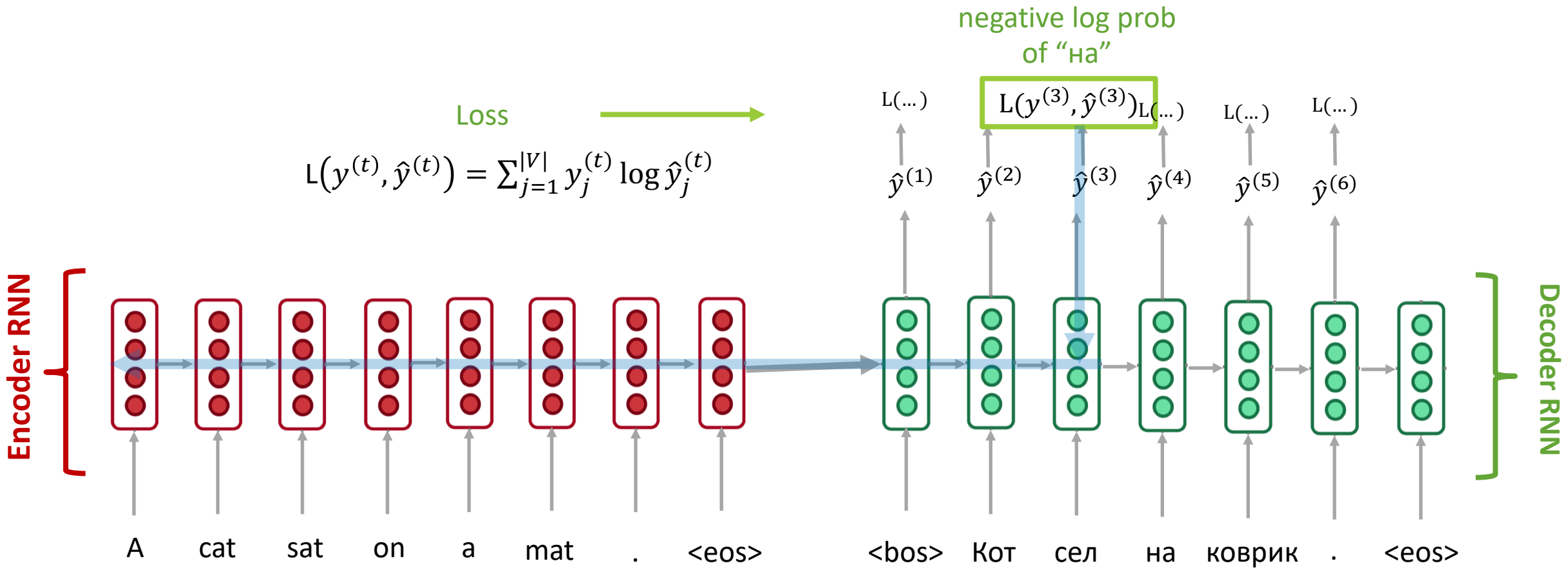

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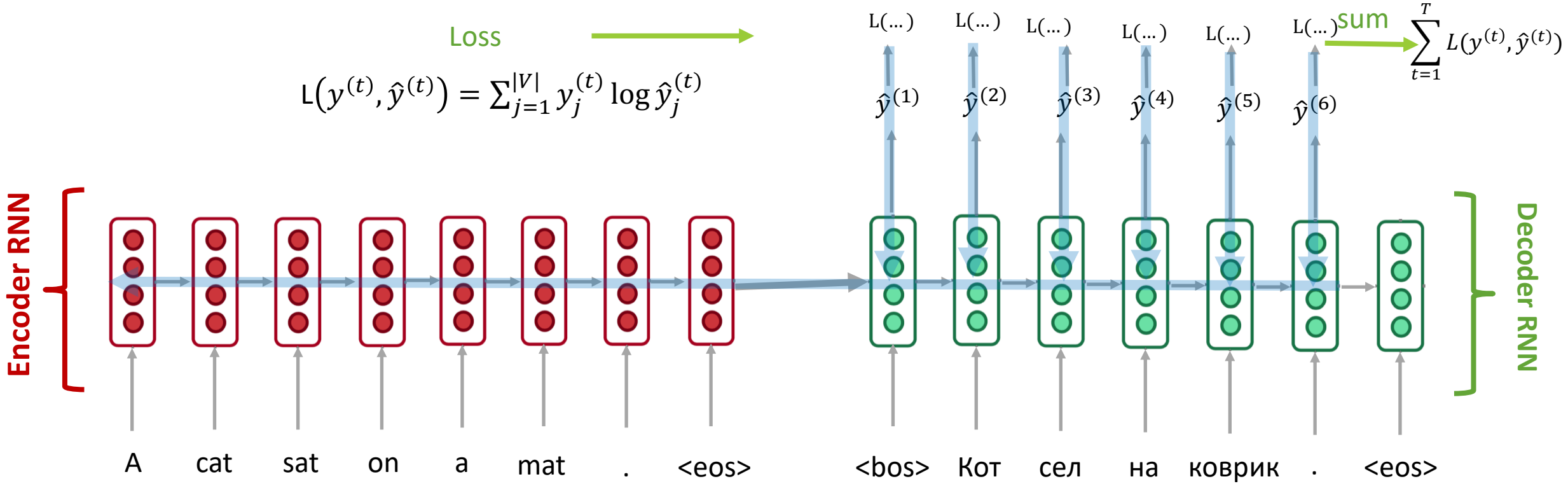
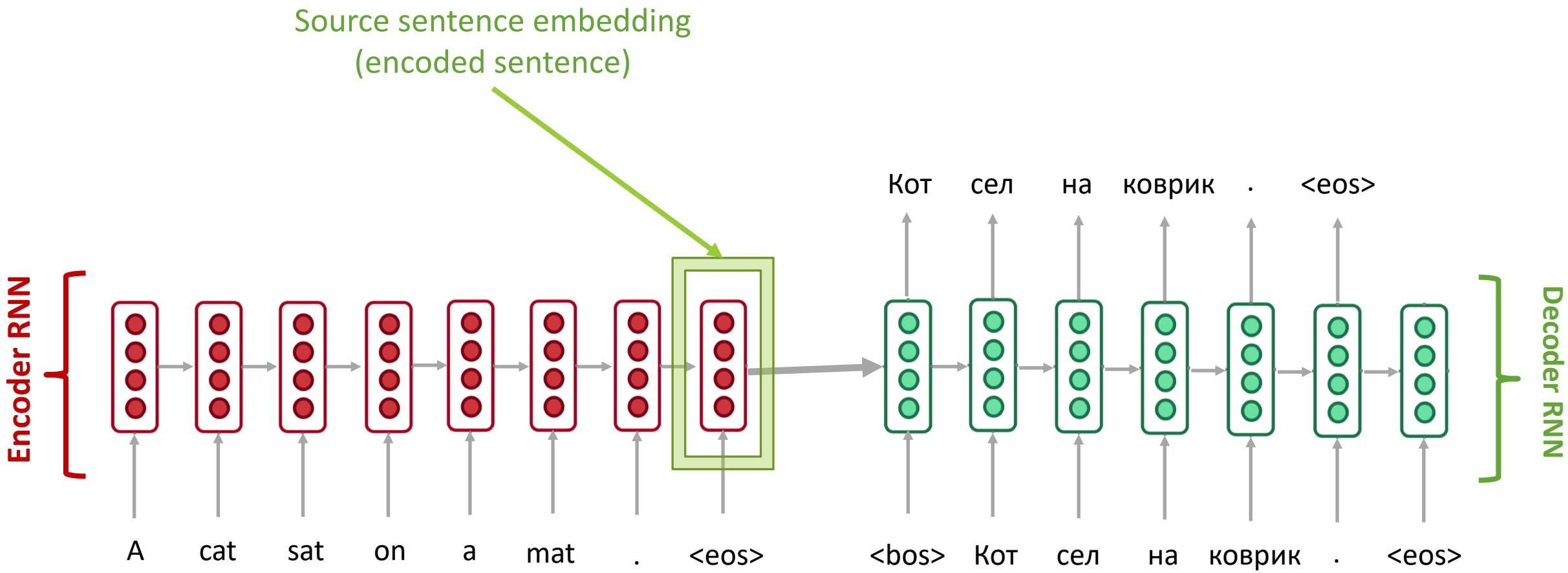


# Let’s look at sentence embedding



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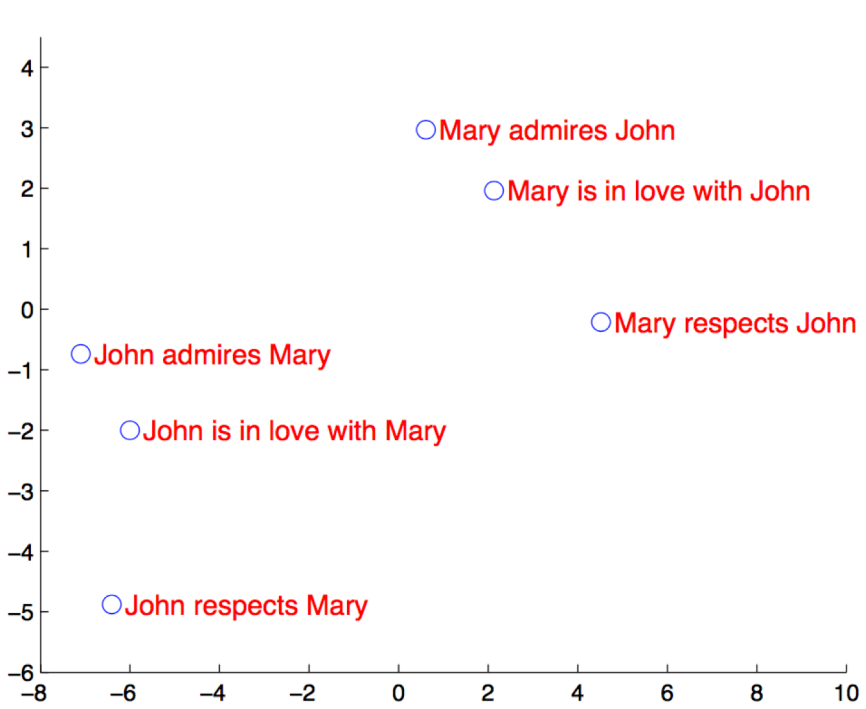
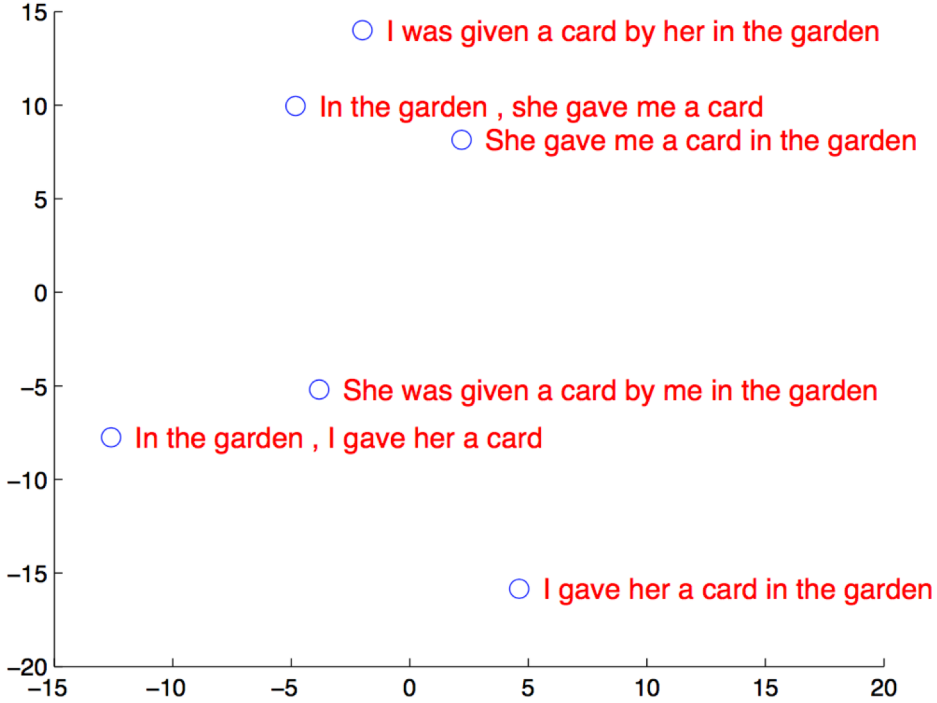
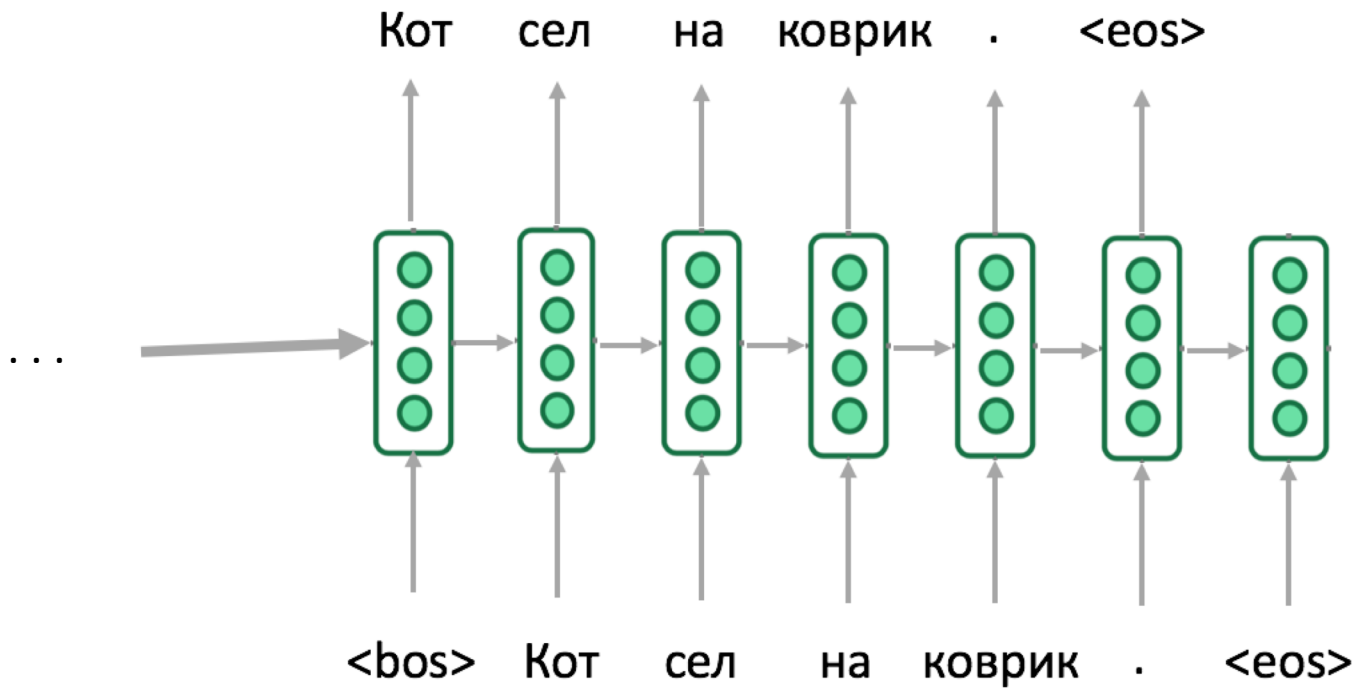



Sutskever et al, NIPS 2014, https://papers.nips.cc/paper/5346-sequence-to-sequence-learning-with-neural-networks.pdf

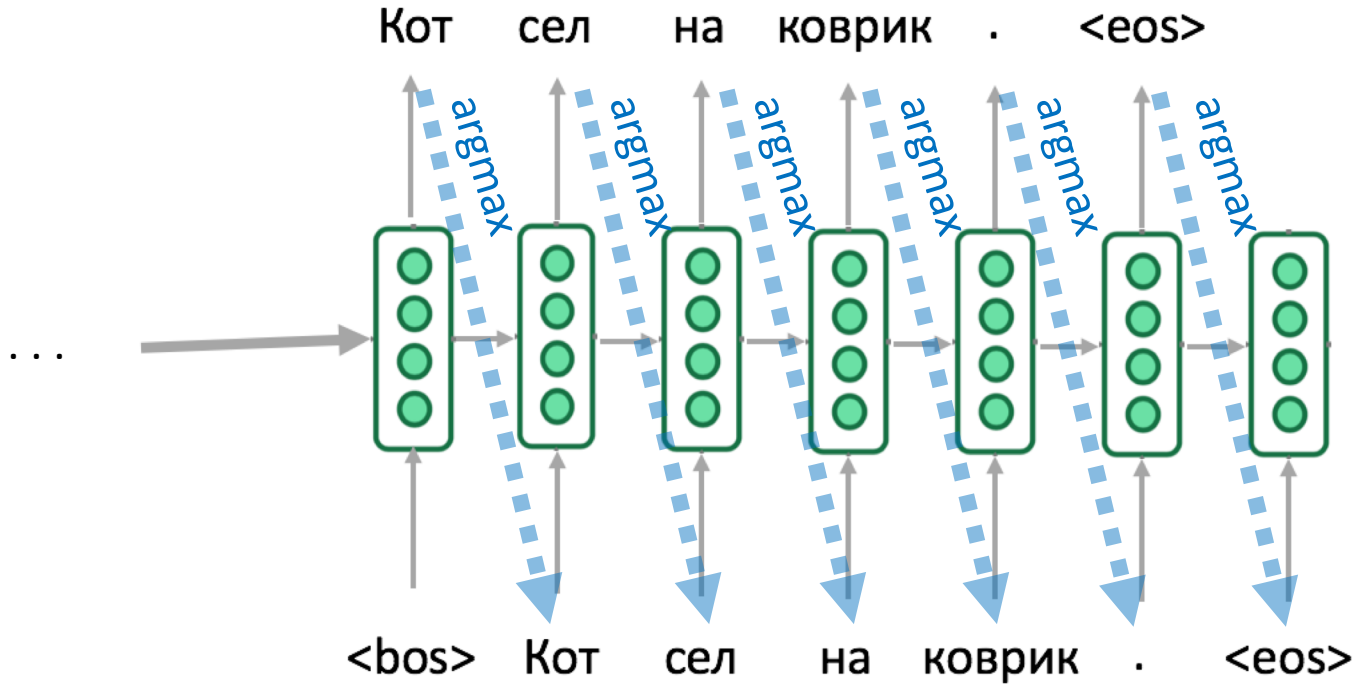
# Decoding

# Decoding: how to generate a sequence?



# Decoding: how to generate a sequence?

## 1. Greedy decoding



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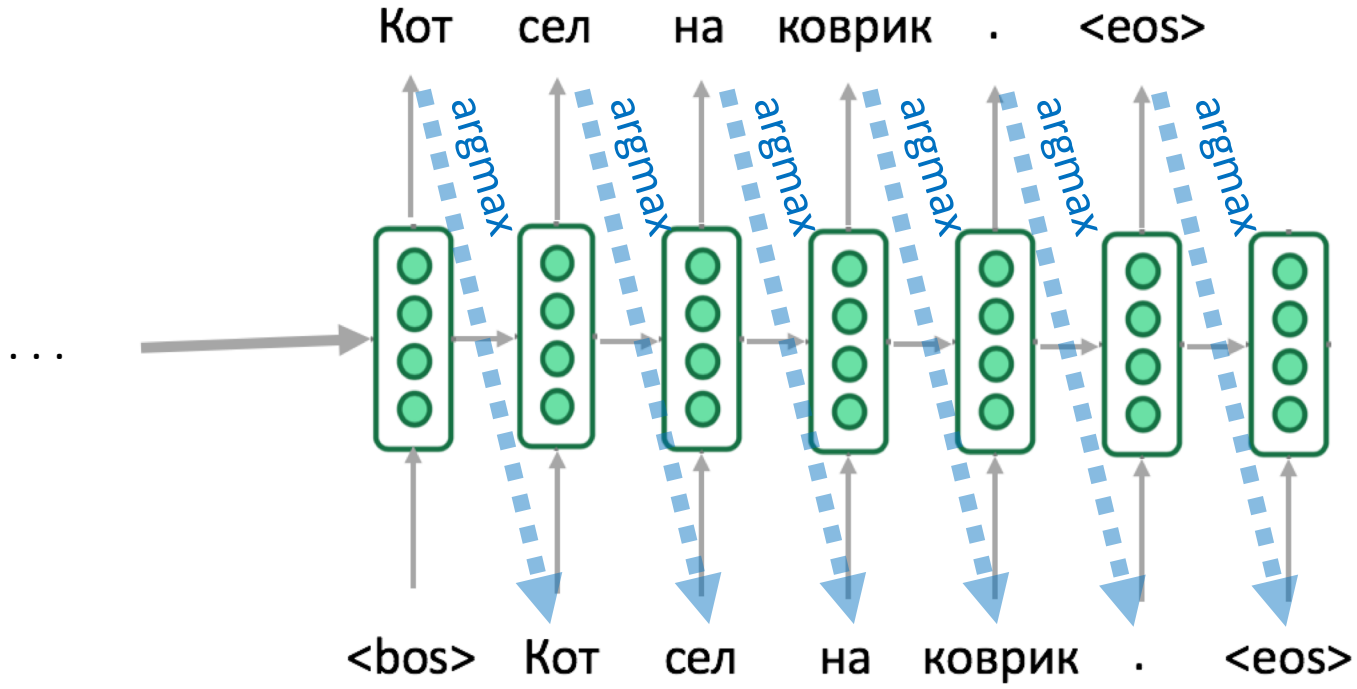
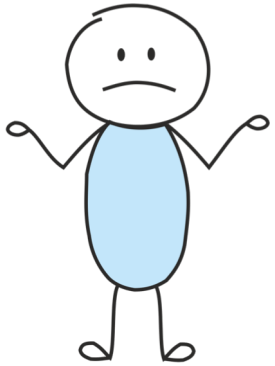



**Is it really good?**



# Decoding: how to generate a sequence?

## 2. Beam search

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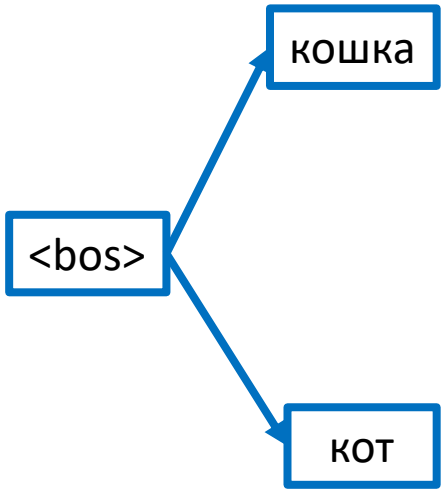

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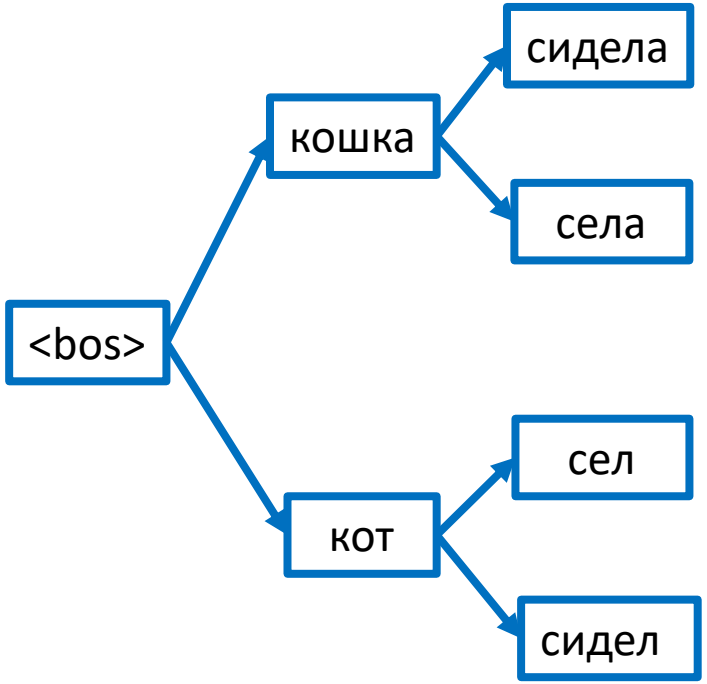

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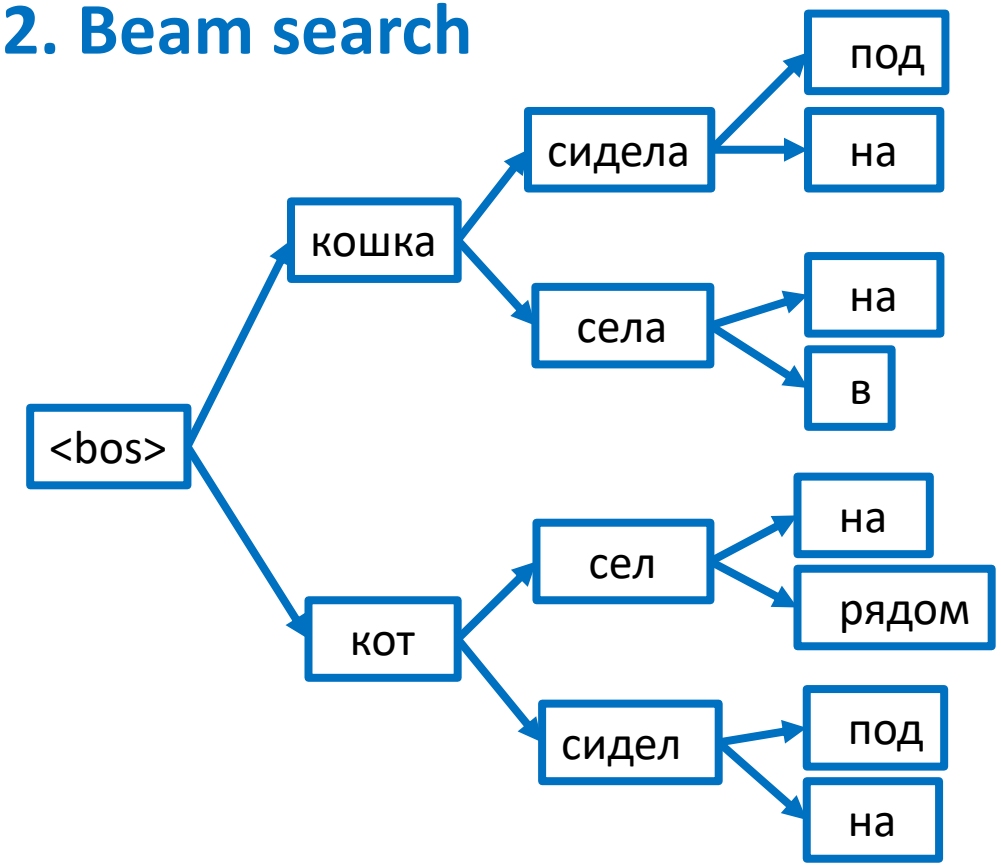

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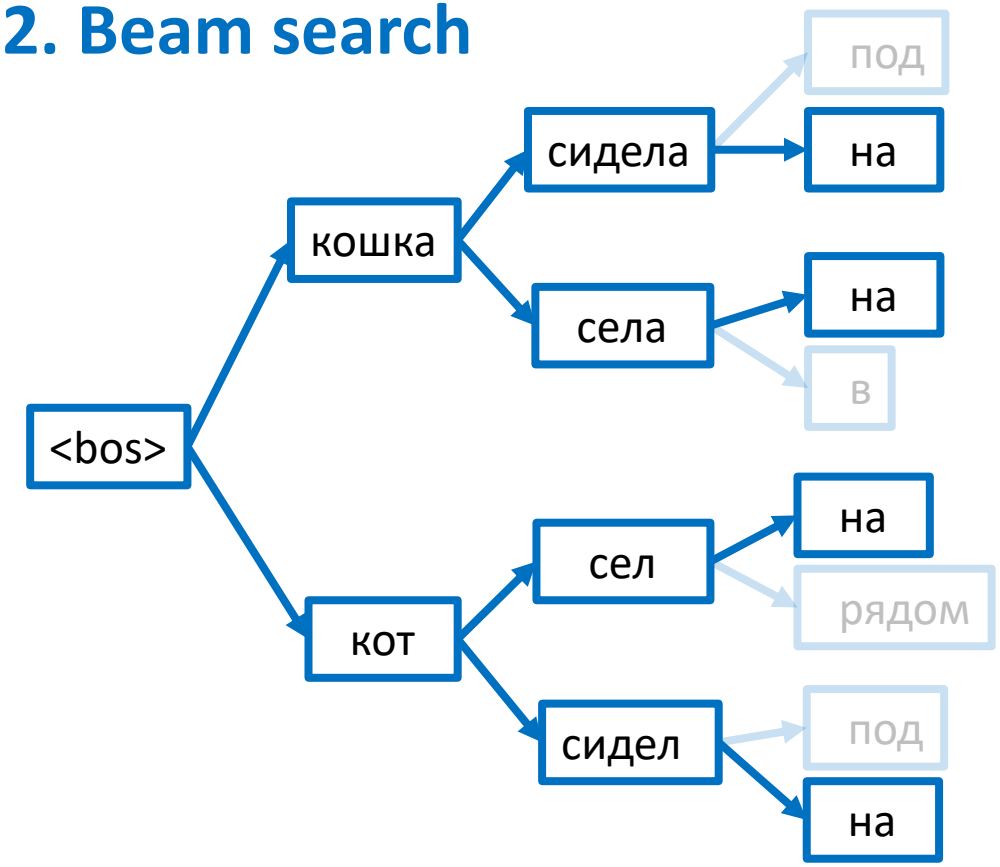

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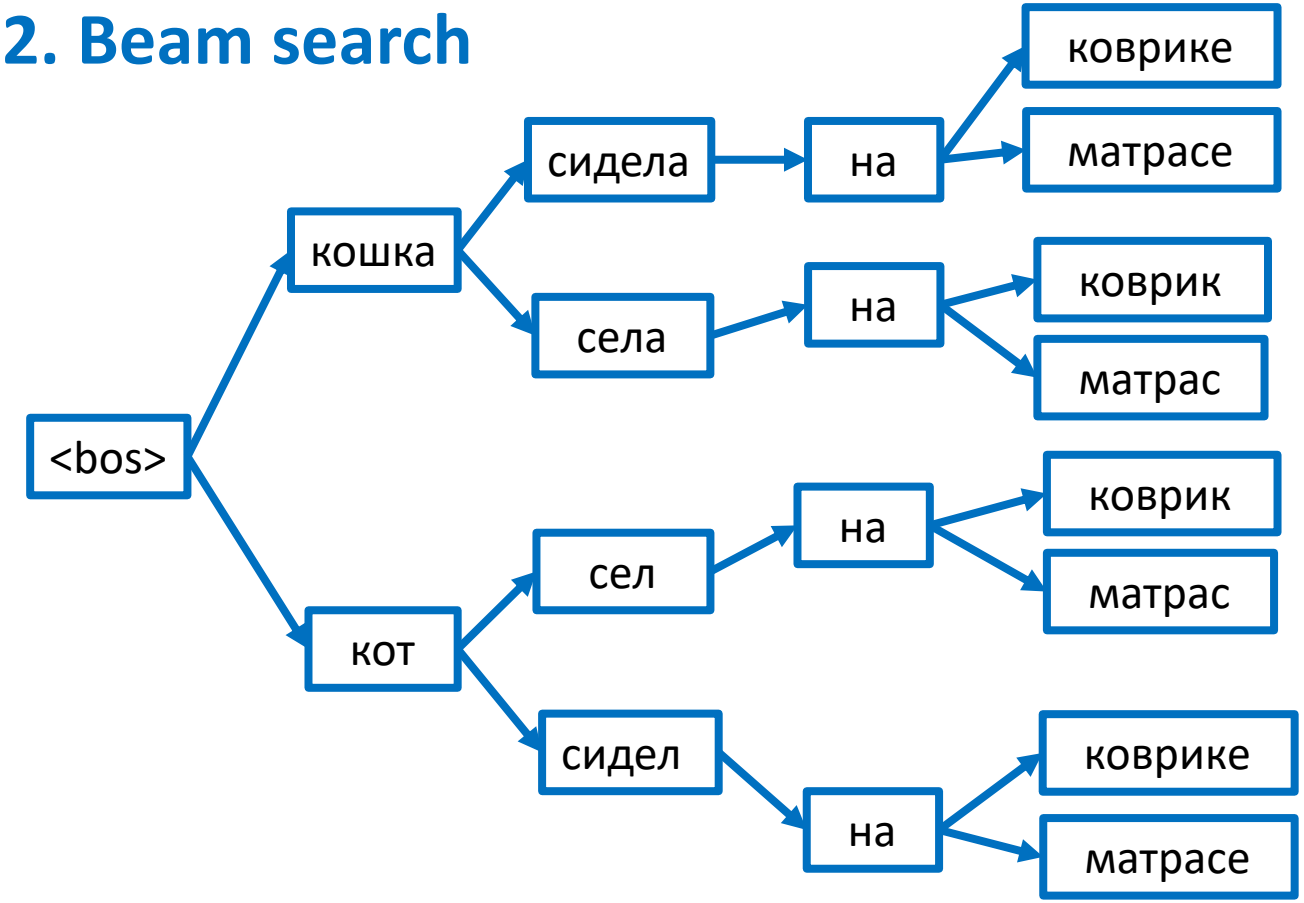

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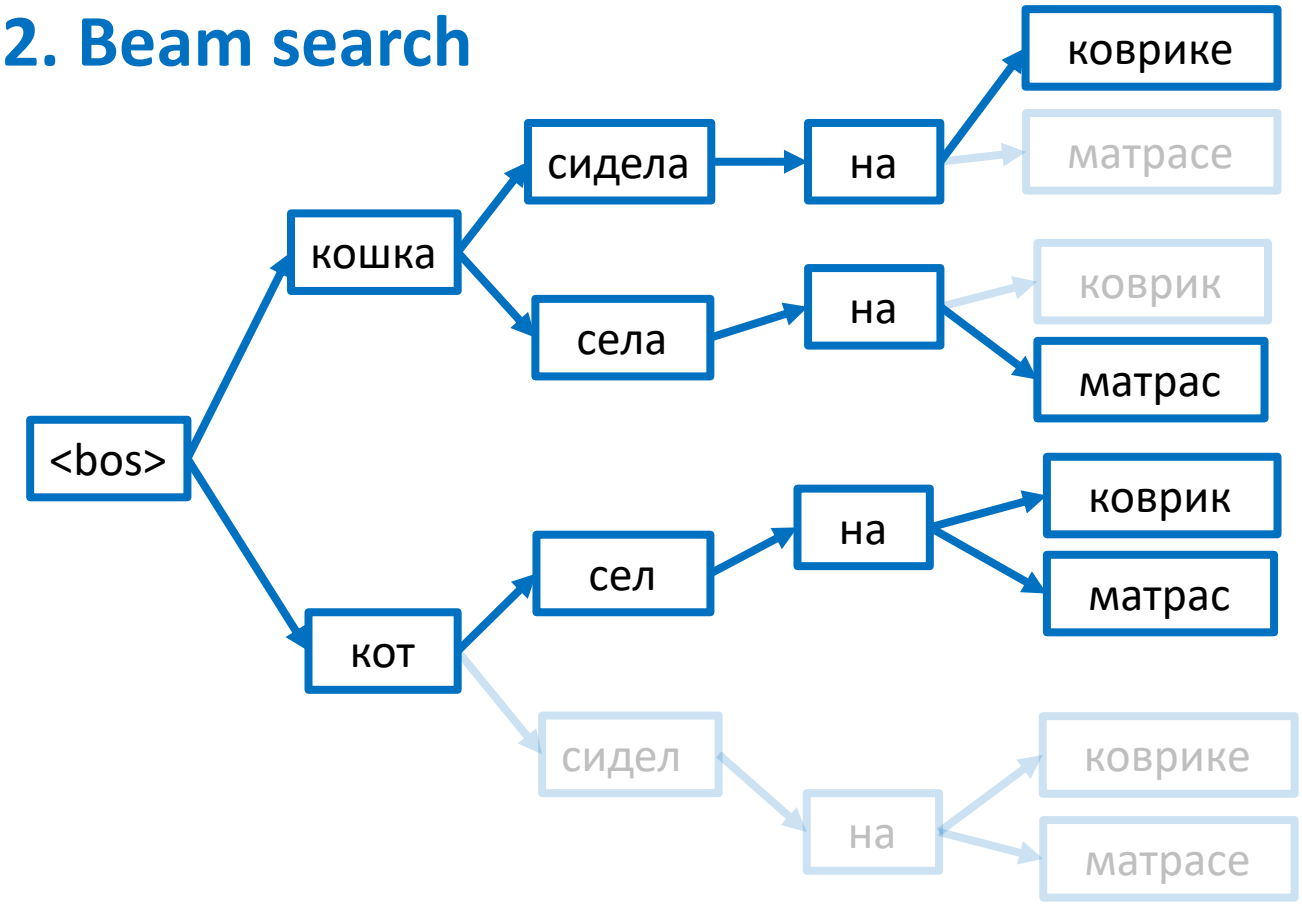

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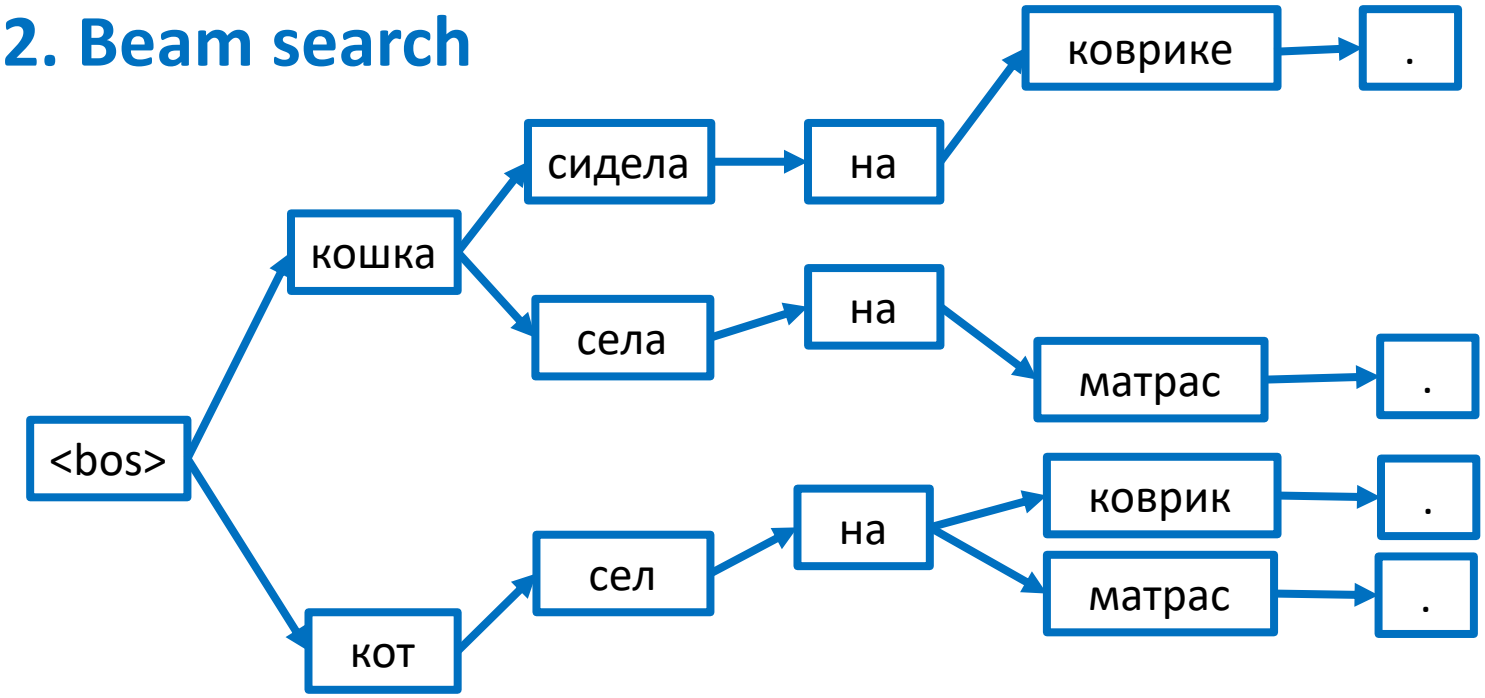

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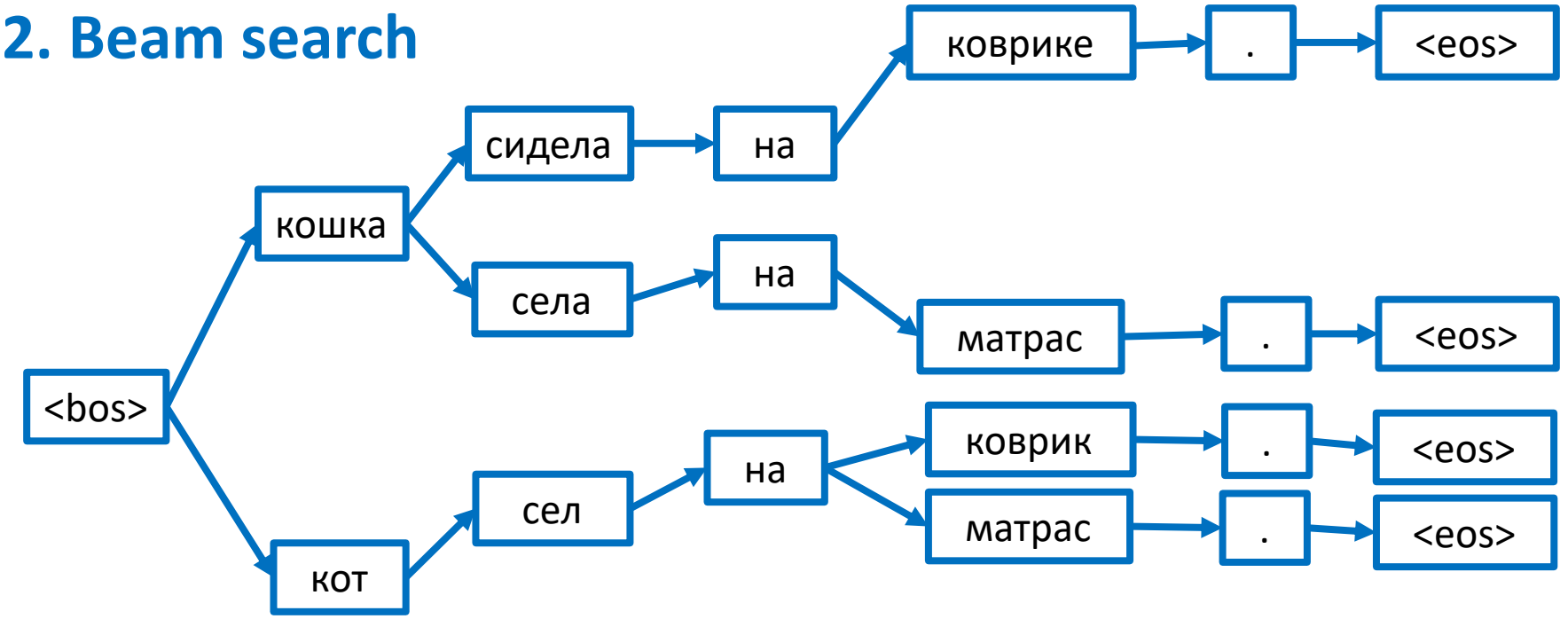

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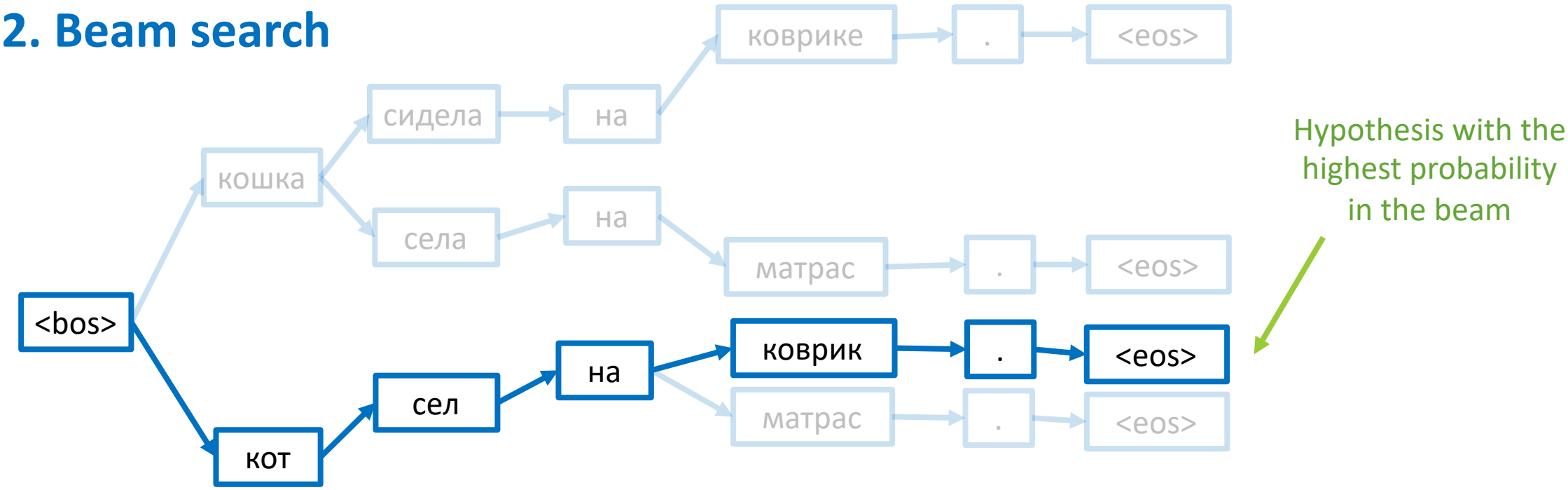
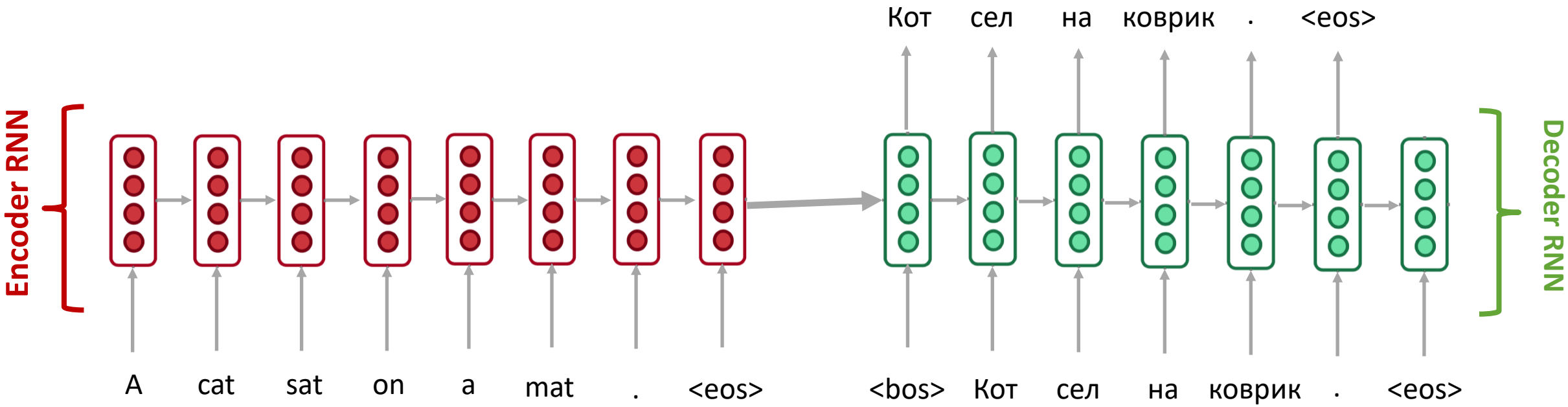


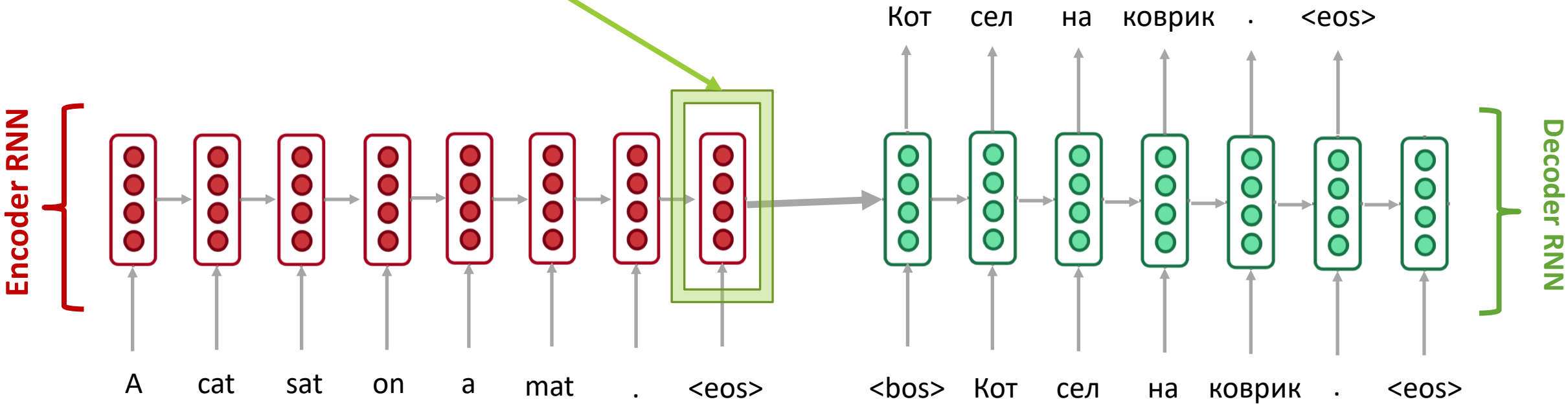
# Attention-based models

# What is the problem with such models?

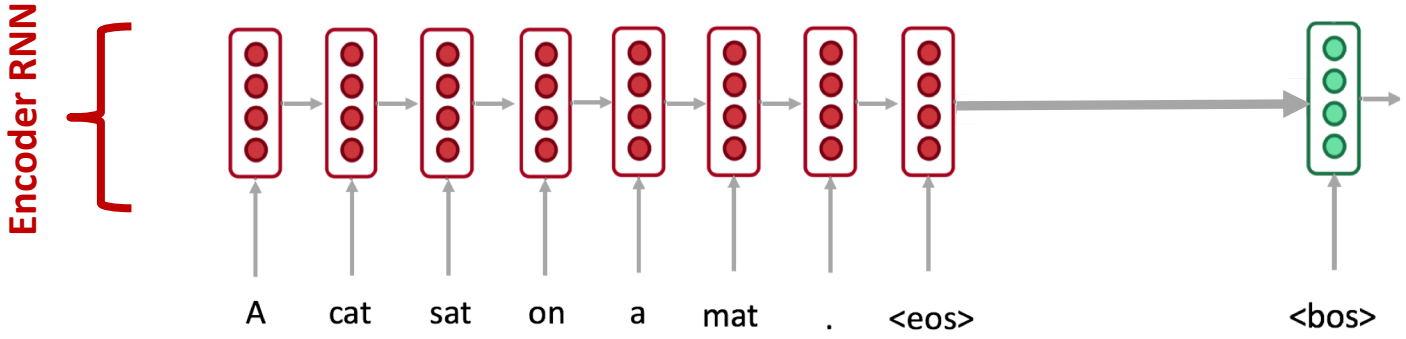


# What is the problem with such models?

**Information bottleneck:**
the model has to put all information about sentence in one vector



# Attention



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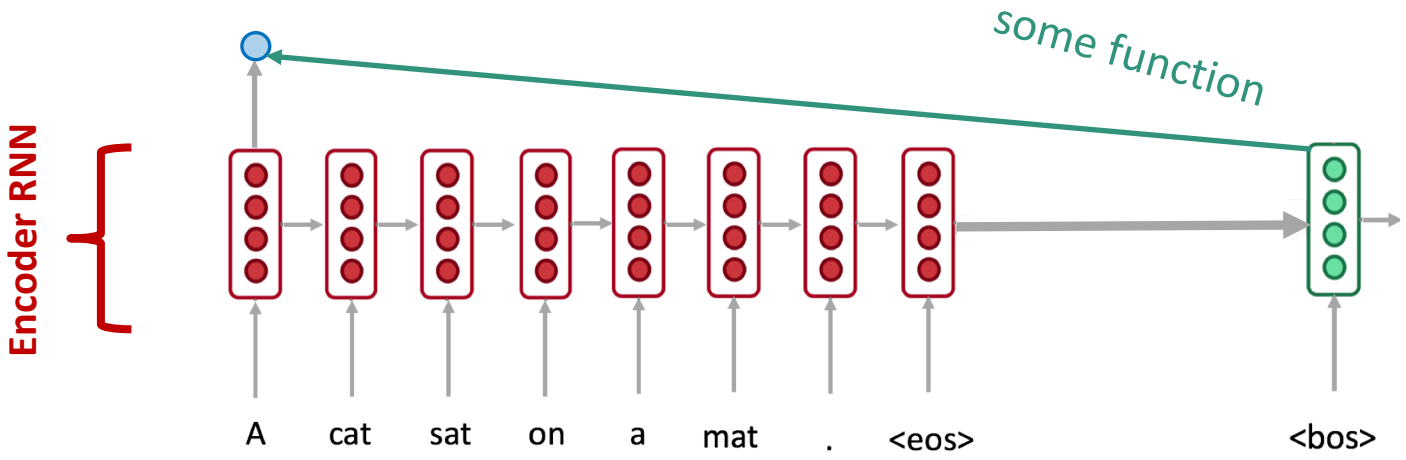

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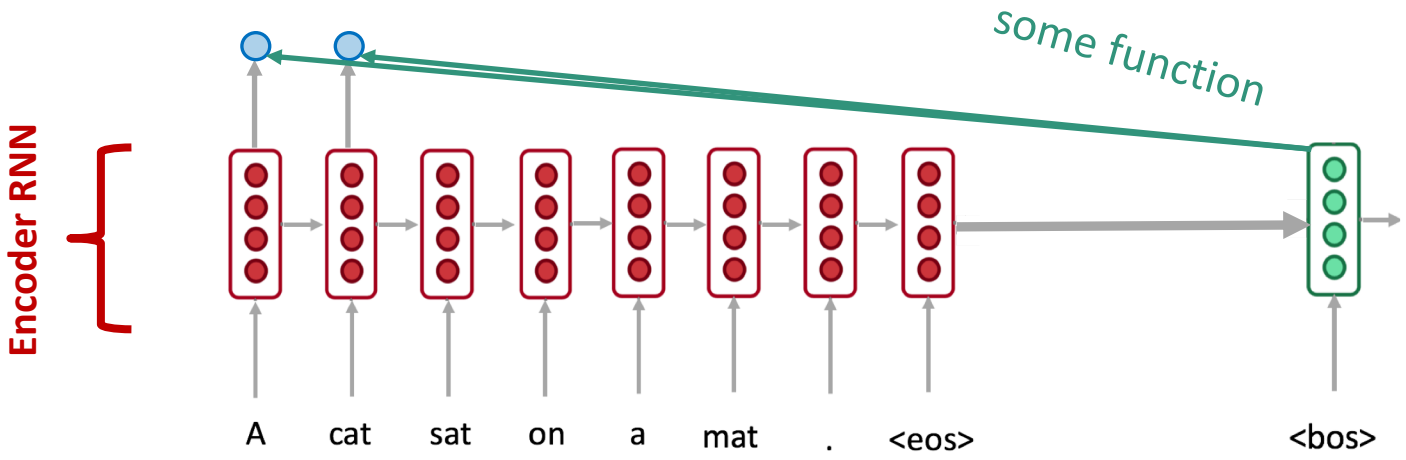

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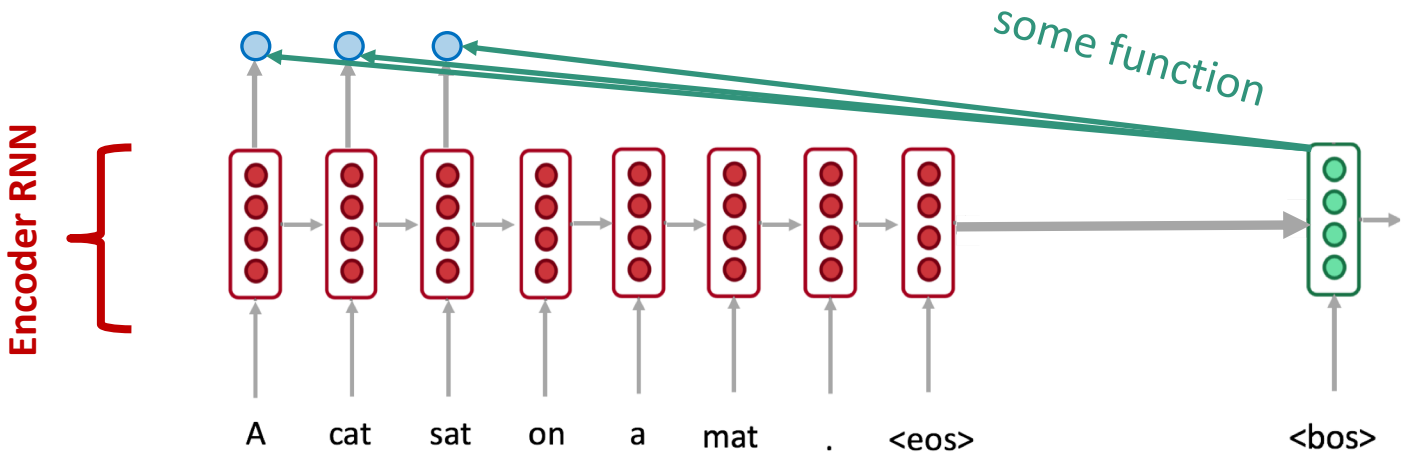

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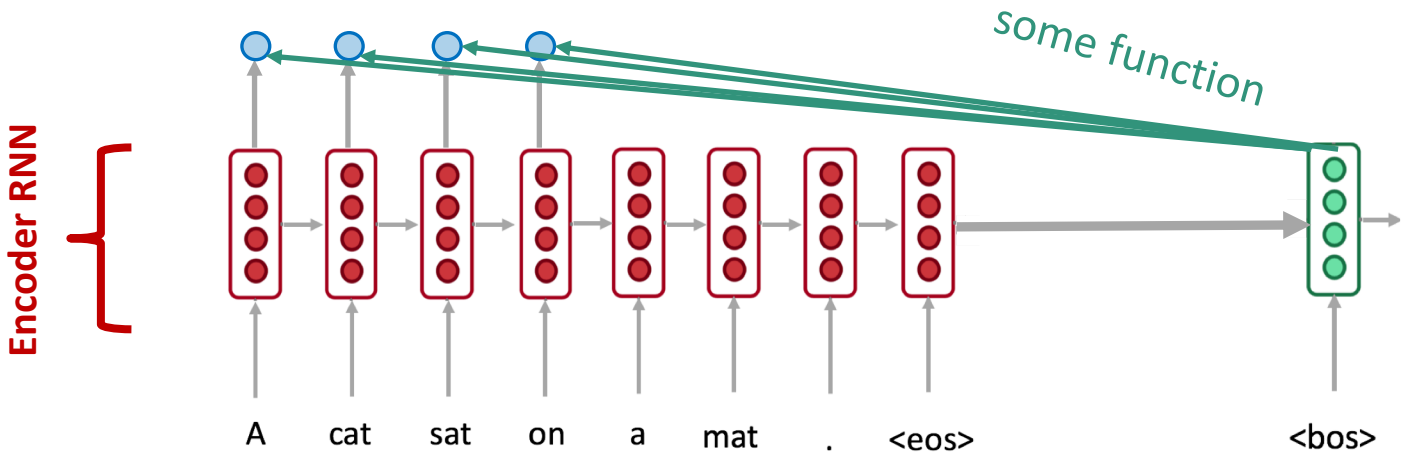

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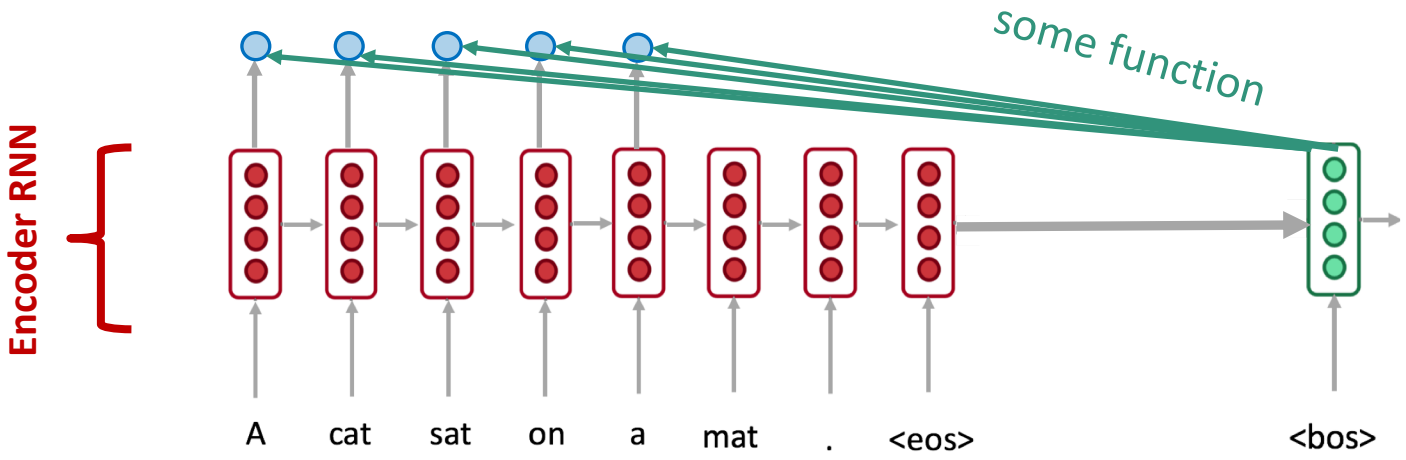
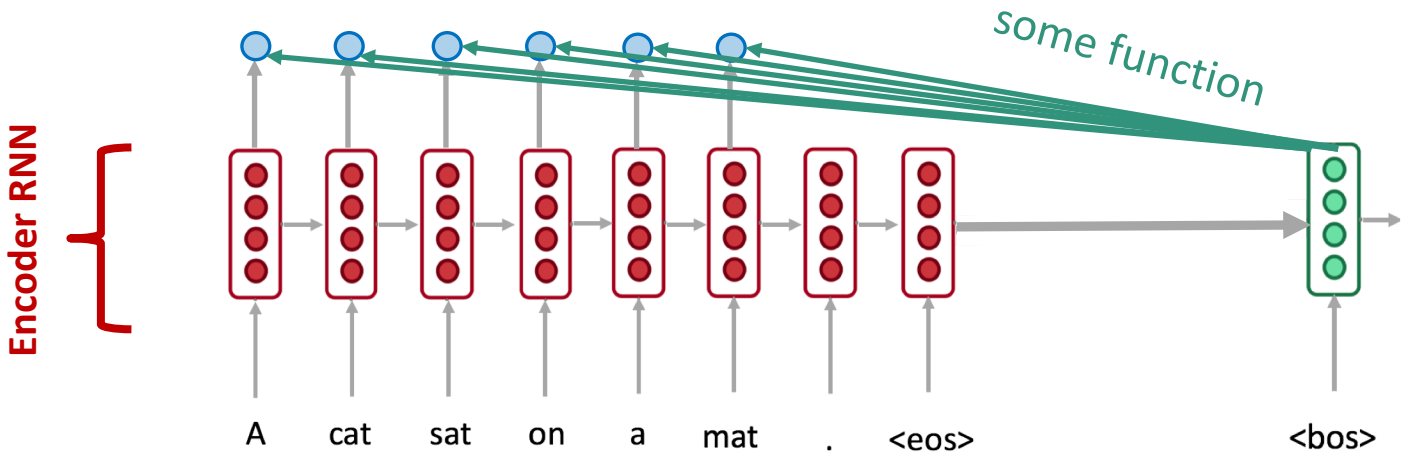


# Attention



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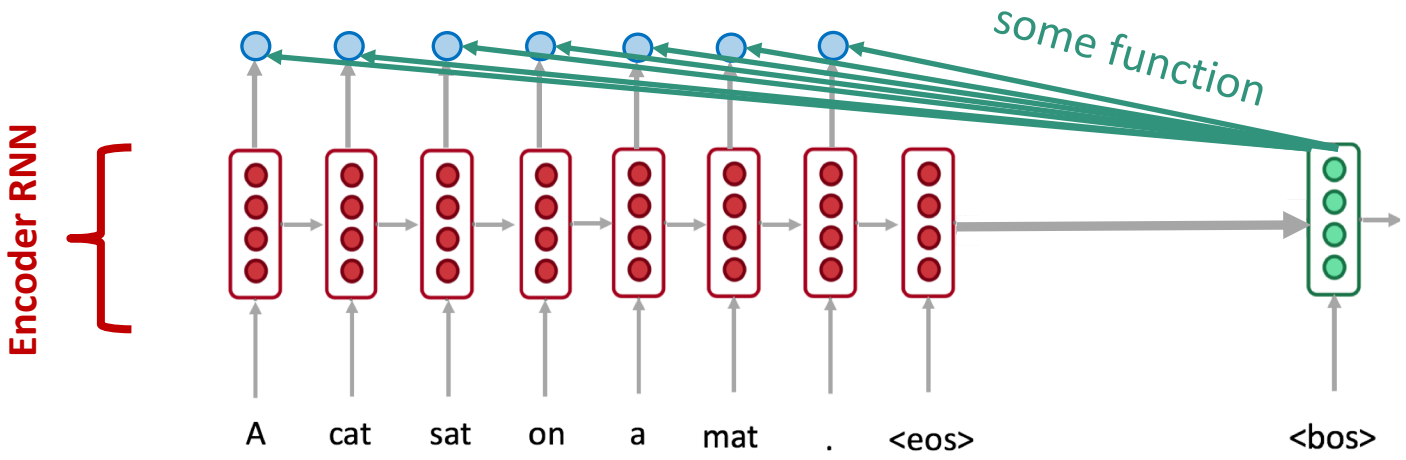

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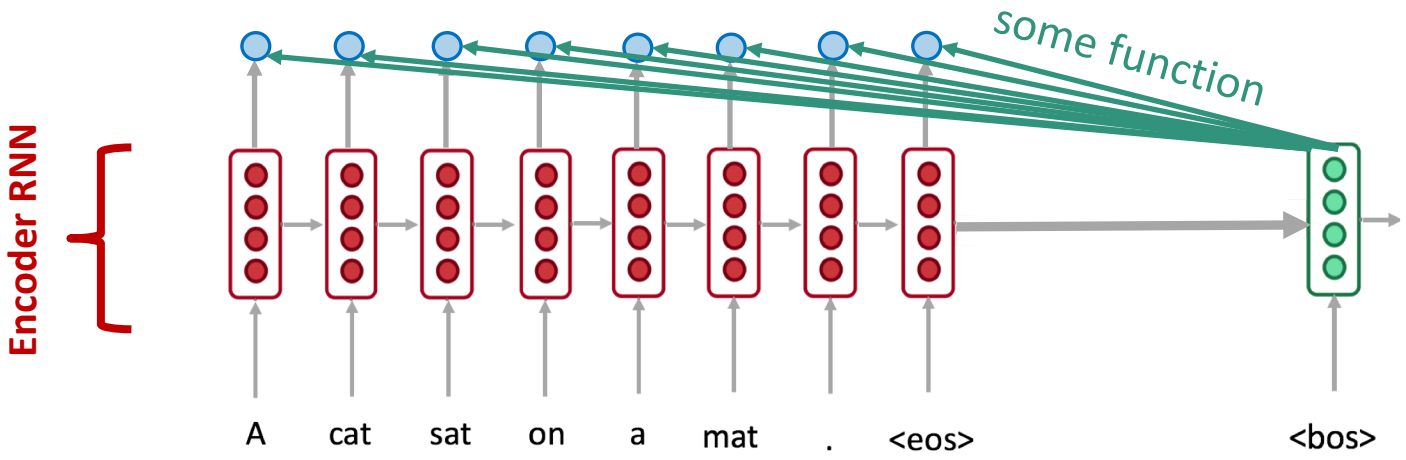

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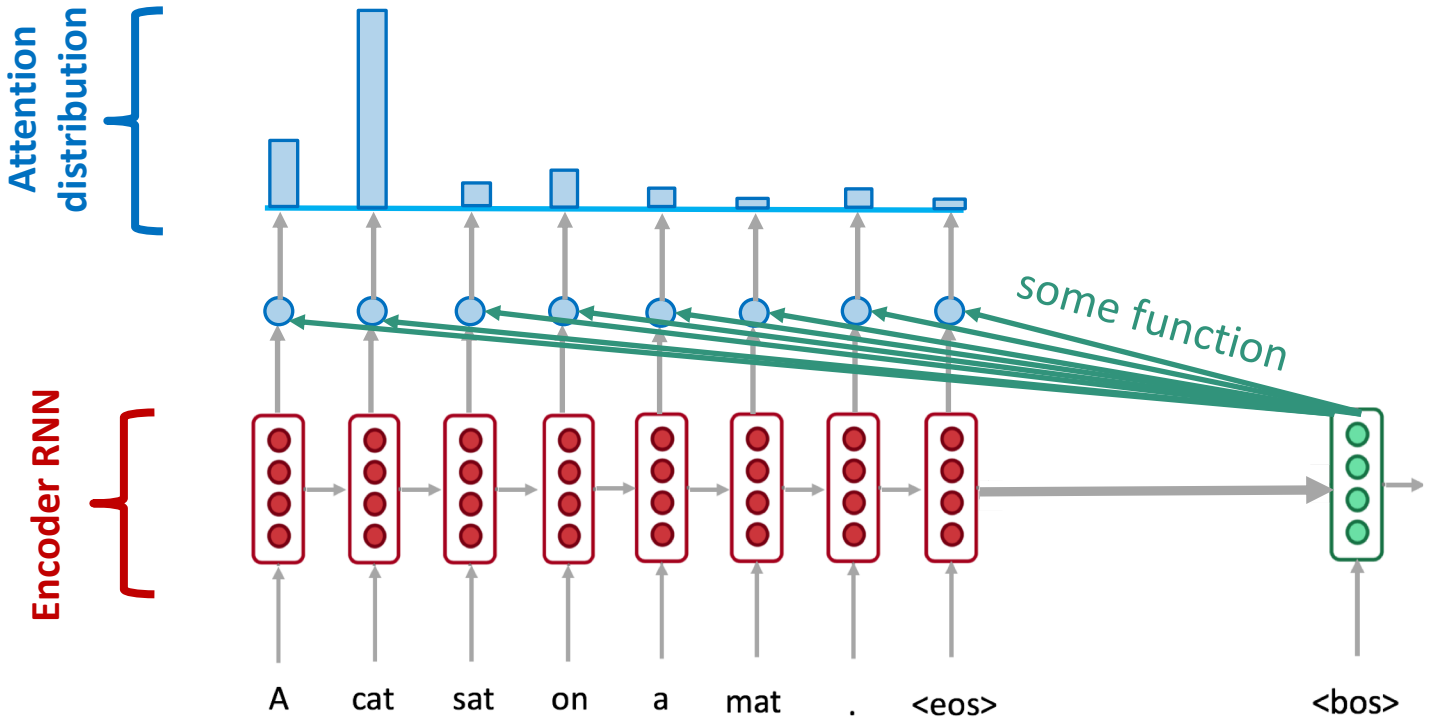
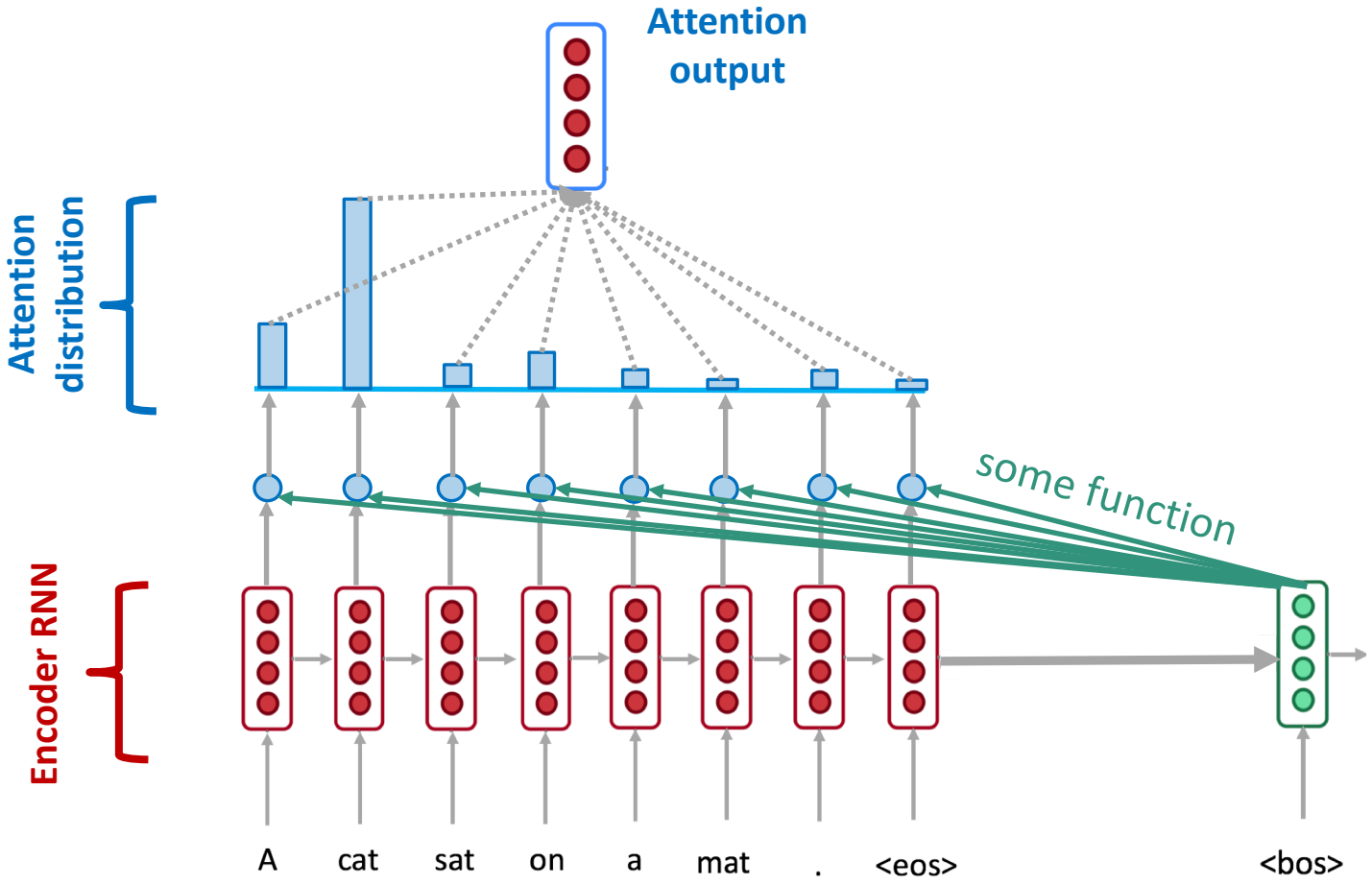

Attention distribution
Encoder RNN
some function
A
cat
sat
on
a
mat
.
<eos>
<bos>

# Attention


Attention output
Attention distribution
some function
Encoder RNN
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<eos>
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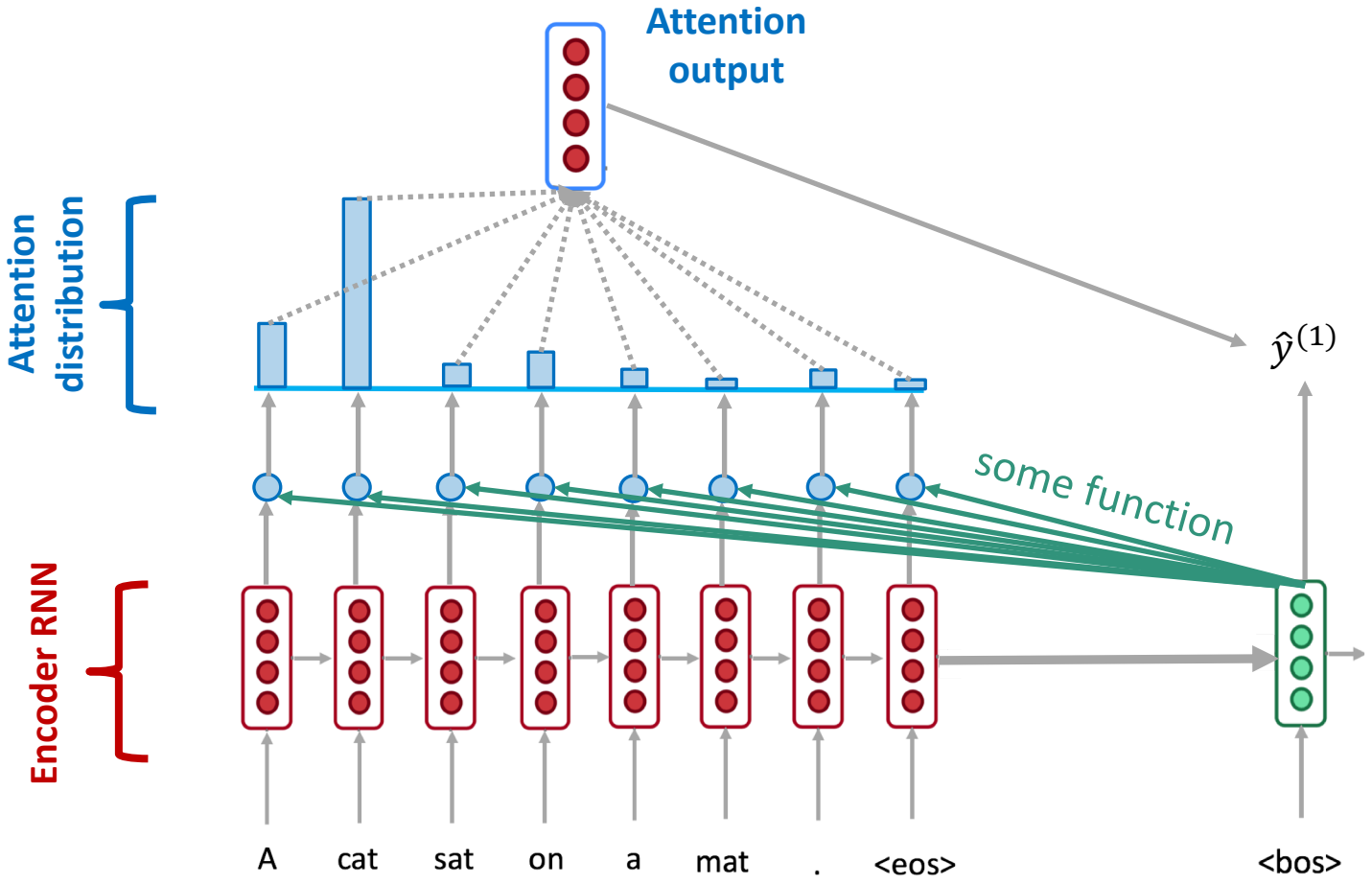
# Attention


Attention output
Attention distribution
$\hat{y}^{(1)}$
some function
Encoder RNN
A
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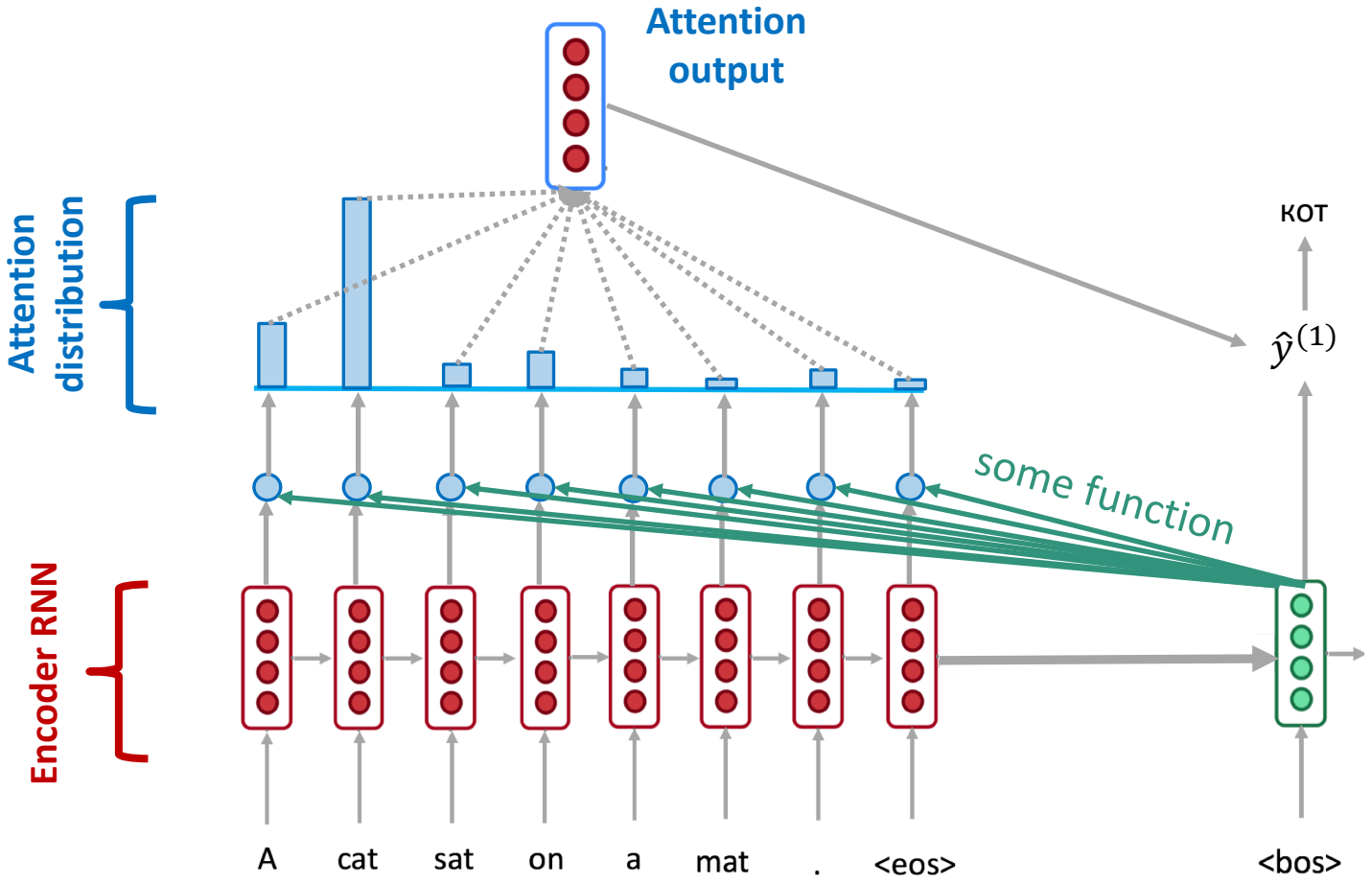
<bos>

# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
$\hat{y}^{(1)}$
A
cat
sat
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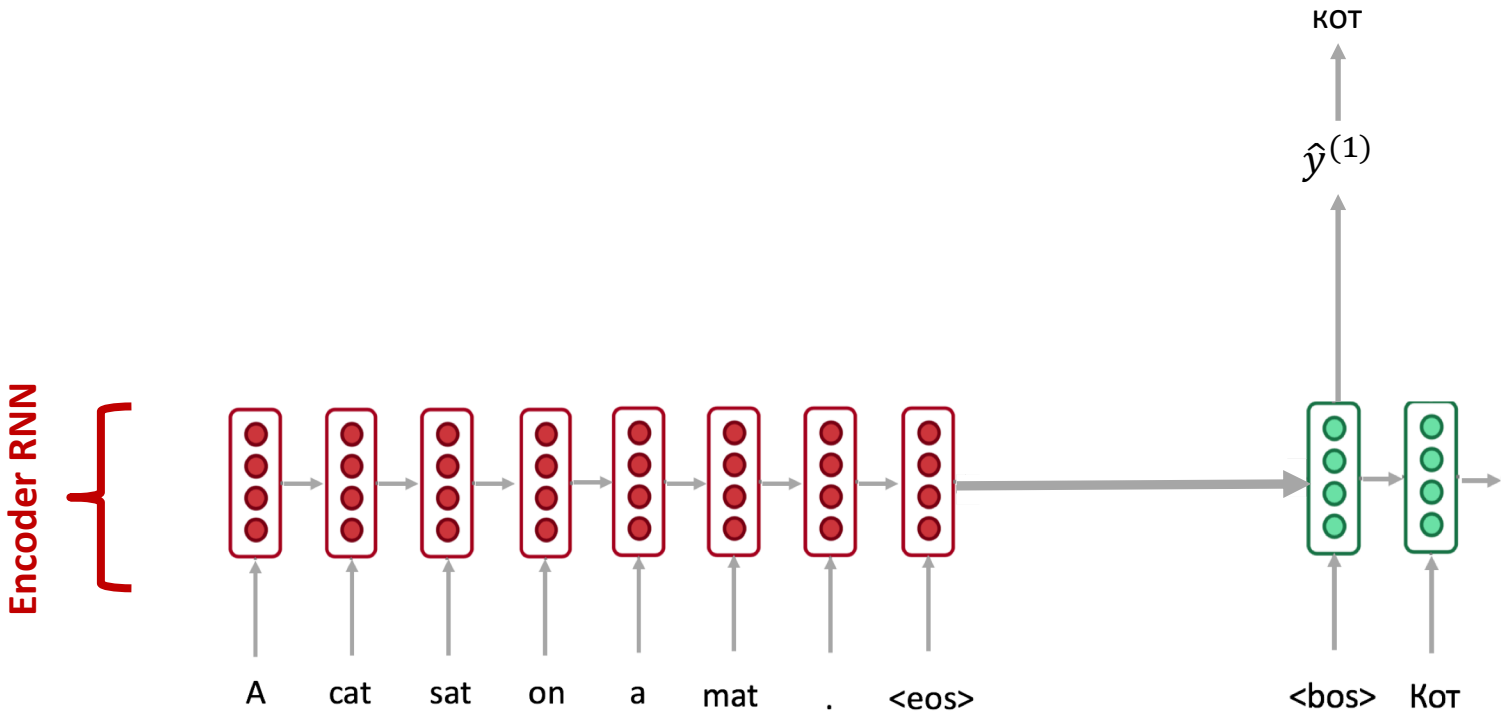
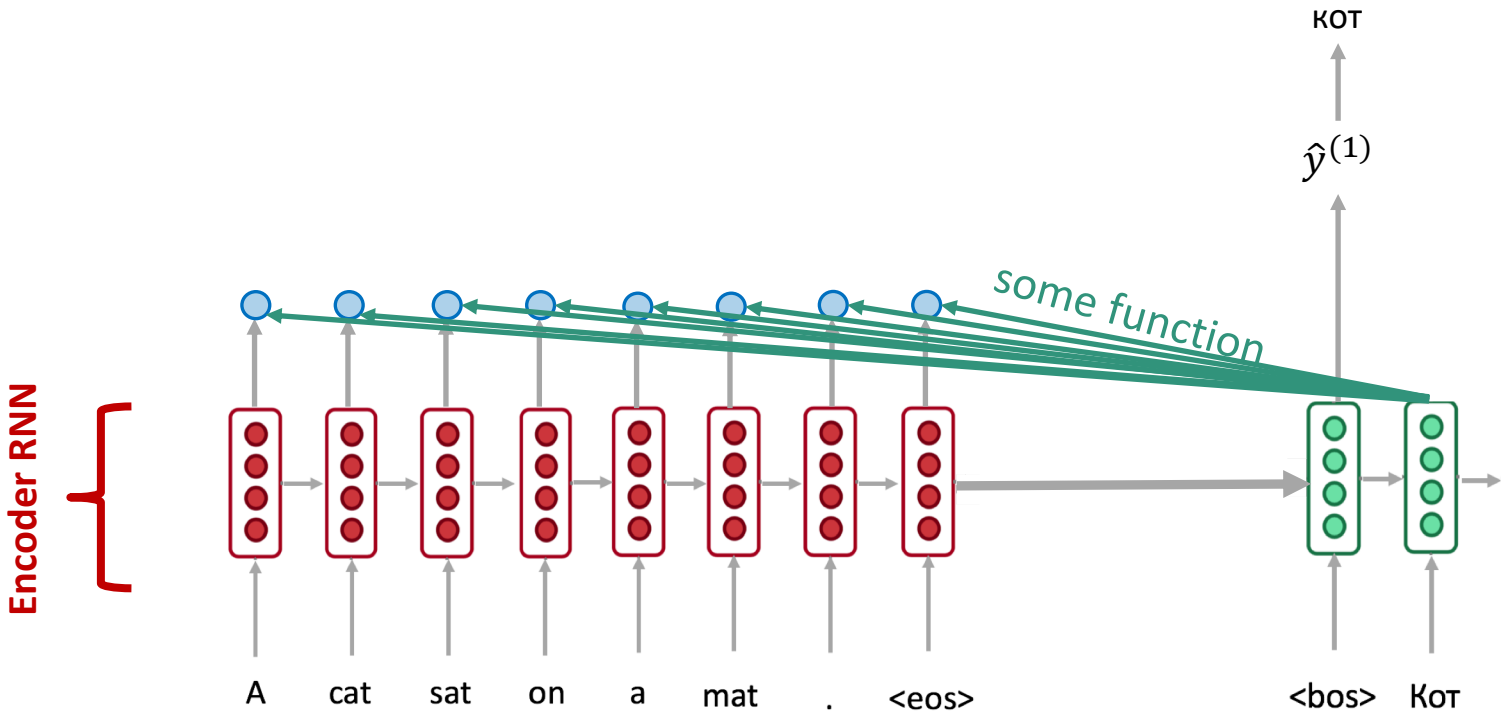
<eos>
<bos>

# Attention



# Attention


кот
$\hat{y}^{(1)}$
some function
Encoder RNN
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cat
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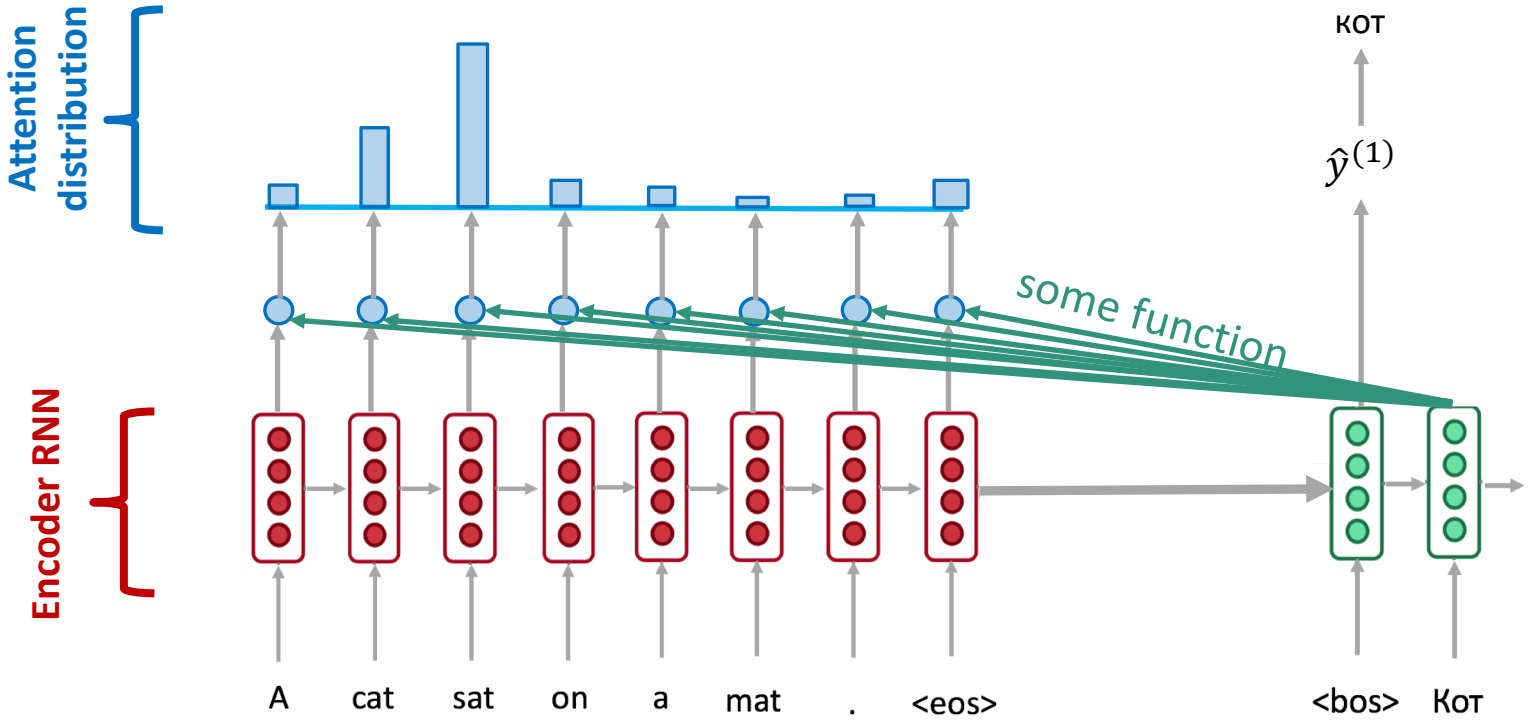
<eos>
<bos>
Кот

# Attention


Attention distribution
Encoder RNN
some function
кот
$\hat{y}^{(1)}$
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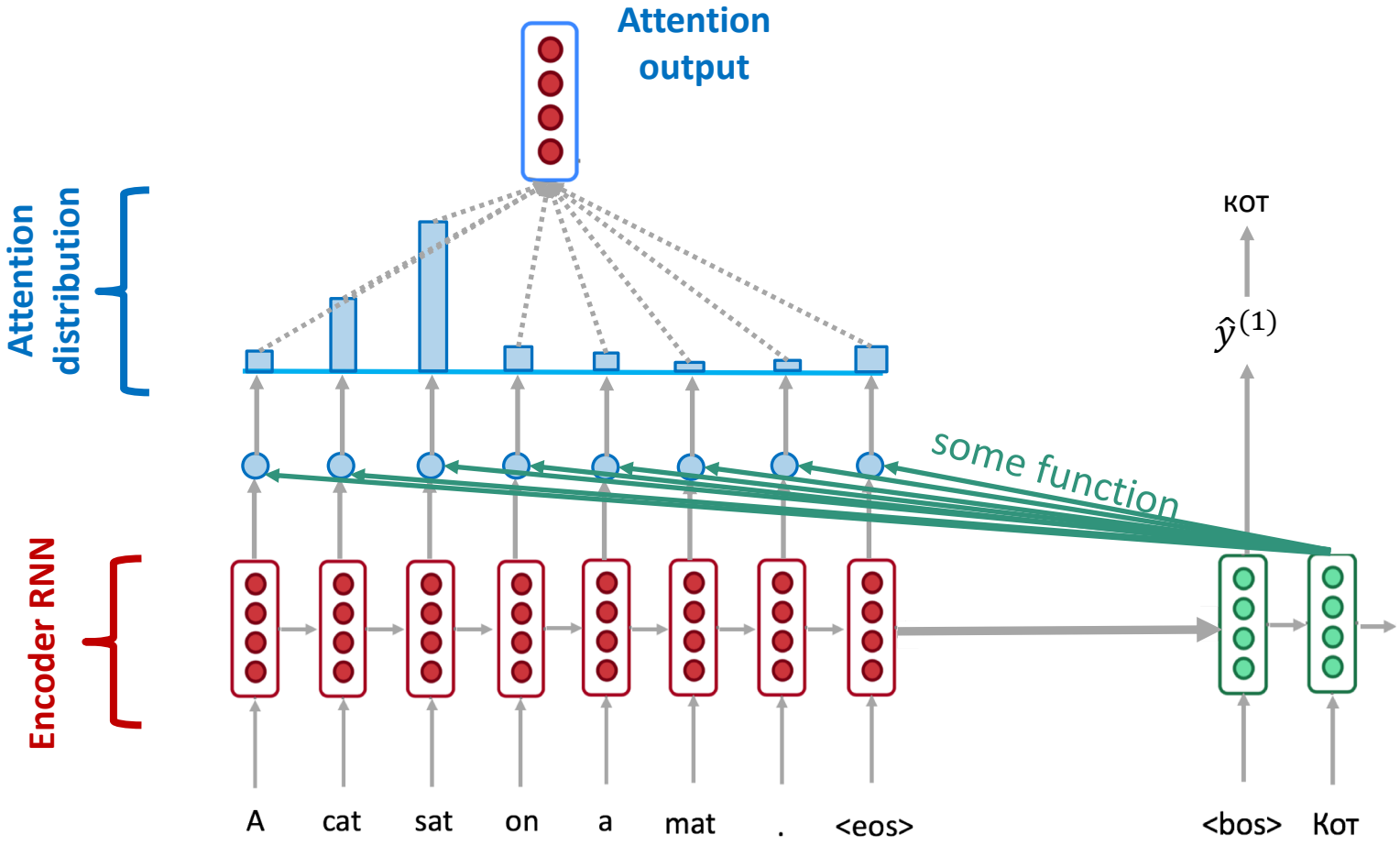
.
<eos>
<bos>
Кот

# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
$\hat{y}^{(1)}$
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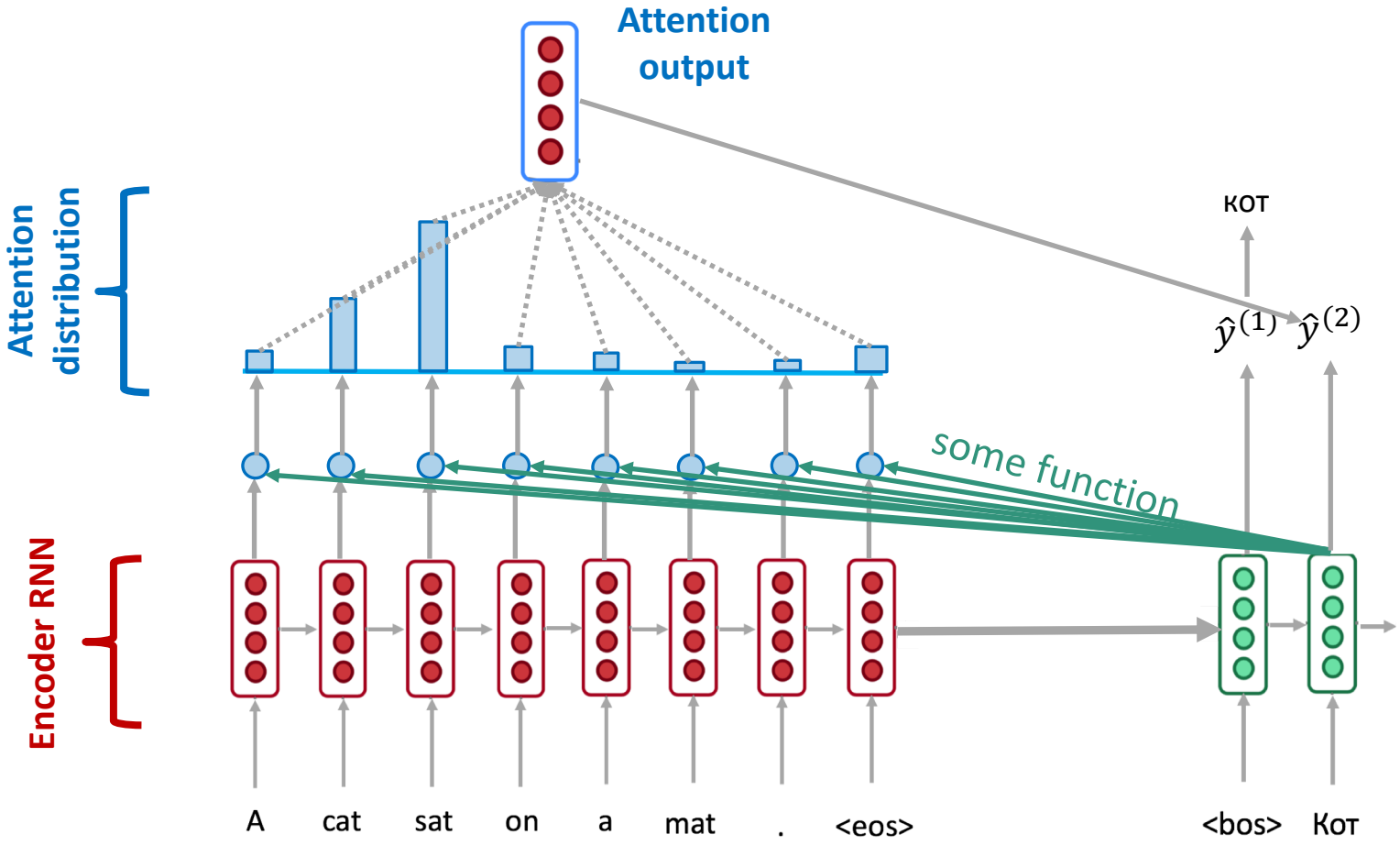
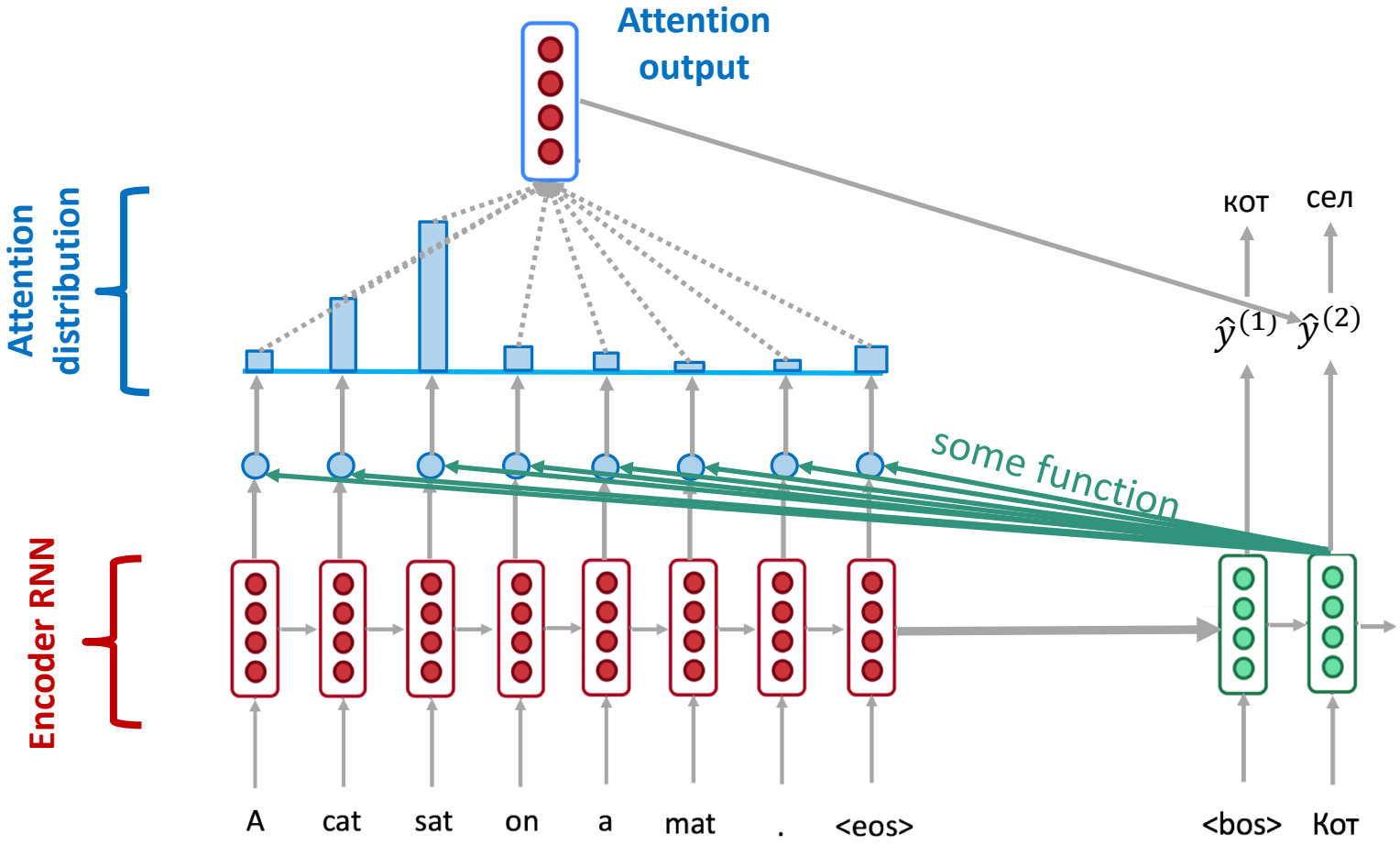
<eos>
<bos>
Кот

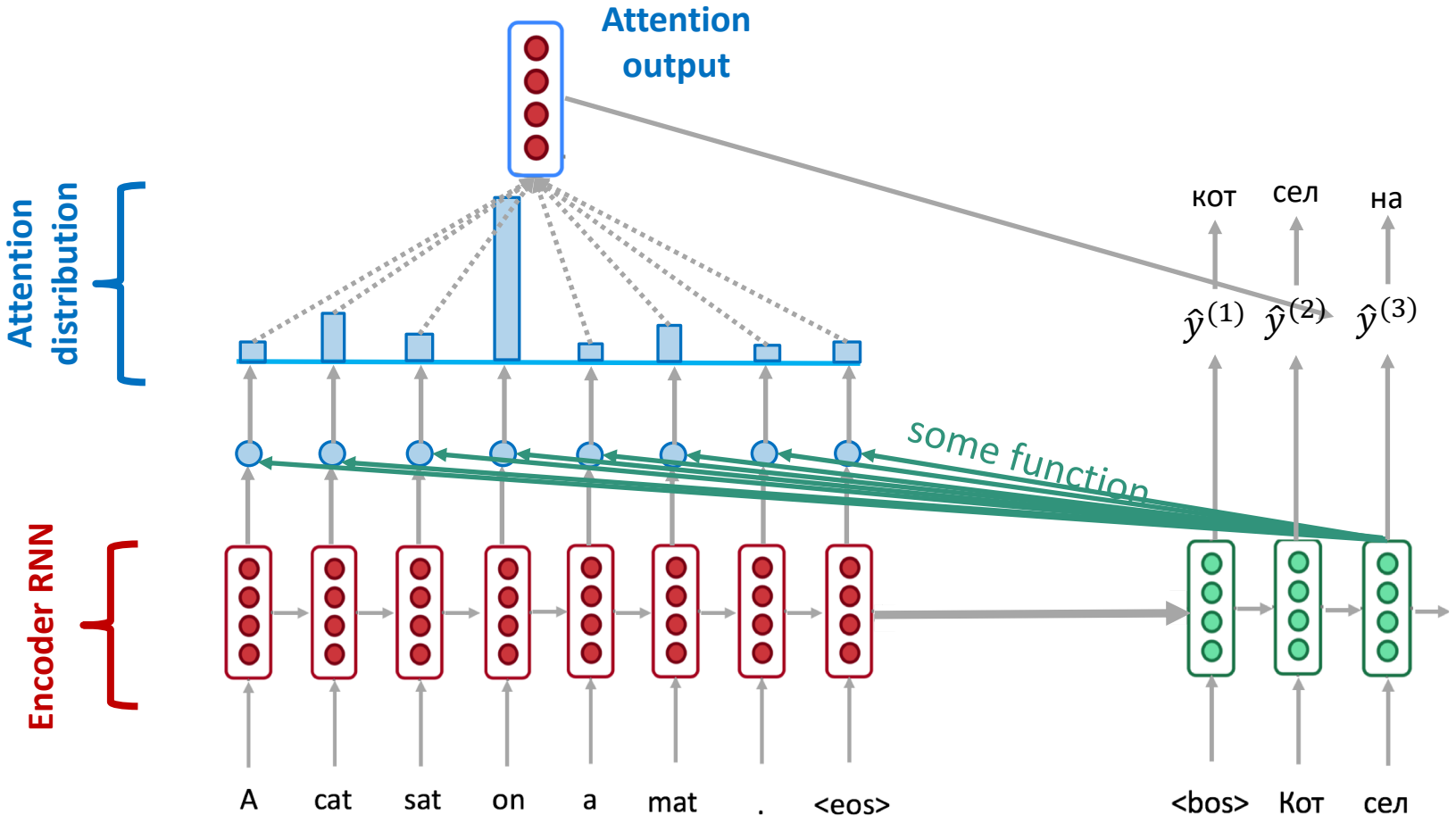
# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
$\hat{y}^{(1)}$
$\hat{y}^{(2)}$
A
cat
sat
on
a
mat
.
<eos>
<bos>
Кот

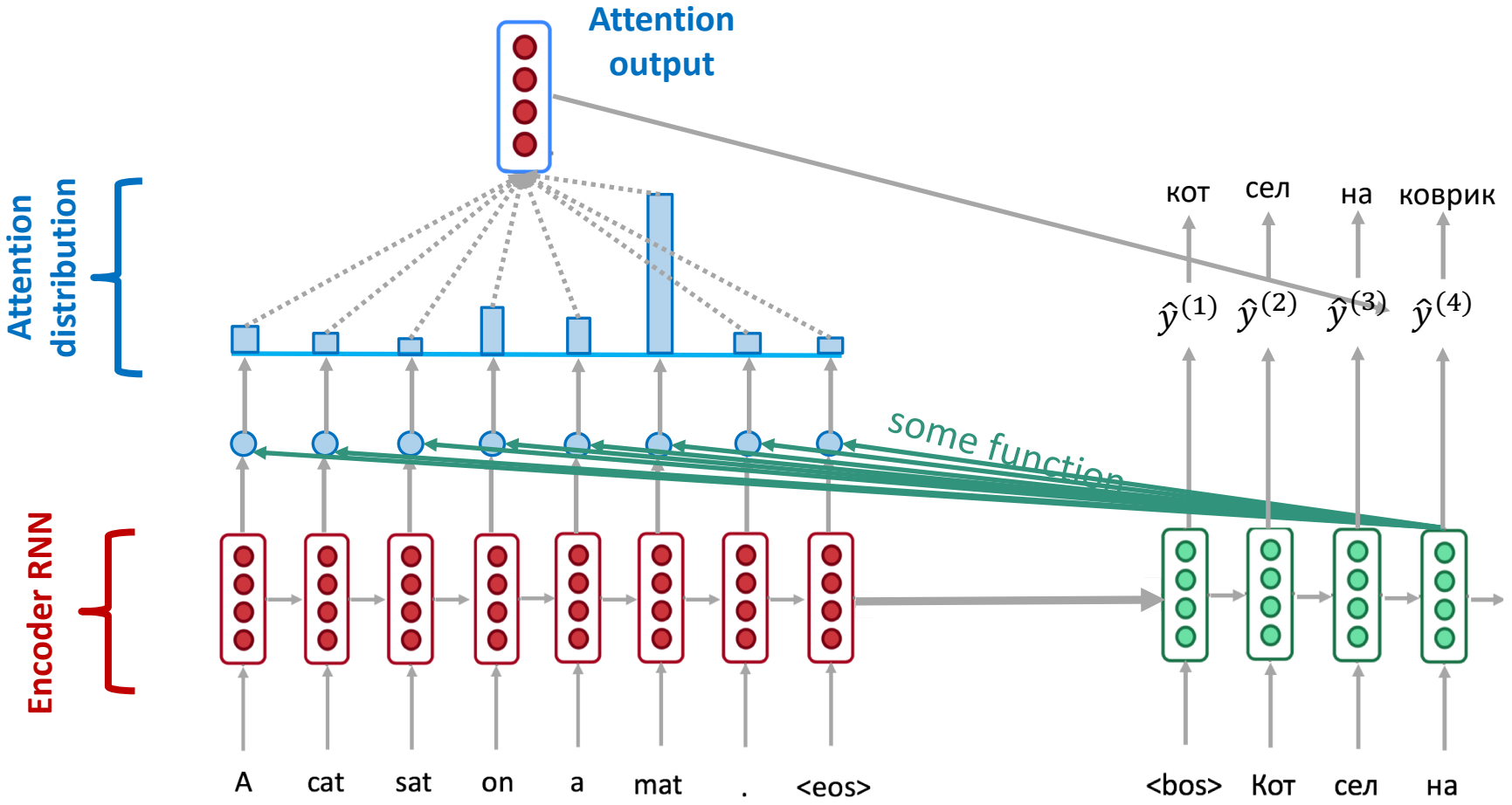
# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
сел
$\hat{y}^{(1)}$
$\hat{y}^{(2)}$
A
cat
sat
on
a
mat
.
<eos>
<bos>
Кот

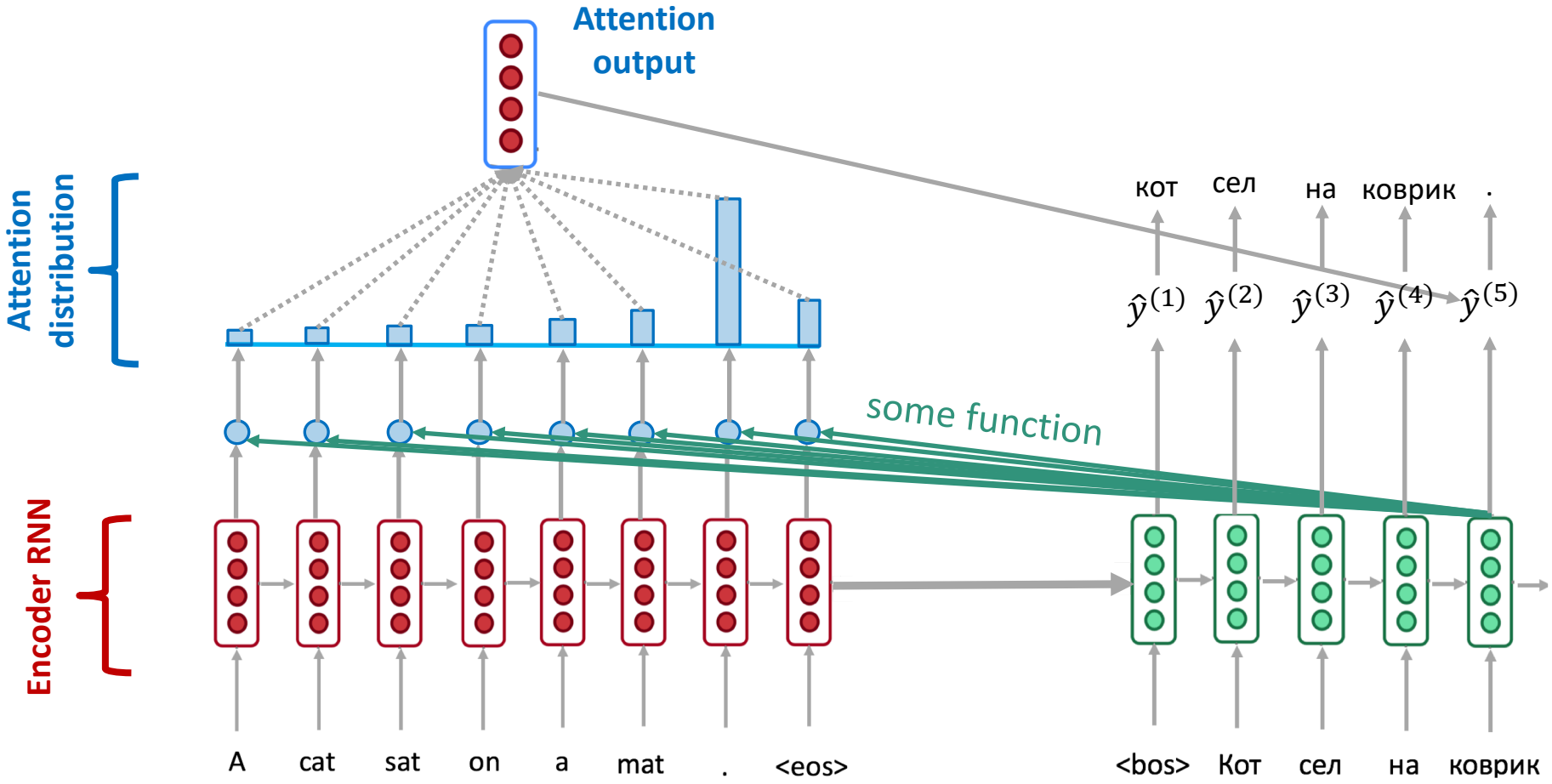
# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
сел
на
$\hat{y}^{(1)}$
$\hat{y}^{(2)}$
$\hat{y}^{(3)}$
A
cat
sat
on
a
mat
.
<eos>
<bos>
Кот
сел

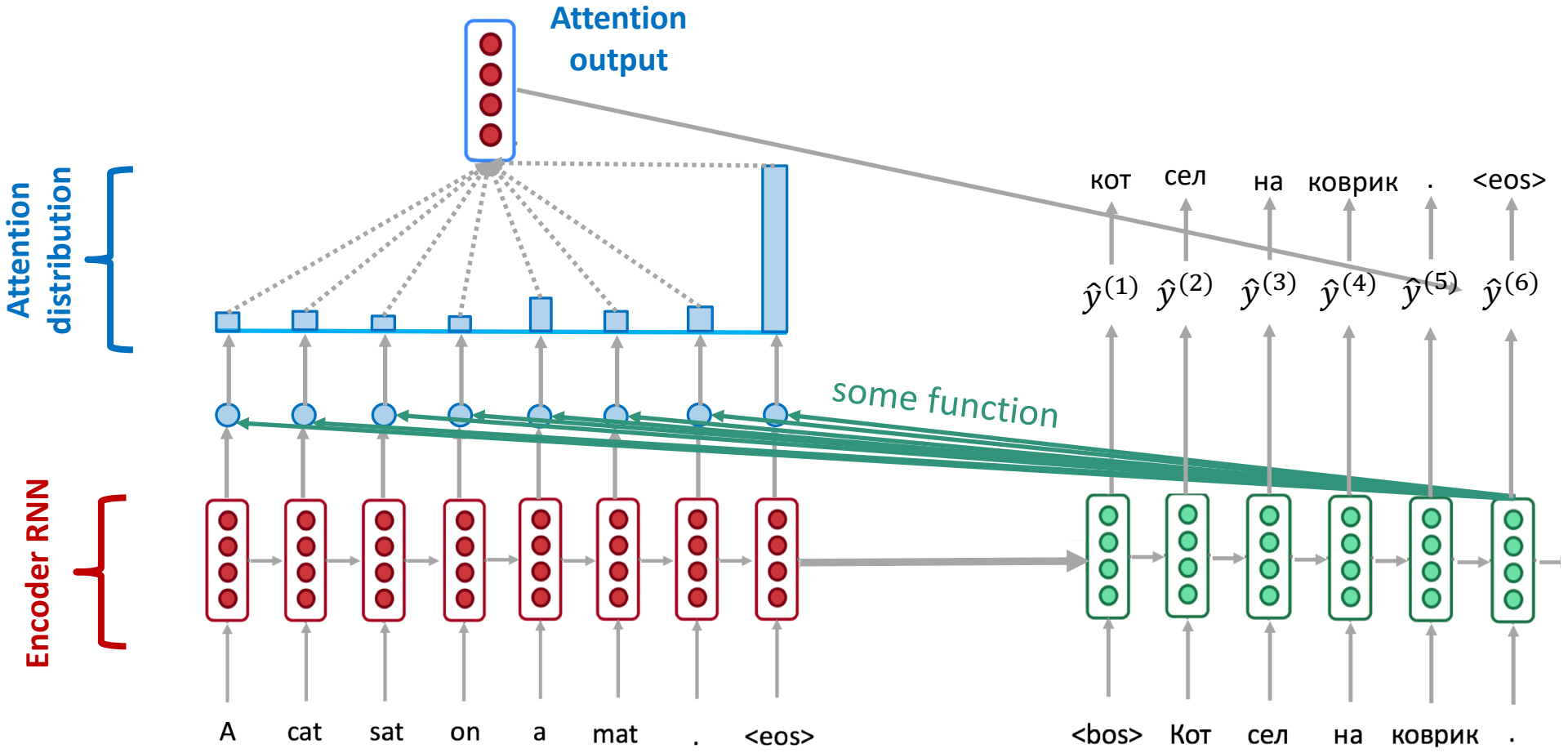
# Attention


Attention output
Attention distribution
Encoder RNN
some function
кот
сел
на
коврик
$\hat{y}^{(1)}$
$\hat{y}^{(2)}$
$\hat{y}^{(3)}$
$\hat{y}^{(4)}$
A
cat
sat
on
a
mat
.
<eos>
<bos>
Кот
сел
на

# Attention


Attention output
Attention distribution
some function
Encoder RNN
A
cat
sat
on
a
mat
.
<eos>
<bos>
Кот
сел
на
коврик
кот
сел
на
коврик
.
$\hat{y}^{(1)}$
$\hat{y}^{(2)}$
$\hat{y}^{(3)}$
$\hat{y}^{(4)}$
$\hat{y}^{(5)}$

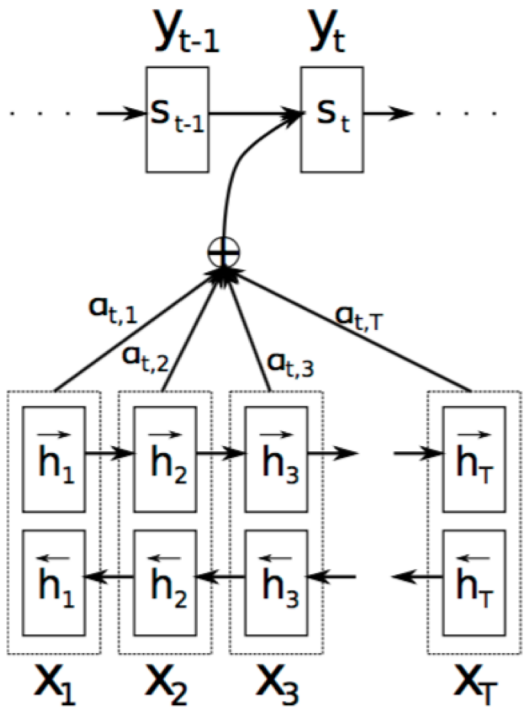
# Attention


Attention output
Attention distribution
some function
Encoder RNN
A cat sat on a mat . <eos>
<bos> Кот сел на коврик .
кот сел на коврик . <eos>
$\hat{y}^{(1)}$ $\hat{y}^{(2)}$ $\hat{y}^{(3)}$ $\hat{y}^{(4)}$ $\hat{y}^{(5)}$ $\hat{y}^{(6)}$

# Neural Machine Translation by Jointly Learn to Align and Translate

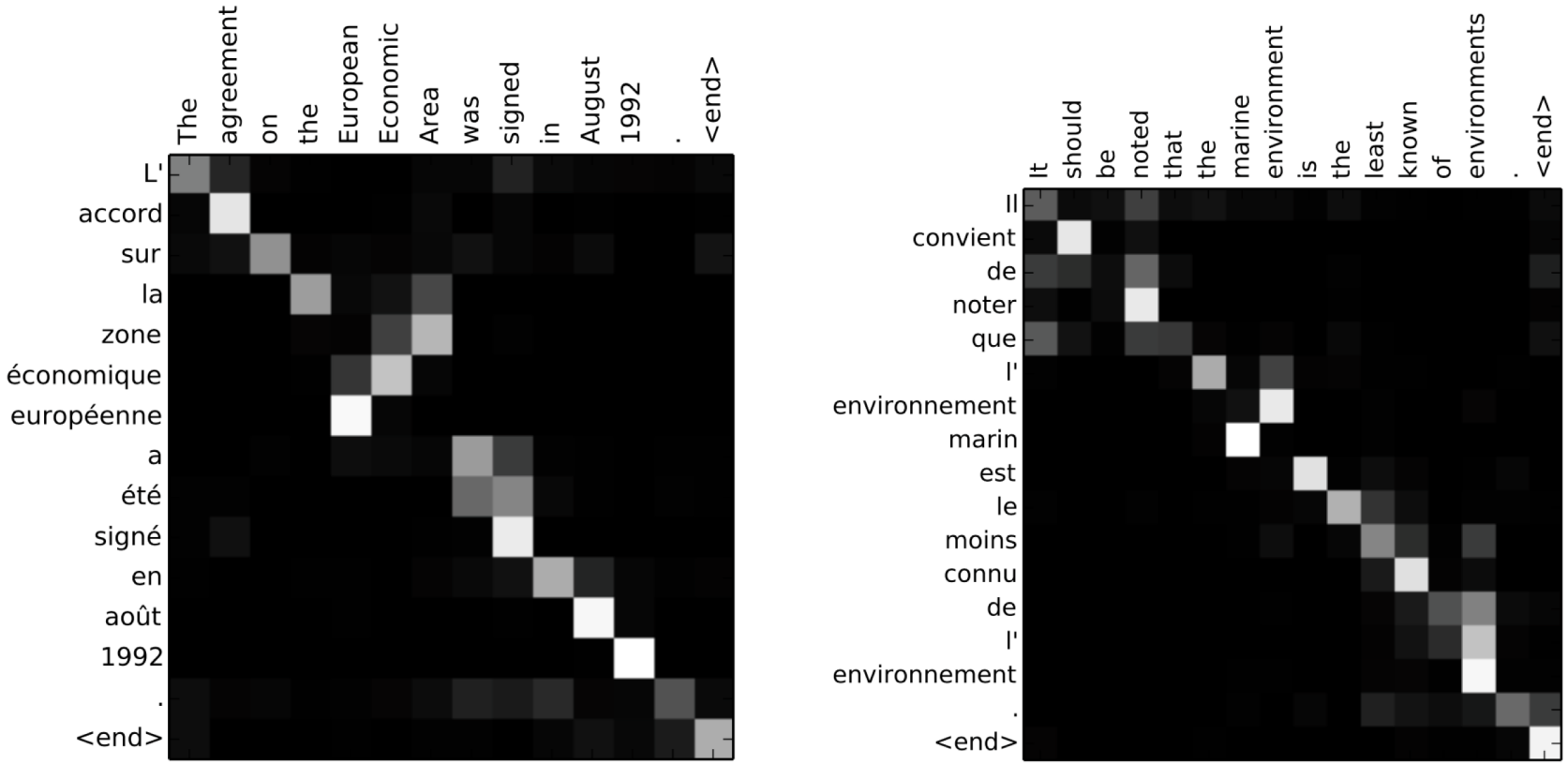
Attention advantages:

- do not have to remember the whole sentence
- no fixed size representation
- similar to humans



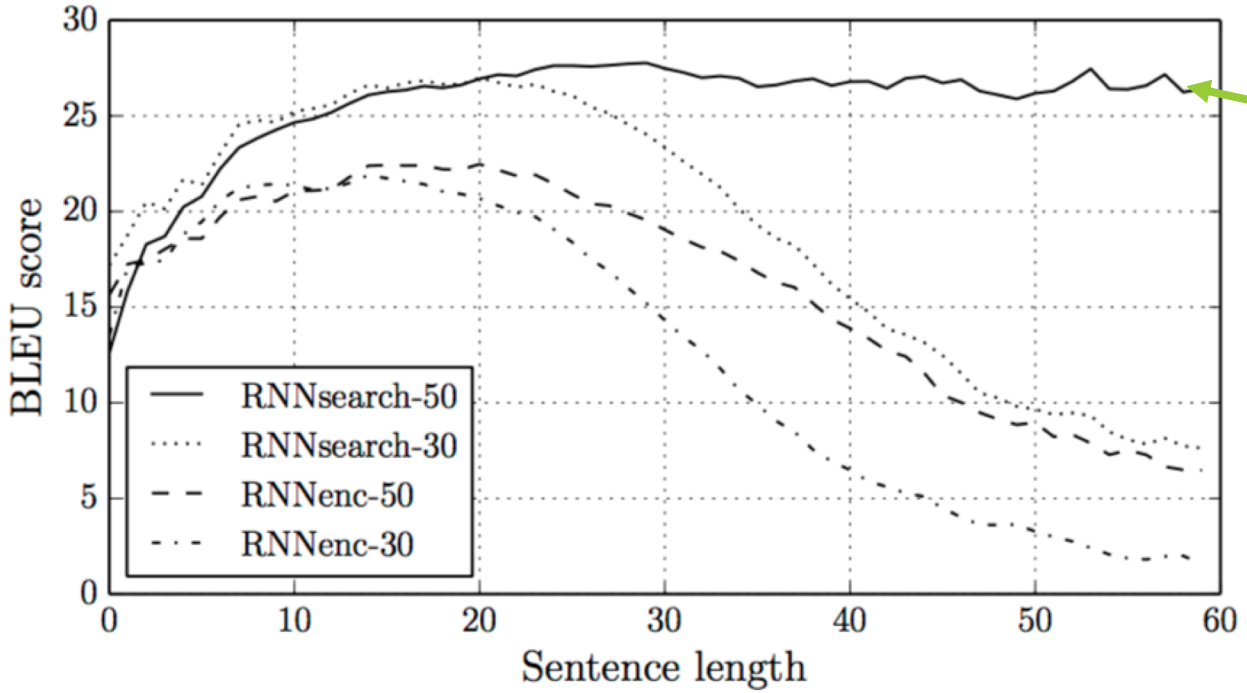

Bahdanau et al, ICLR 2015, https://arxiv.org/pdf/1409.0473.pdf

# Neural Machine Translation by Jointly Learn to Align and Translate




Bahdanau et al, ICLR 2015, https://arxiv.org/pdf/1409.0473.pdf

# Neural Machine Translation by Jointly Learn to Align and Translate




No performance drop for long sentences!

Bahdanau et al, ICLR 2015, https://arxiv.org/pdf/1409.0473.pdf

# Attention score: how to compute it?

- dot product
- bilinear function
- multi-layer perceptron
- any, literally, ANY function you can imagine

$$f_{att}(h_i, s_j) = h_i^\top s_j$$

$$f_{att}(h_i, s_j) = h_i^\top \mathbf{W}_a s_j$$

$$f_{att}(\mathbf{h}_i, \mathbf{s}_j) = \mathbf{v}_a^\top \tanh(\mathbf{W}_a[\mathbf{h}_i; \mathbf{s}_j])$$

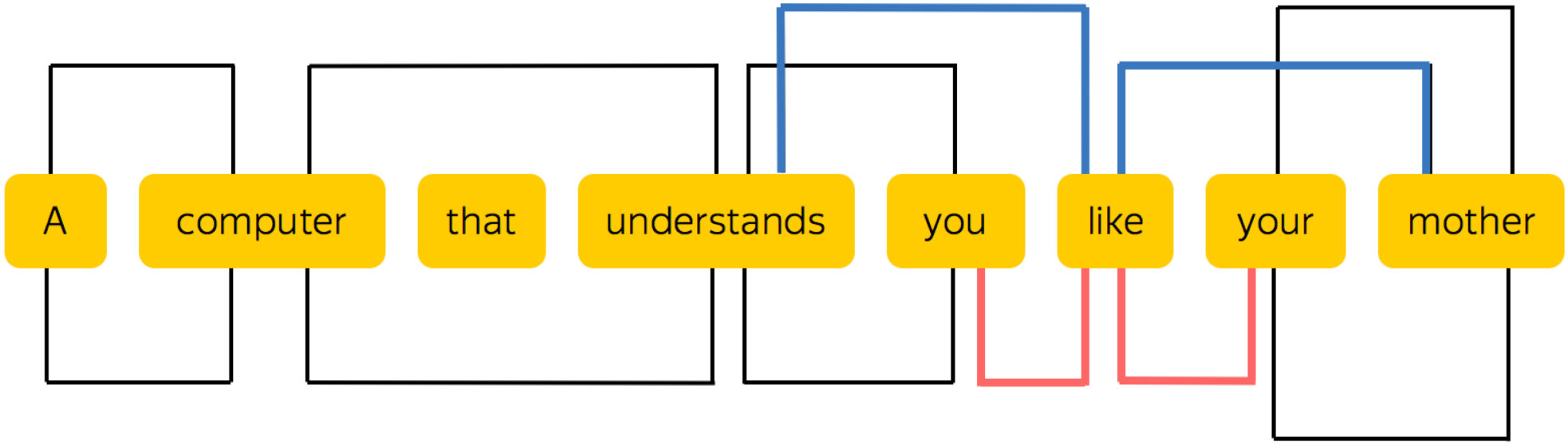
$$z(c, m, q) = \left[c, m, q, c \circ q, c \circ m, |c - q|, |c - m|, c^T W^{(b)} q, c^T W^{(b)} m\right]$$

$$G(c, m, q) = \sigma\left(W^{(2)} \tanh\left(W^{(1)} z(c, m, q) + b^{(1)}\right) + b^{(2)}\right)$$

A bit crazy, huh?

# Transformer: Attention is all you need!

# Self-Attention

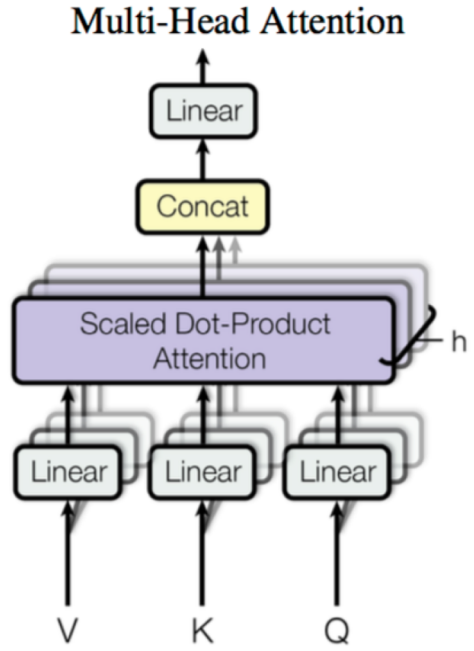



(example and picture from David Talbot)

# Multi-Head Attention

- Gender agreement
- Case government
- Lexical preferences
- ...

(example from David Talbot)



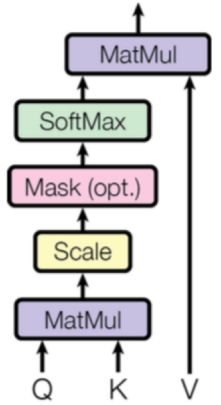

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_\text{h})W^O$$

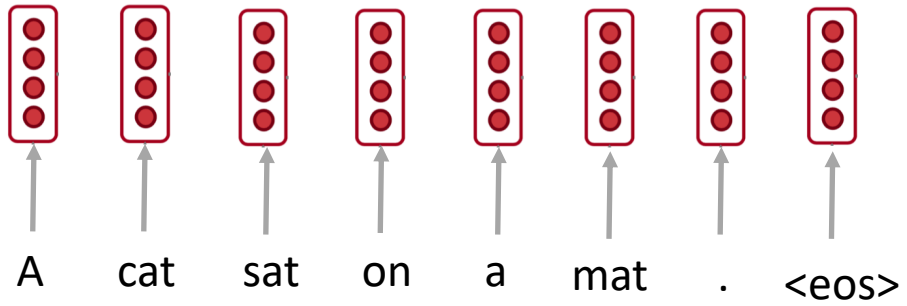
$$\text{where head}_\text{i} = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

# Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

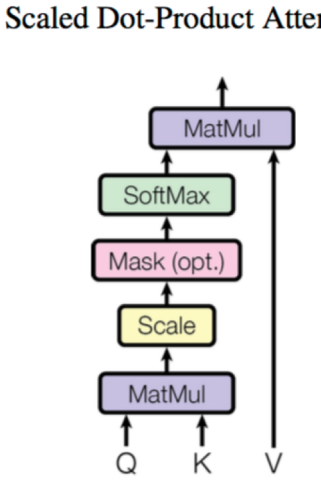
Scaled Dot-Product Attention

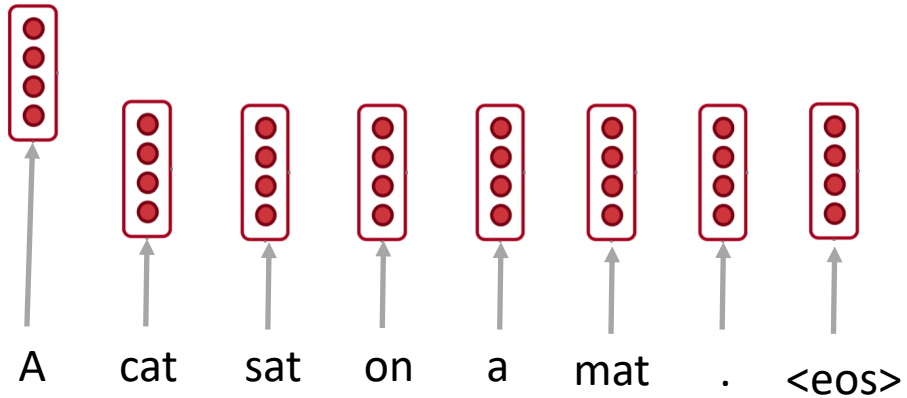


# Self-Attention

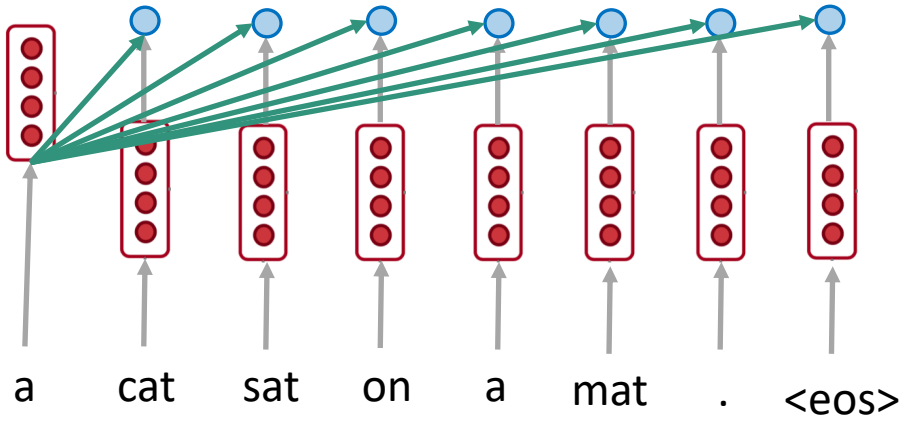
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

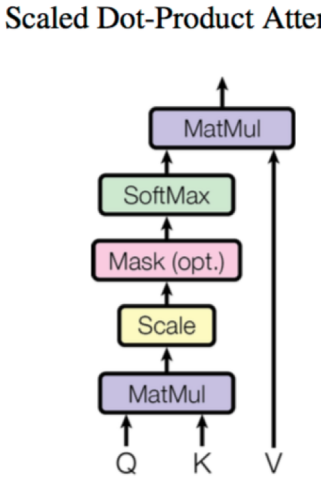


# Self-Attention

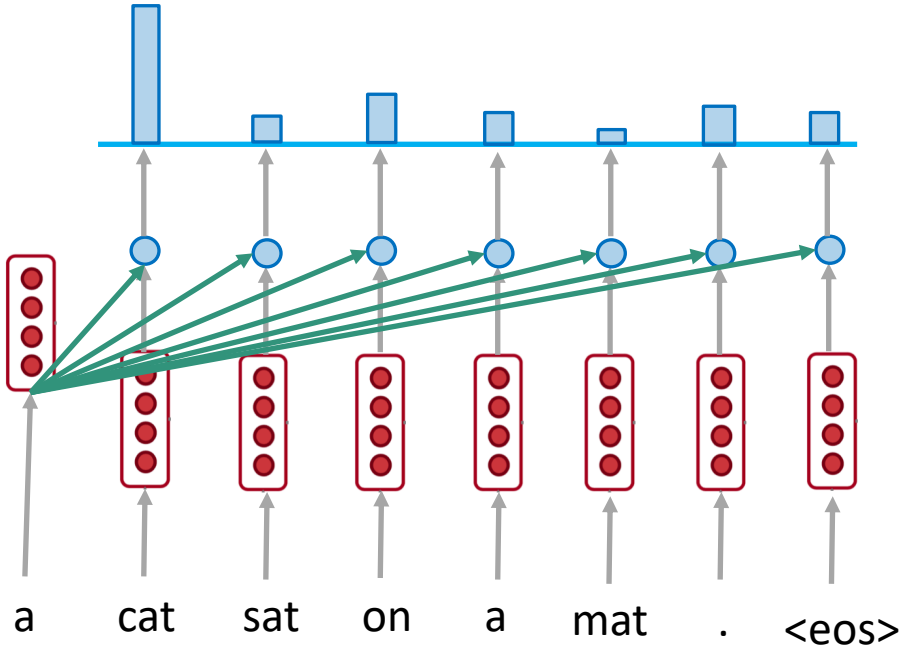
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

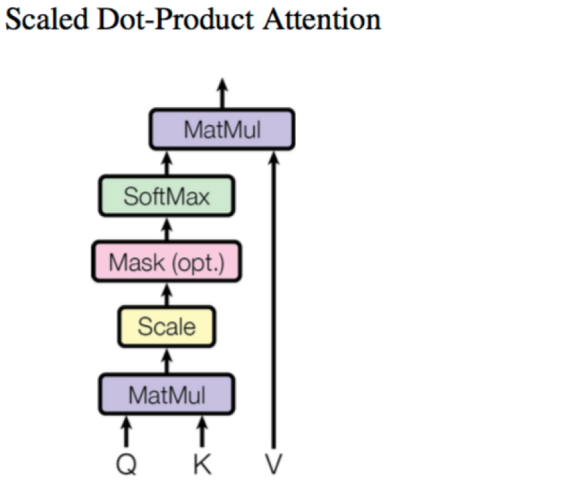


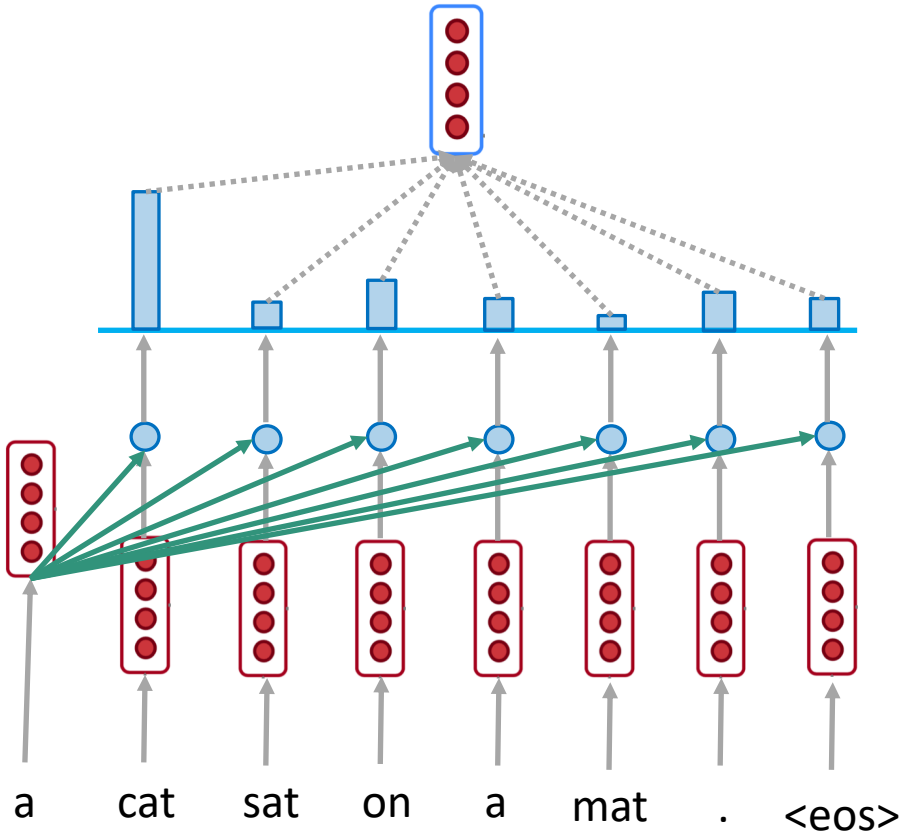
# Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



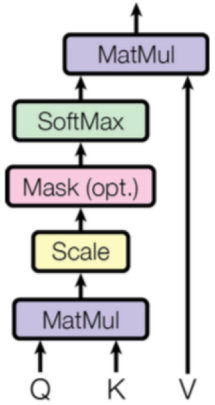



# Self-Attention



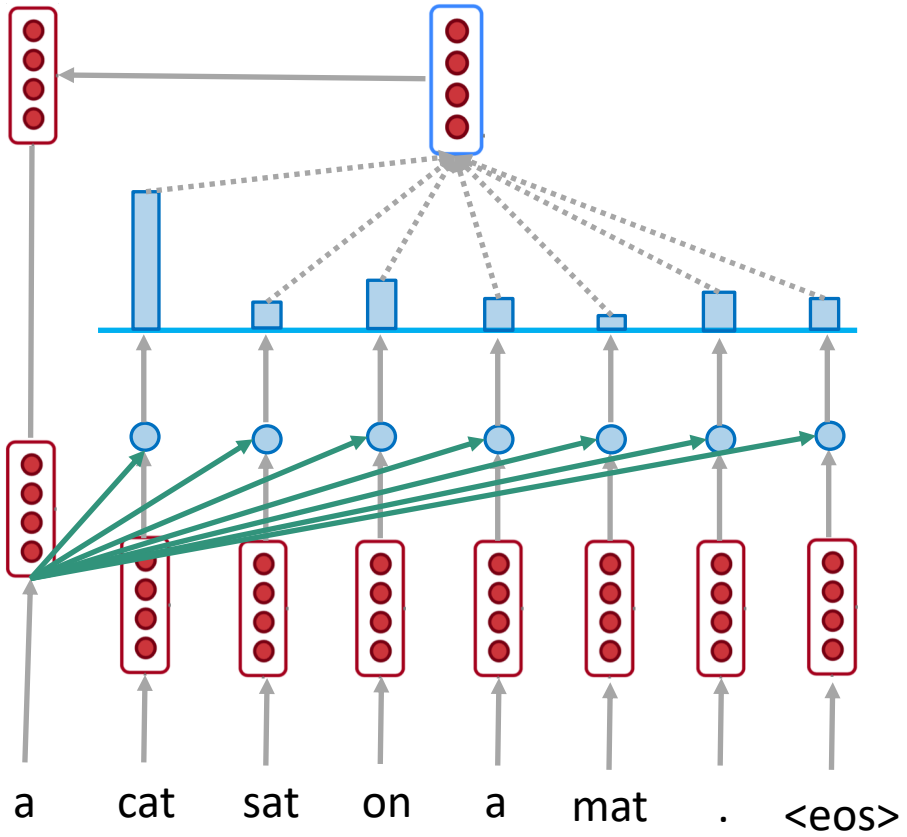

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Scaled Dot-Product Attention



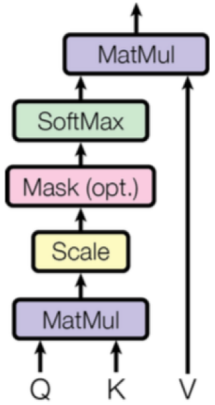
# Self-Attention

update representation for the word ”a”



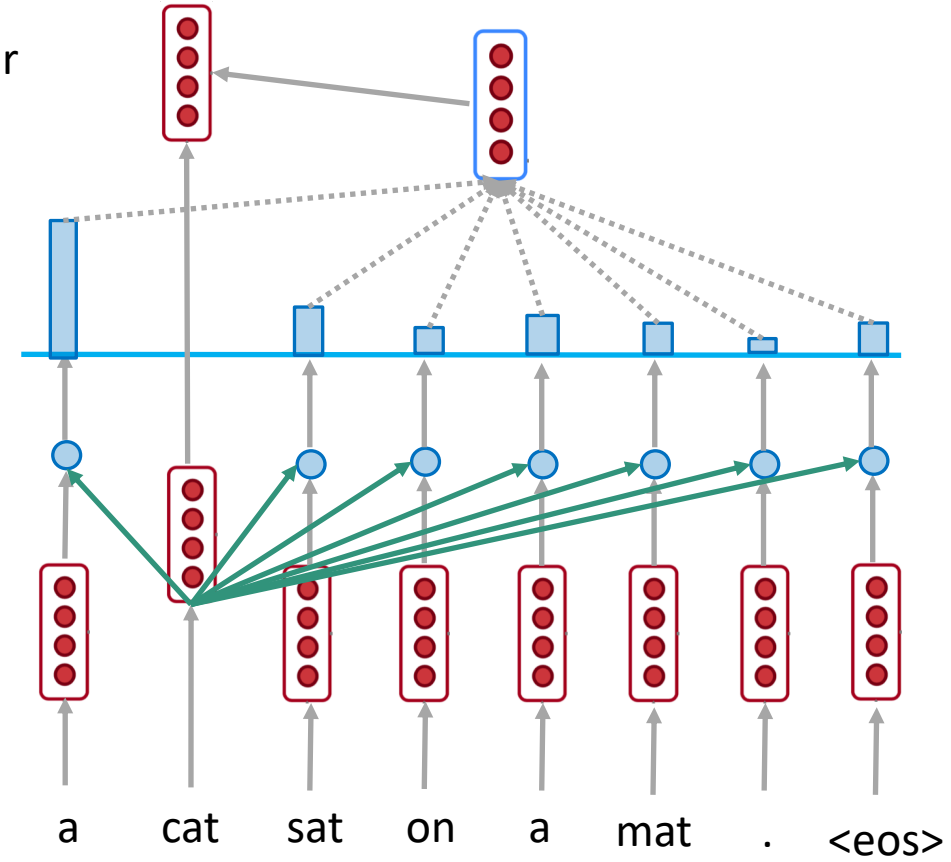

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Scaled Dot-Product Attention



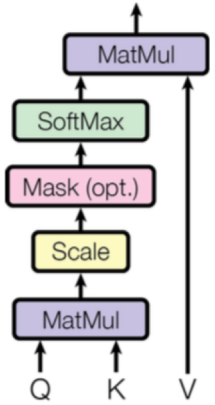
# Self-Attention

update representation for the word ”cat”



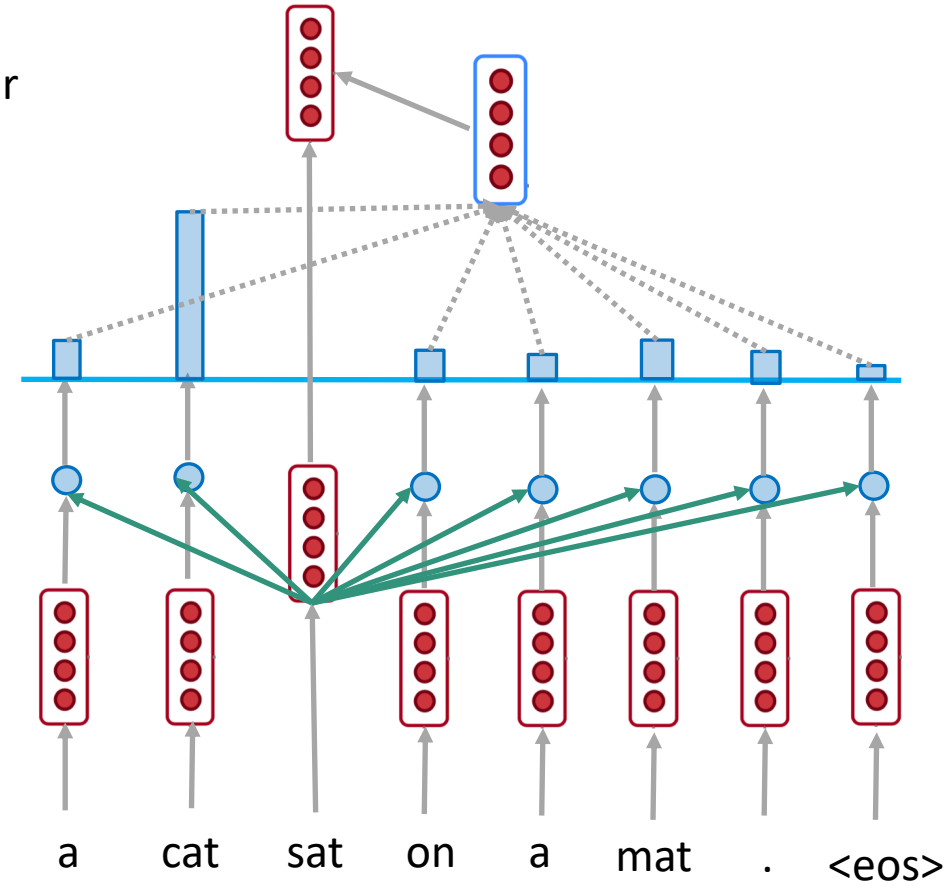

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Scaled Dot-Product Attention



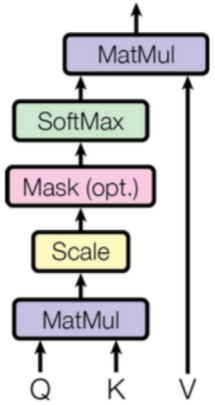
# Self-Attention

update representation for the word ”sat”



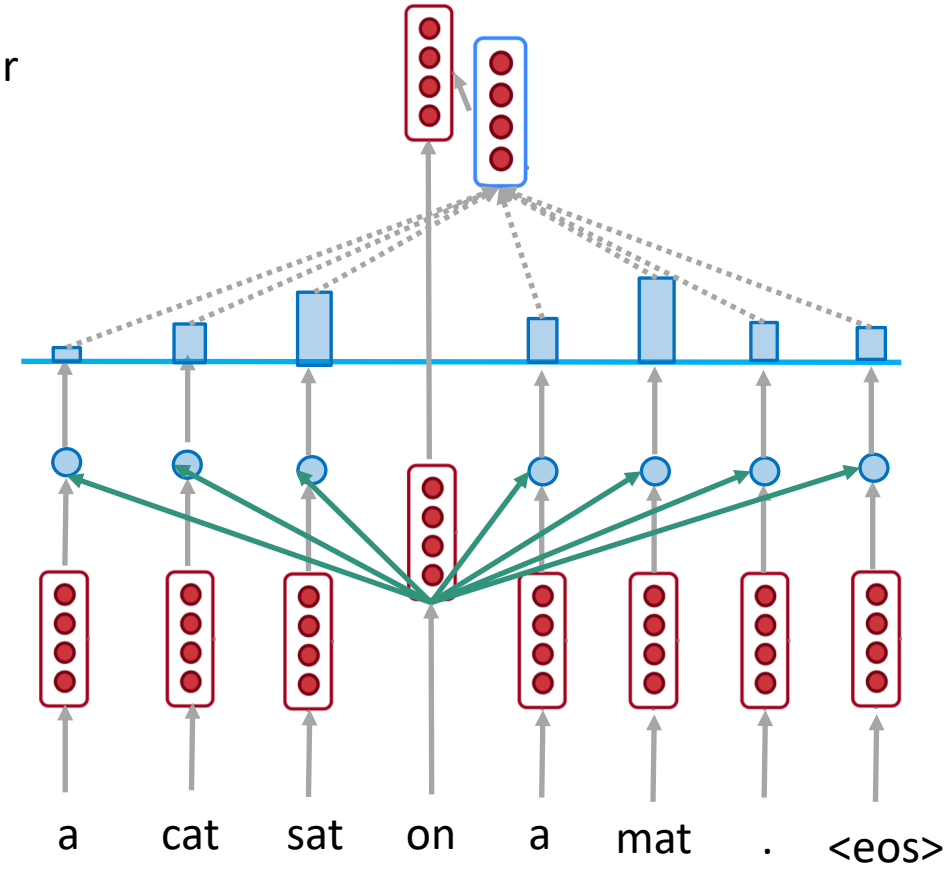

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Scaled Dot-Product Attention



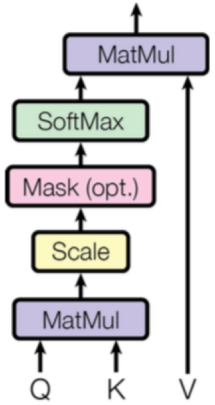
# Self-Attention

update representation for the word ”on”




$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Scaled Dot-Product Attention



# Self-Attention

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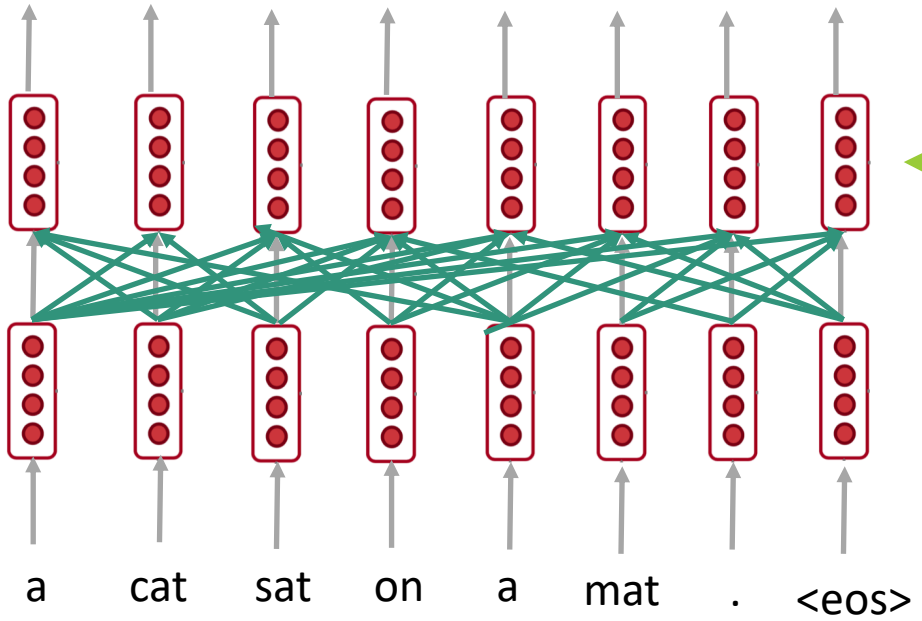
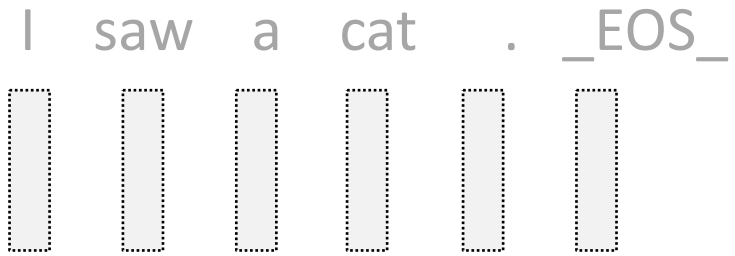
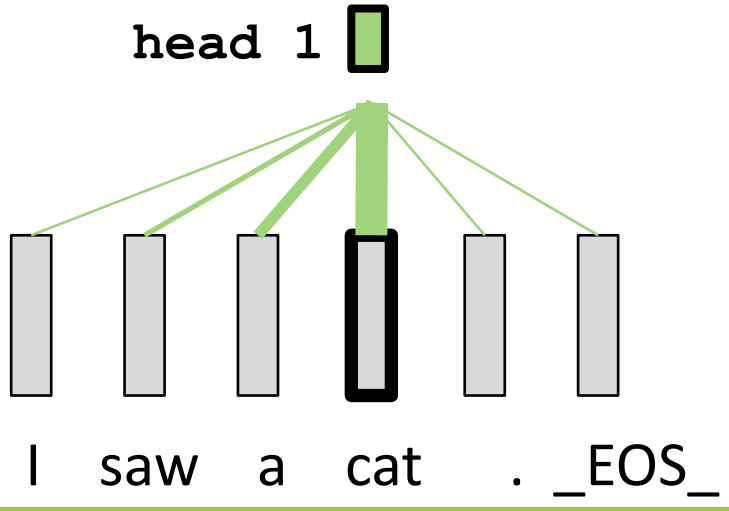


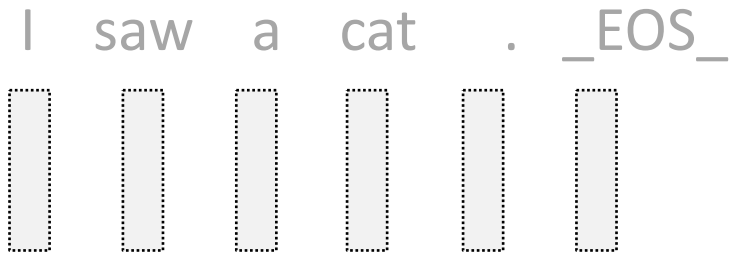

Updated representations for each word (one layer)

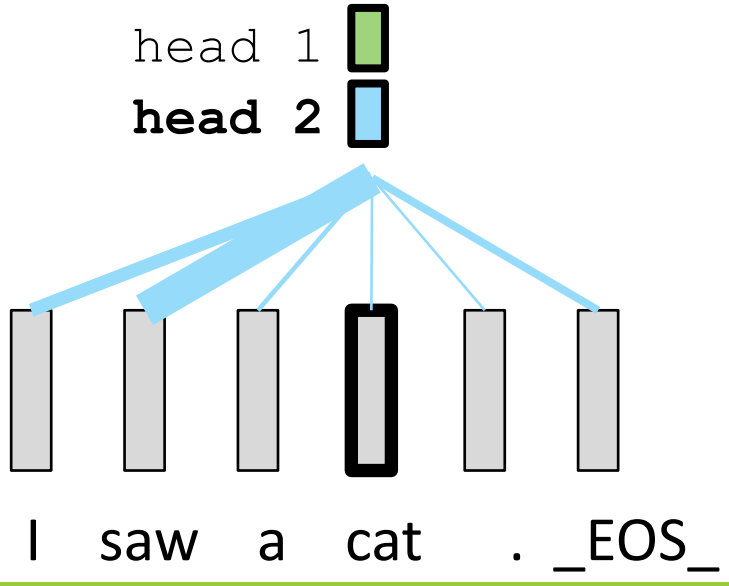
# Multi-Head Attention



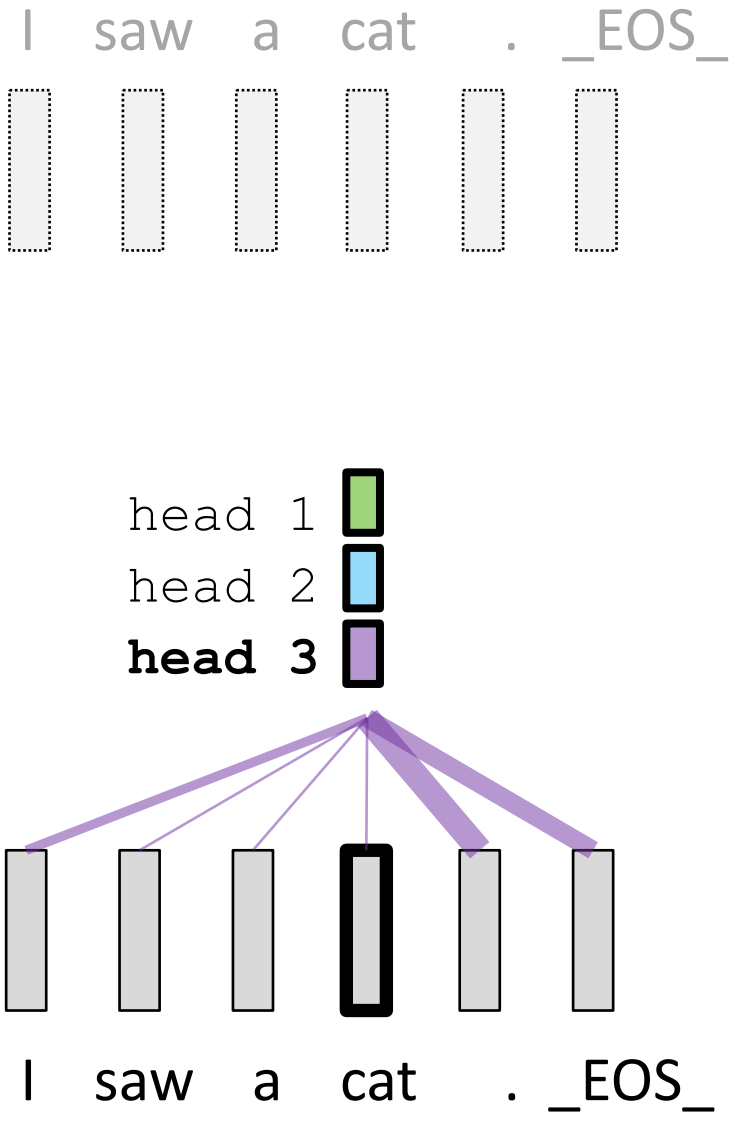



# Multi-Head Attention

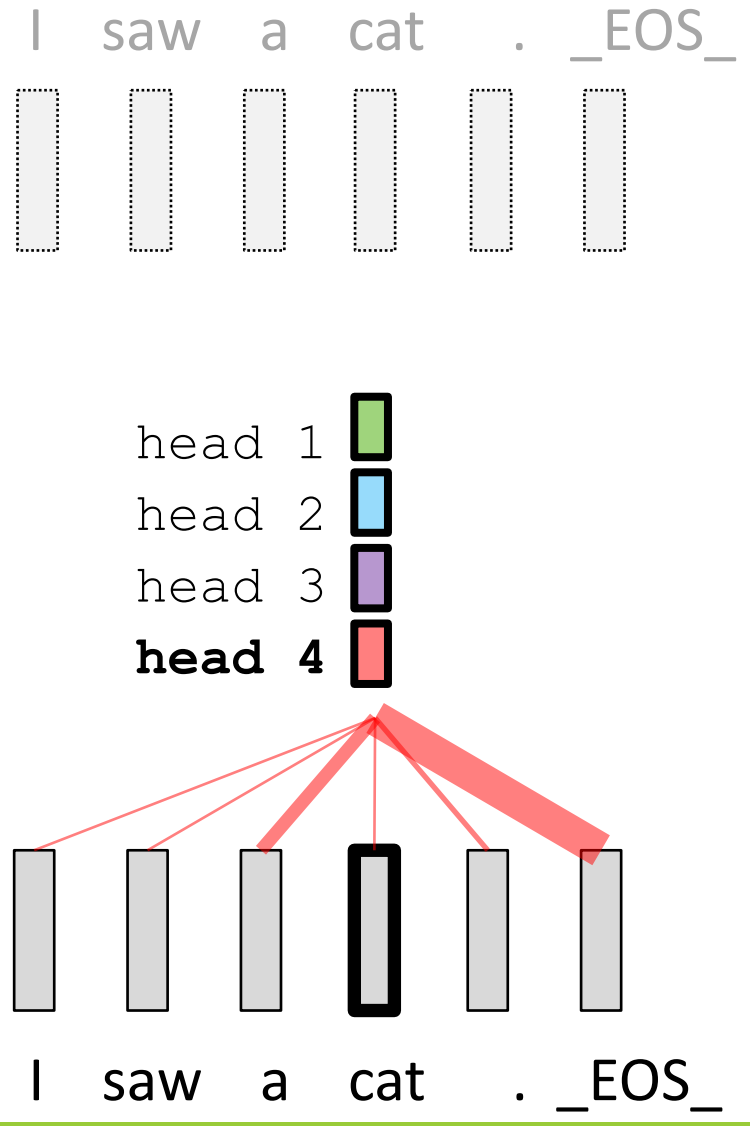


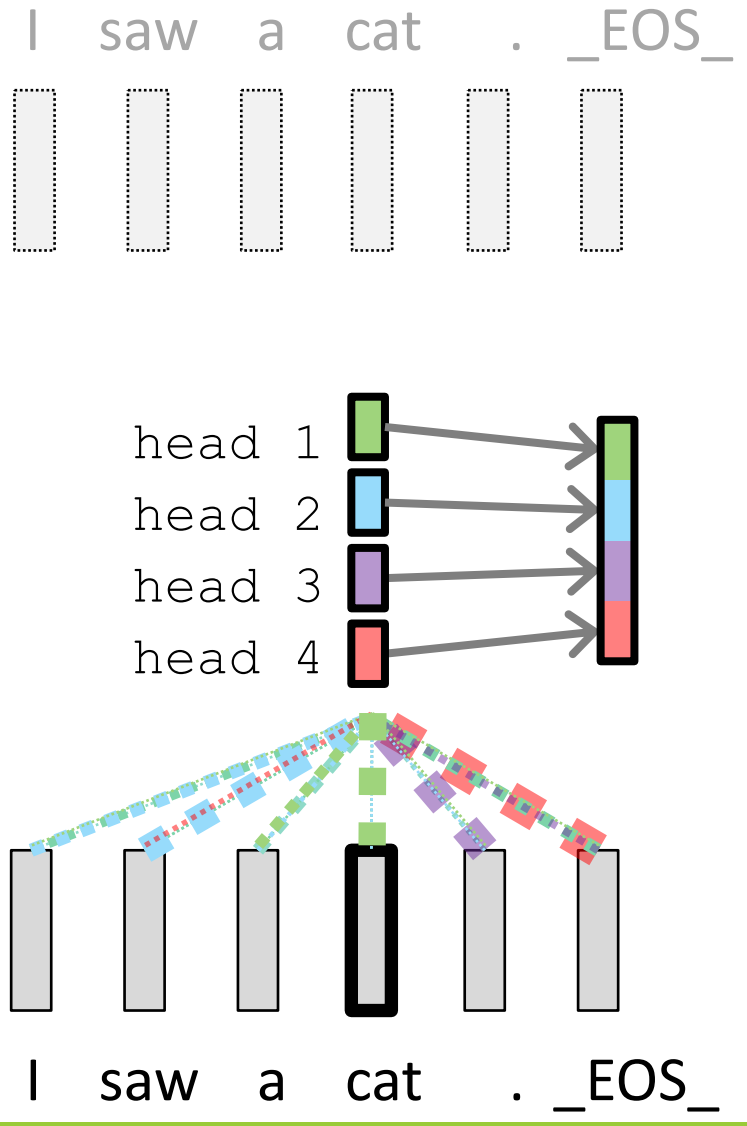
# Multi-Head Attention



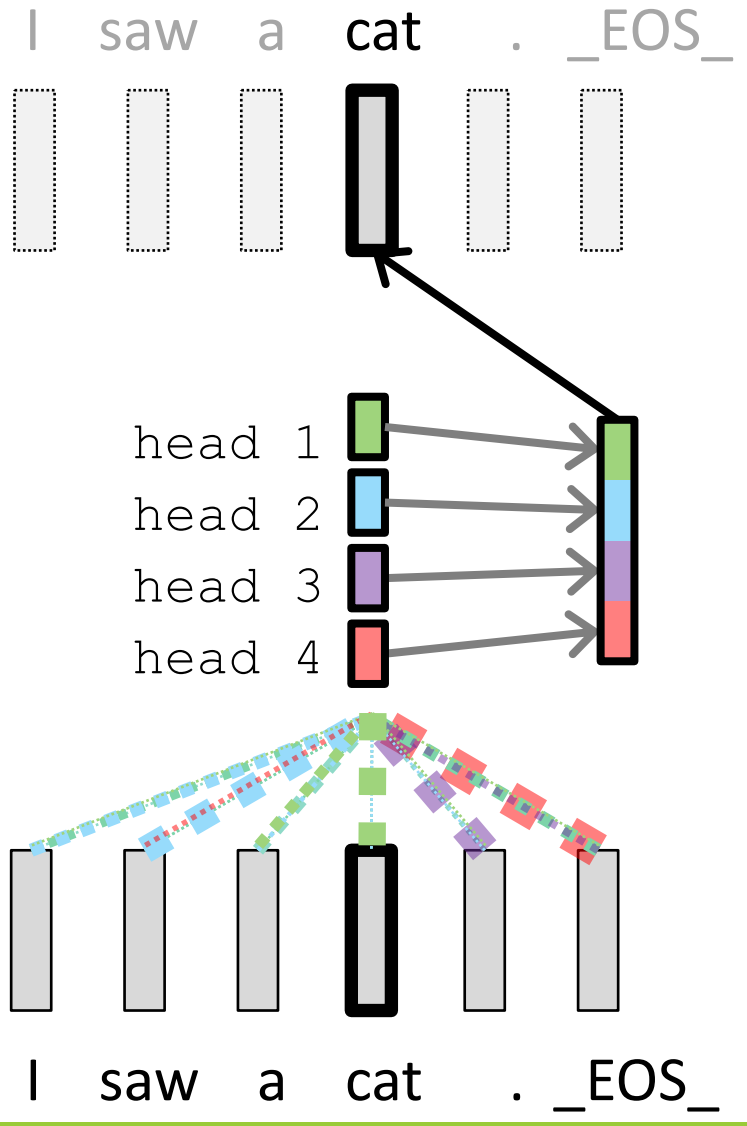
# Multi-Head Attention



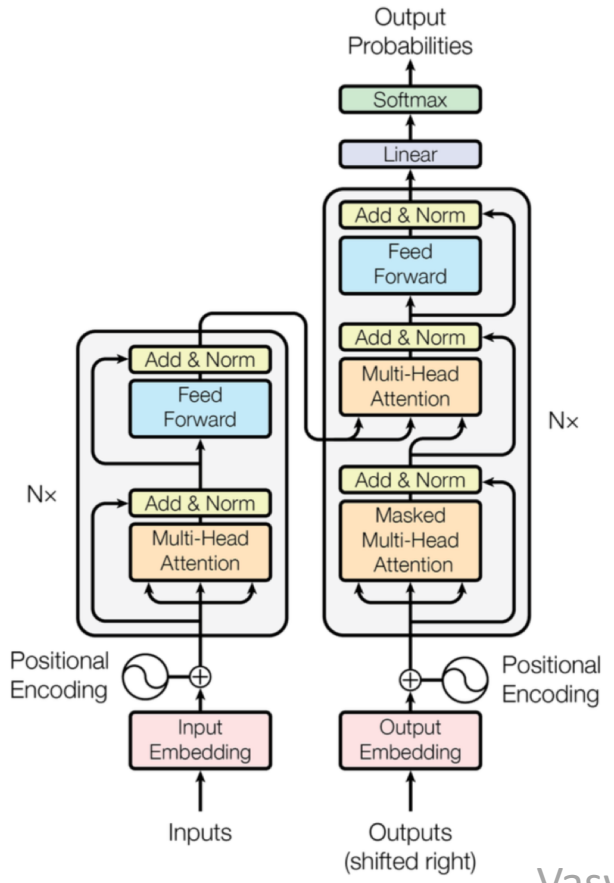
# Multi-Head Attention

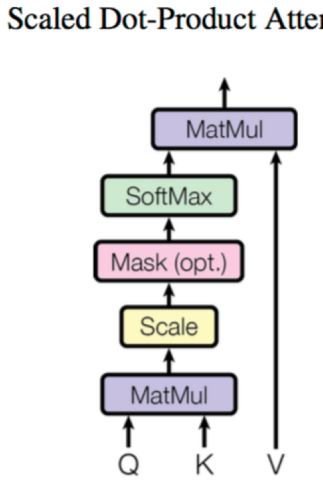


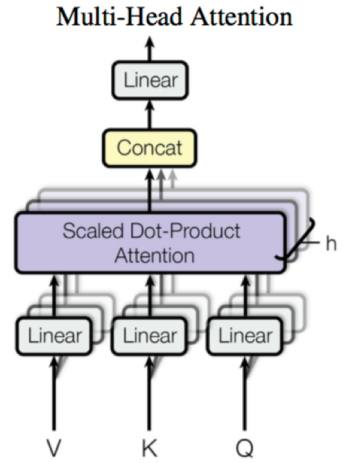
# Multi-Head Attention



# Transformer



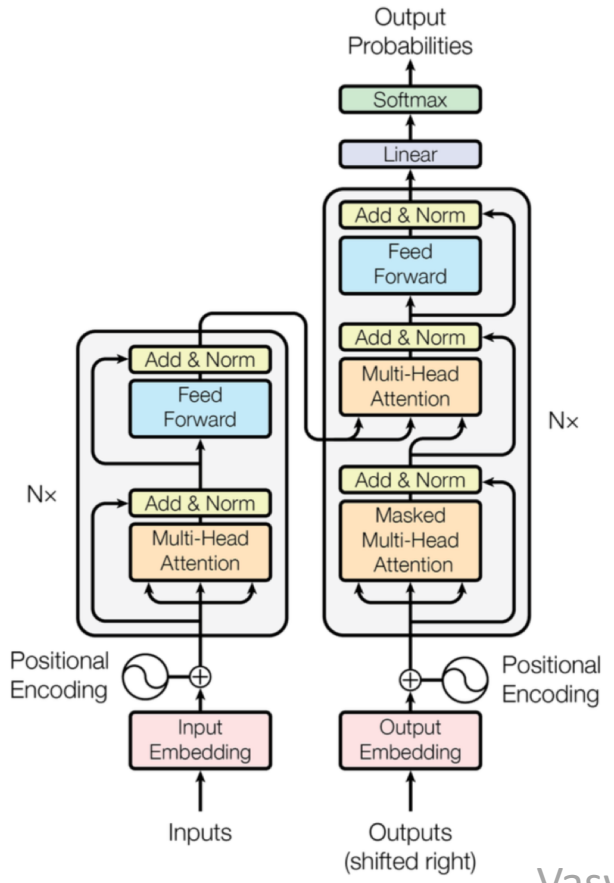




$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

# Transformer




Advantages:

- No recurrence: parallel encoding
- Fast training: both encoder and decoder are parallel
- No long connections: O(1) for all tokens
- Three attentions: the model does not have to remember too much
- Multi-head attention allows to pay attention to different aspects

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

# Transformer

I arrived at the

I arrived at the bank after crossing the…

… river? …road?

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

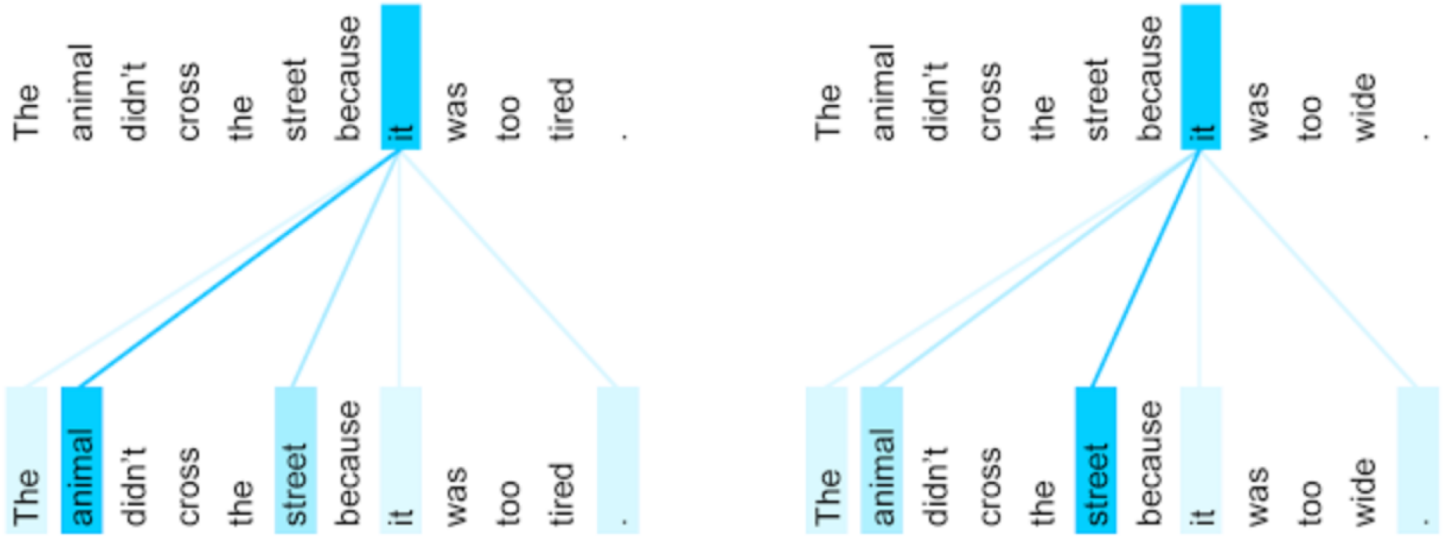
# Transformer

*The animal didn't cross the street because it was too tired.*
*L'animal n'a pas traversé la rue parce qu'il était trop fatigué.*

*The animal didn't cross the street because it was too wide.*
*L'animal n'a pas traversé la rue parce qu'elle était trop large.*

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

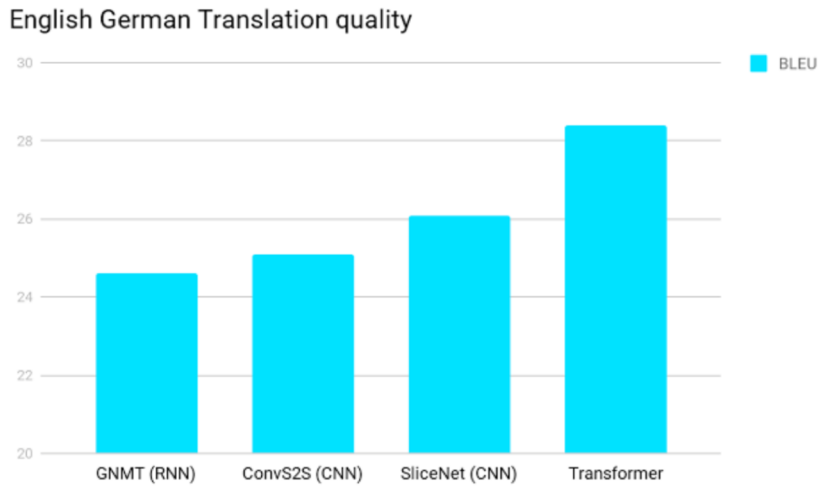
# Transformer



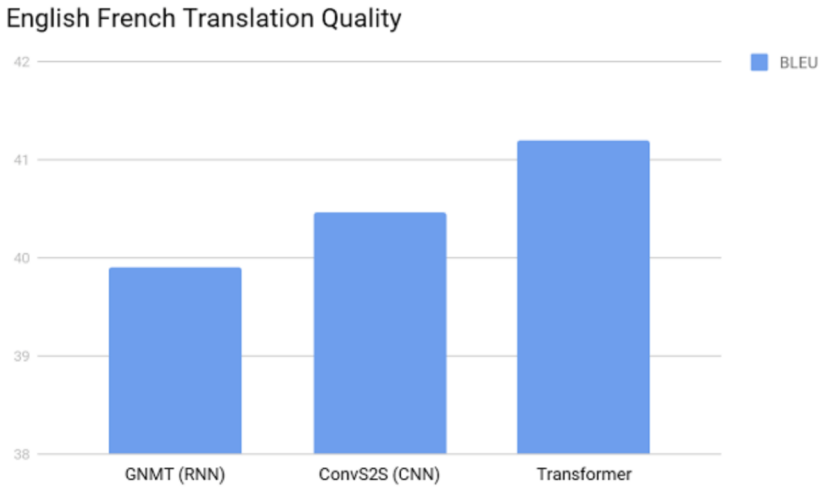

The encoder self-attention distribution for the word “it” from the 5th to the 6th layer of a Transformer trained on English to French translation (one of eight attention heads).

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

# Transformer




BLEU scores (higher is better) of single models on the standard WMT newstest2014 English to German translation benchmark.




BLEU scores (higher is better) of single models on the standard WMT newstest2014 English to French translation benchmark.

Vaswani et al, NIPS 2017, https://papers.nips.cc/paper/7181-attention-is-all-you-need

# What do encoder self-attention heads learn?

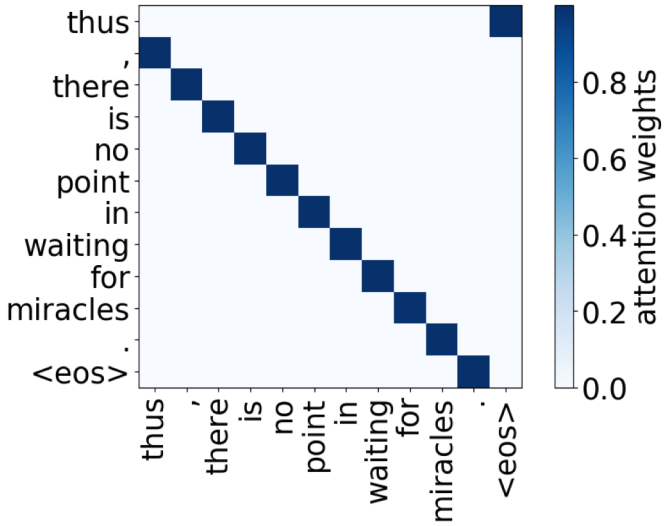
# Analyzing Multi-Head Self-Attention:
## Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned

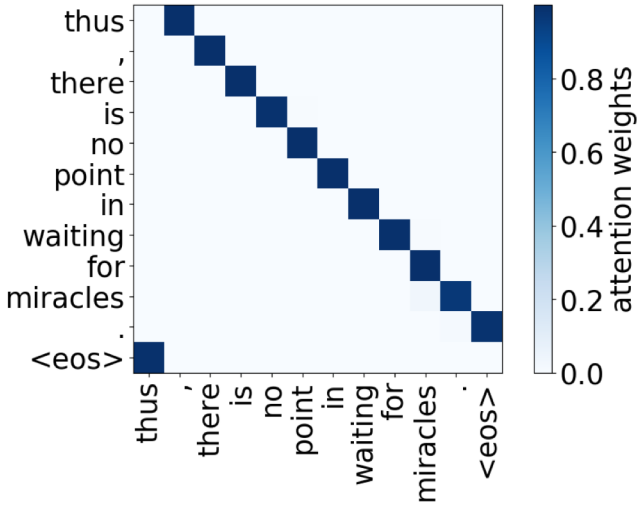
- Only a small subset of heads are important for translation;
- Important heads have one or more specialized and interpretable functions in the model;
- The vast majority of heads can be removed from the trained model without seriously affecting performance.

Voita et al, ACL 2019, https://www.aclweb.org/anthology/P19-1580/
Blog post: https://lena-voita.github.io/posts/acl19_heads.html

# Role 1: positional heads

- at least 90% of the time maximum attention weight is assigned to a specific relative position (in practice either -1 or +1)

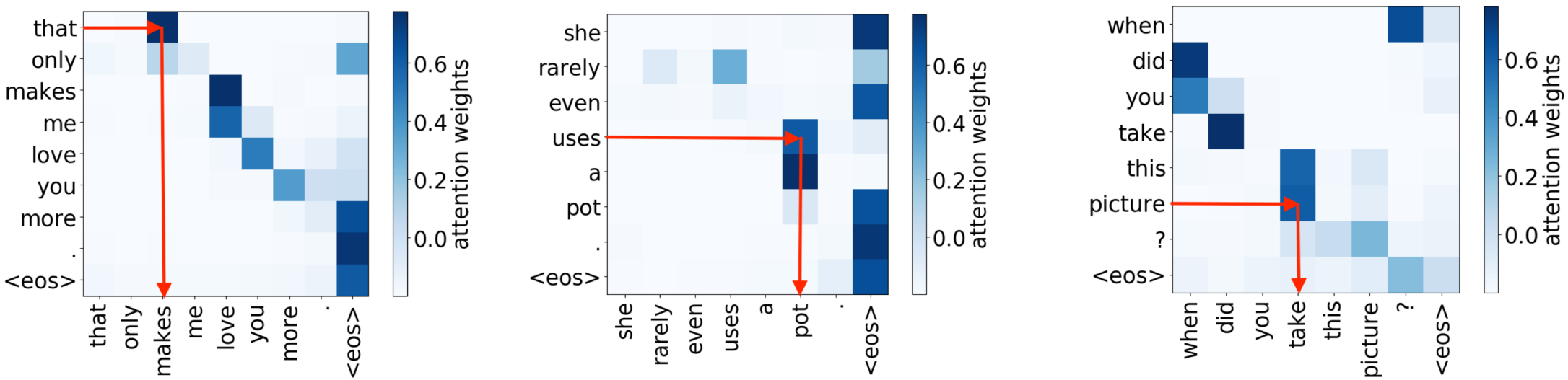



Voita et al, ACL 2019, https://www.aclweb.org/anthology/P19-1580/
Blog post: https://lena-voita.github.io/posts/acl19_heads.html

# Role 2: syntactic heads

- heads which track syntactic relations

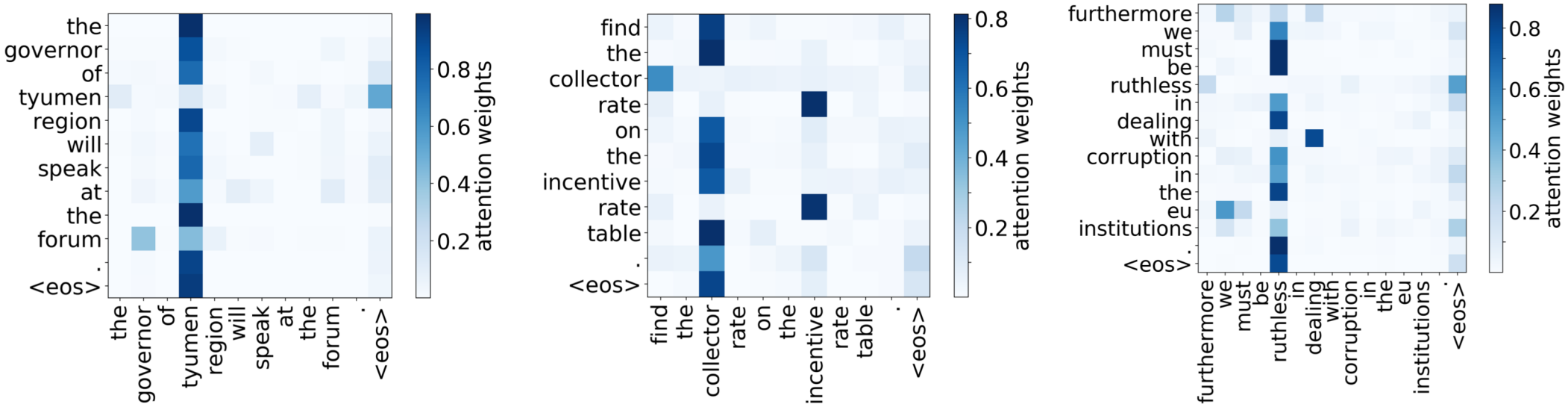



Voita et al, ACL 2019, https://www.aclweb.org/anthology/P19-1580/
Blog post: https://lena-voita.github.io/posts/acl19_heads.html

# Role 3: attention to rare tokens

- In more than 50% of the cases, maximum attn weight points to one of the two least frequent tokens

Models trained on WMT: (a) EN-RU, (b) EN-DE, (c) EN-FR




Voita et al, ACL 2019, https://www.aclweb.org/anthology/P19-1580/
Blog post: https://lena-voita.github.io/posts/acl19_heads.html

# Head roles: different language pairs

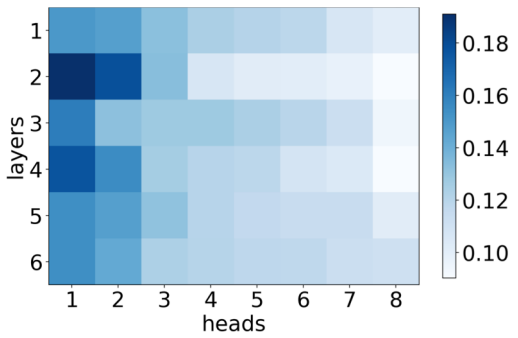
positional

syntactic

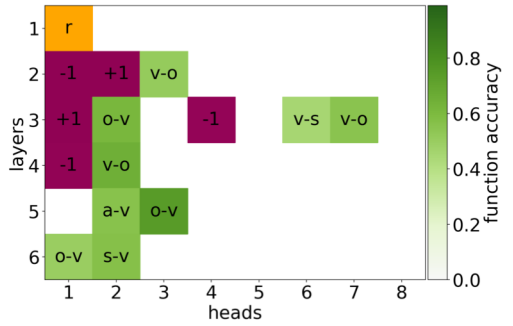
rare tokens

(EN-DE)

Relevance

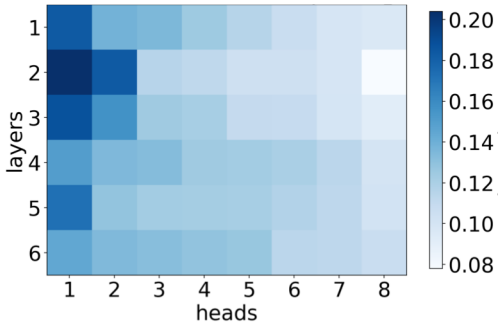



Functions

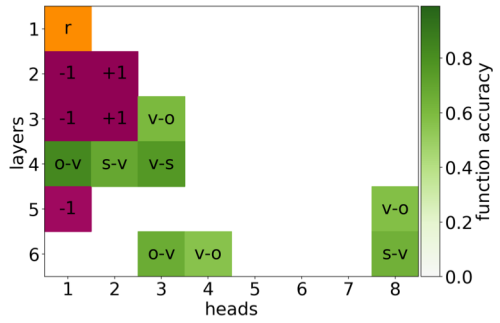



(EN-RU)

Relevance

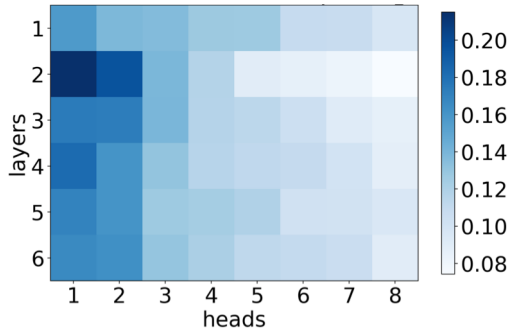



Functions

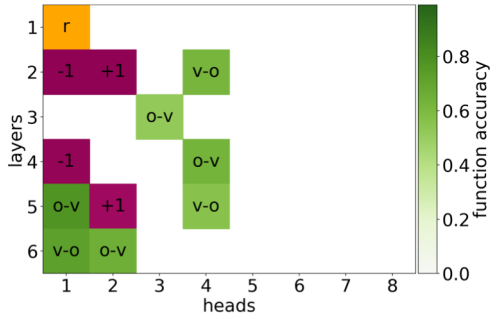



(EN-FR)

Relevance

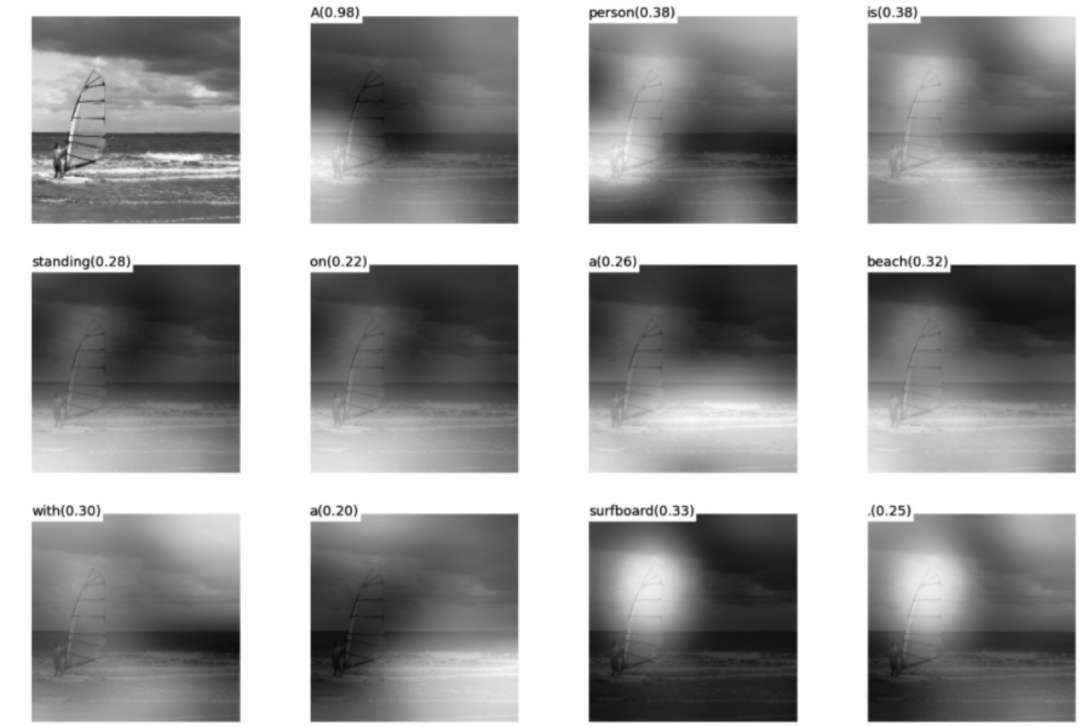



Functions



# Attention:
# Other use cases

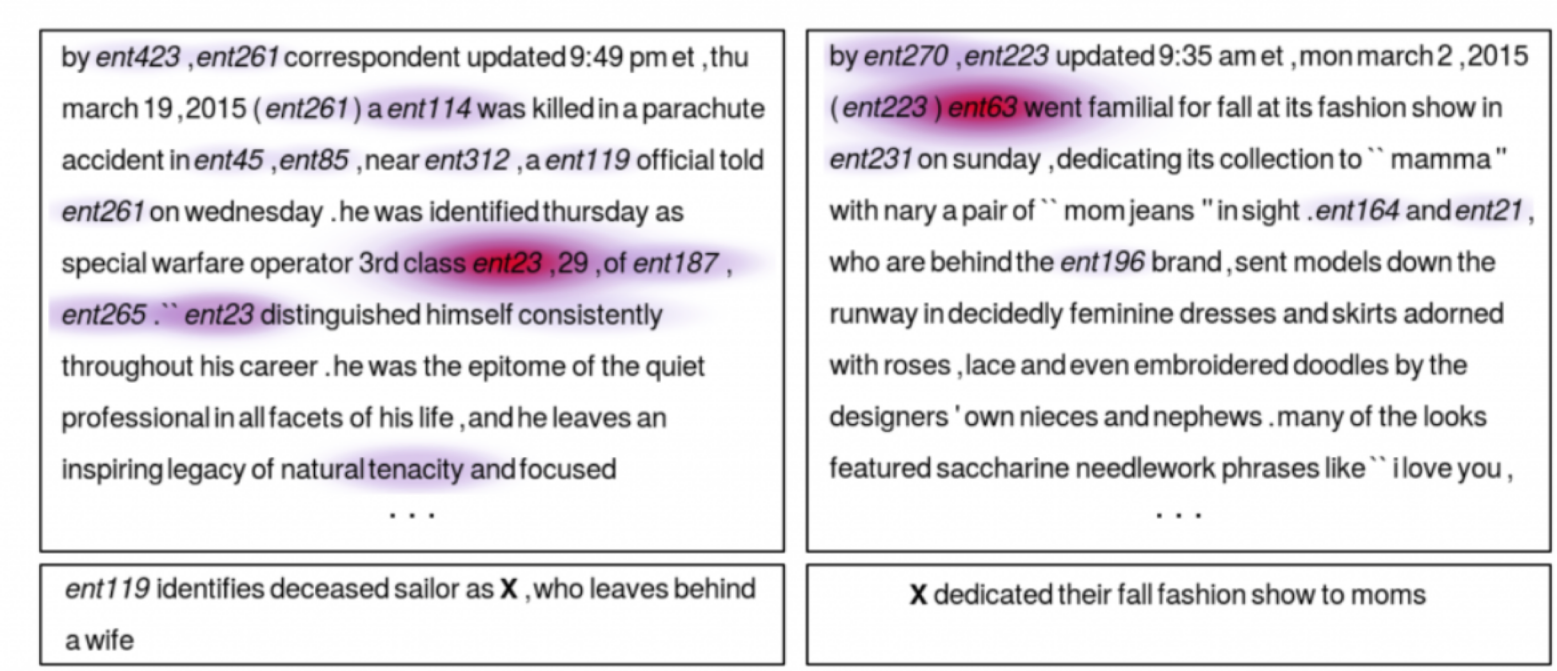
# Image caption generation




(b) A person is standing on a beach with a surfboard.

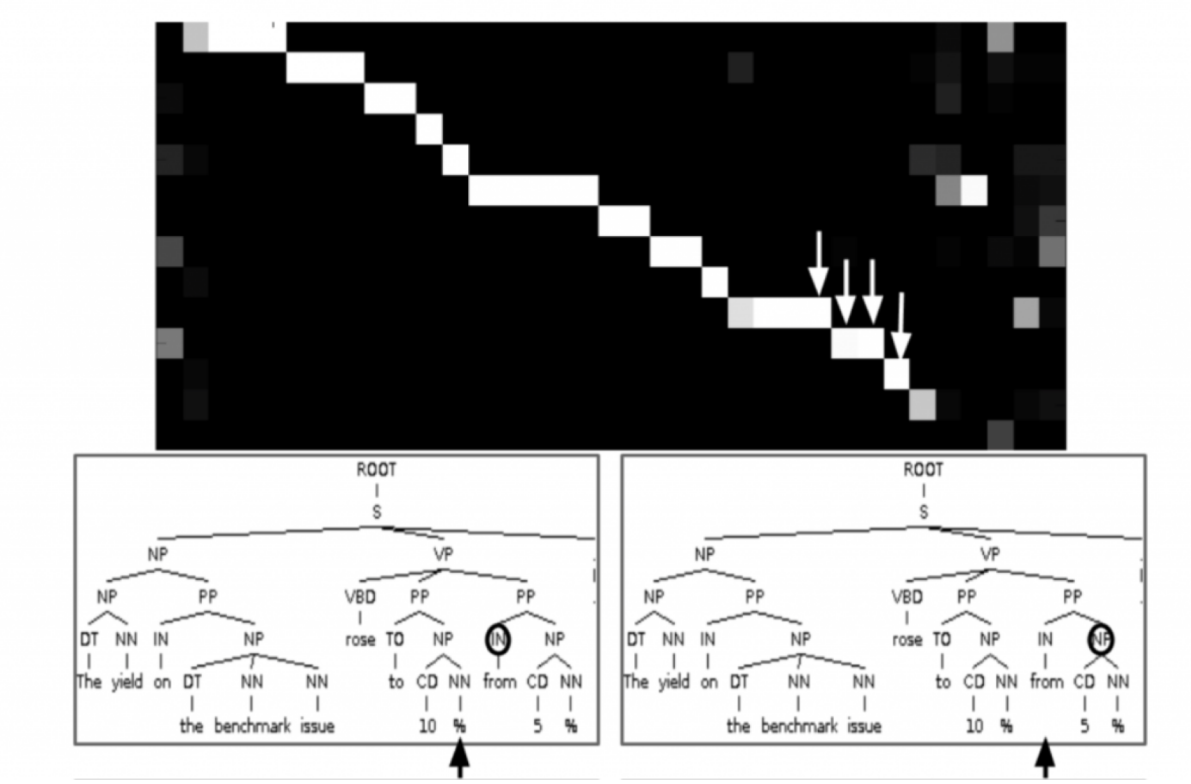
Use a Convolutional Neural Network to “encode” the image, and a Recurrent Neural Network with attention mechanisms to generate a description.

# Question answering task



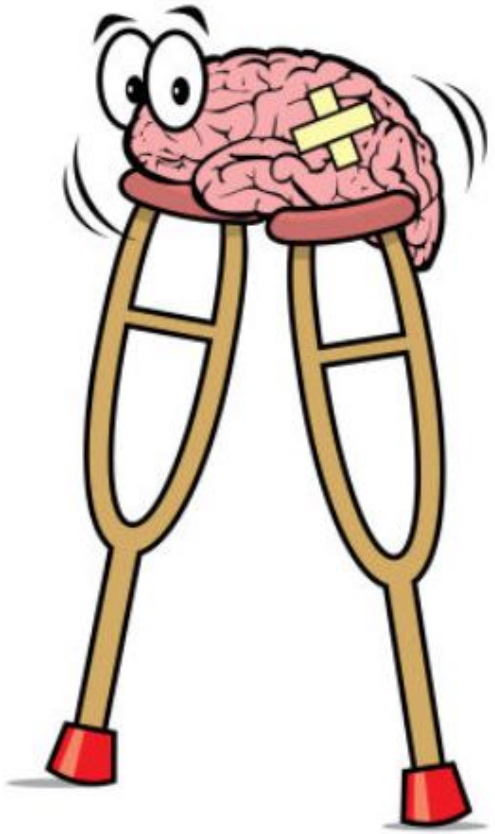

Use a RNN to read a text, read a (synthetically generated) question, and then produce an answer.

# Generate sentence parse trees

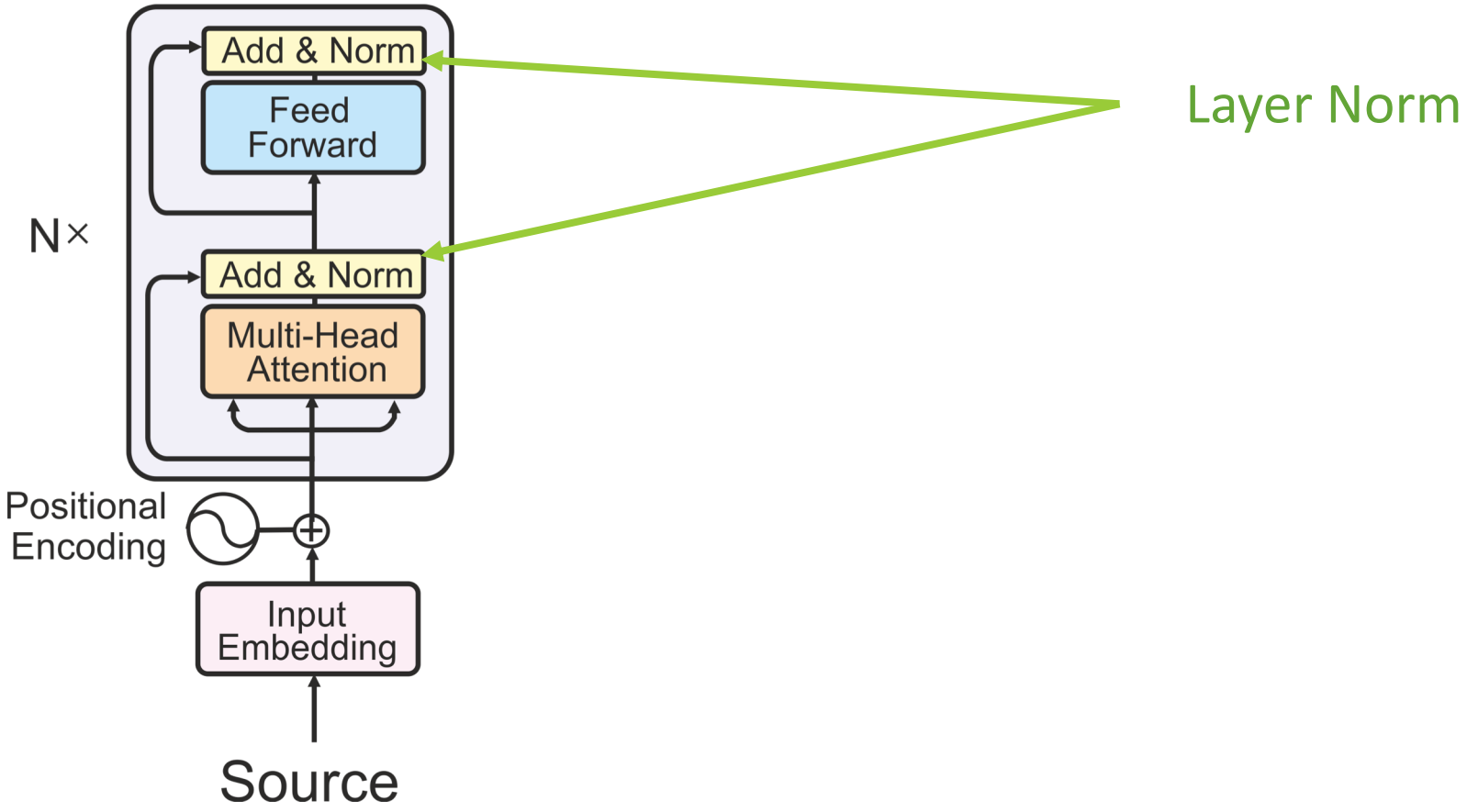



Use a Recurrent Neural Network with attention mechanism to generate sentence parse trees

# Hack of the day



# Layer Norm


Add & Norm
Feed Forward
N×
Add & Norm
Multi-Head Attention
Positional Encoding
Input Embedding
Source
Layer Norm

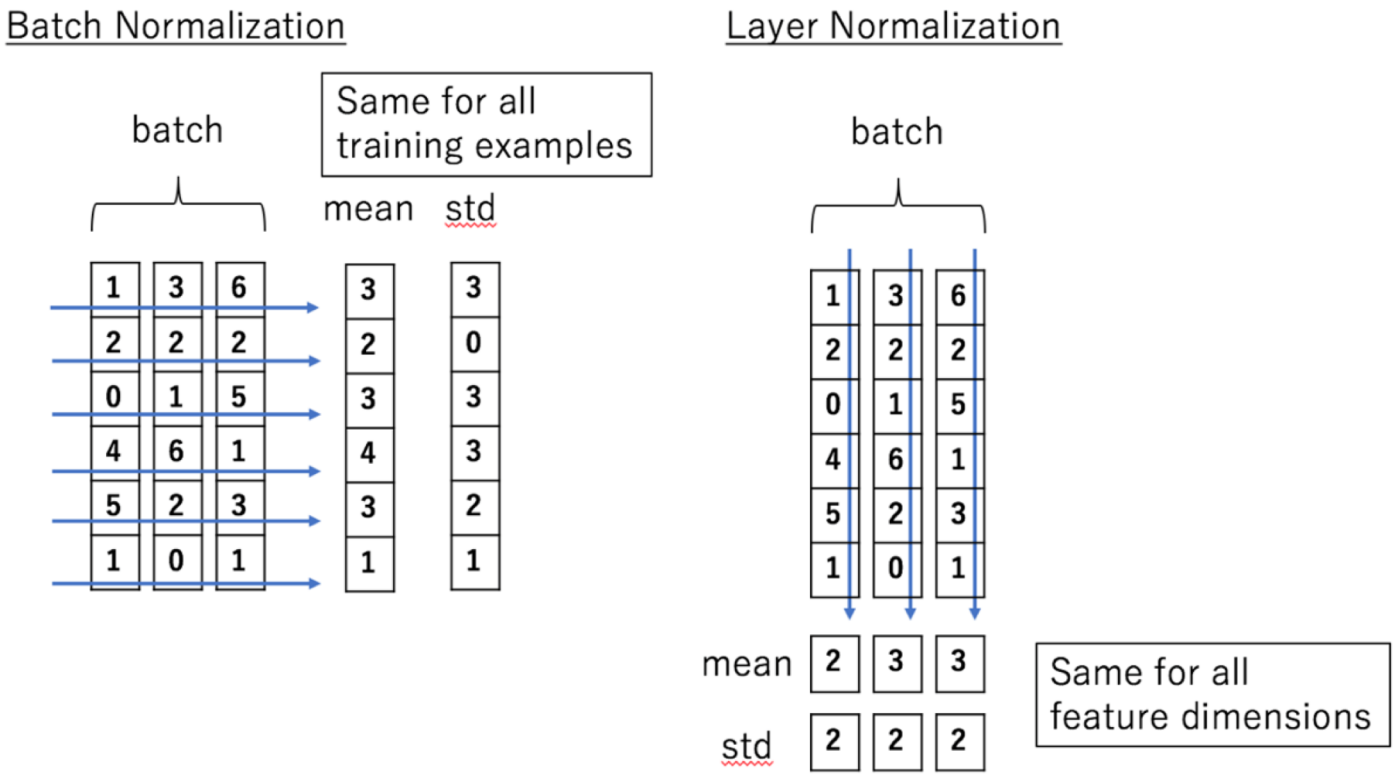
# Layer norm

**Batch normalization**

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_{ij}$$
$$\sigma_j^2 = \frac{1}{m}\sum_{i=1}^{m}(x_{ij} - \mu_j)^2$$
$$\hat{x_{ij}} = \frac{x_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

**Layer normalization:**

$$\mu_i = \frac{1}{m}\sum_{j=1}^{m} x_{ij}$$
$$\sigma_i^2 = \frac{1}{m}\sum_{j=1}^{m}(x_{ij} - \mu_i)^2$$
$$\hat{x_{ij}} = \frac{x_{ij} - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}$$




Paper: Ba et al, 2016, https://arxiv.org/abs/1607.06450

Pictures: http://mlexplained.com/2018/01/13/weight-normalization-and-layer-normalization-explained-normalization-in-deep-learning-part-2/

# That’s all for today!

## Coming soon:

- EM, Machine Translation (specific)
- Structured prediction
- Speech
- Dialogue systems
- and much, much more

Sincerely yours,

Yandex Research